Quantitative Aptitude & Probability

A Placement Preparation Booklet

Counting · Numbers · Arithmetic · Algebra · Geometry · Data Interpretation · Reasoning · Probability · Quant Interviews

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How to use this booklet

This edition has ten parts:

- **Part I – Counting fundamentals.** From the multiplication principle to Catalan numbers, now with "Why it works" boxes that explain the reasoning behind each formula.
- **Part II – Advanced counting.** Recurrences, generating functions, Burnside's lemma, bijections, Stirling numbers and partitions.
- **Part III – Number sense.** Exam strategy, speed maths and estimation, number system, remainders and cyclicity.
- **Part IV – Arithmetic aptitude.** Percentages, profit & loss, interest and financial maths, ratio & partnership, averages, mixtures & alligation, time & work, speed–distance, trains, boats, races, clocks and calendars.
- **Part V – Algebra.** Linear equations and word problems, quadratics, inequalities and optimisation, indices and logarithms, sequences and series.
- **Part VI – Geometry and mensuration.** Triangles, circles and polygons, areas and volumes, coordinate geometry, trigonometry.
- **Part VII – Data interpretation and statistics.** Tables and charts, caselets and Venn logic, descriptive statistics.
- **Part VIII – Logical reasoning.** Ranking, seating, multi-attribute puzzles, blood relations, syllogisms, data sufficiency, series, coding, directions, cubes and dice, formal logic and weighing.
- **Part IX – Probability and quant interviews.** Bayes, expectation tricks, Markov chains, geometric probability, optimal stopping, variance and correlation, the normal distribution and CLT, conditional expectation, random walks, trading and option-pricing questions, maths refreshers and brainteasers.
- **Part X – Mock tests and formula sheets.** Two timed papers and a last-minute revision sheet.

Each chapter follows the same rhythm: idea → why it works → worked examples → traps → exercises. **All exercise solutions are at the end**, organised by chapter. Answers were cross-checked by computer wherever a check was practical (brute-force enumeration, exact fractions, or simulation for the probability puzzles); the simplest one-line results were worked by hand, and each seating/floor puzzle was checked to have exactly one solution.

If you only have a week: do Chapters 18–19, then Percentages to Trains in Part IV, then the first mock test. Use the errors you make in the mock to choose what to read next.

Box legend: Key idea Why it works Common trap Speed shortcut .

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Part I

Counting Fundamentals

1. The Master Decision: What Kind of Problem Is This?

Most counting mistakes in an exam are not arithmetic mistakes. They come from using the *wrong model*. Before writing a single formula, answer these questions:

Question to ask	**What it decides**
Does order matter?	Arrangement (permutation) vs selection (combination)
Can things repeat?	n^r vs ${}^nP_r$; stars & bars vs plain ${}^nC_r$
Are objects / boxes identical or distinct?	Which distribution formula (Chapter 6 table)
Is there an "at least / at most / not"?	Complement or inclusion–exclusion
Is it a circle?	Divide by rotations (and reflections for necklaces)
Does each step's *number* of options stay fixed?	If not, split into cases

The swap test for order. Swap two of the chosen items. If you get a *different* outcome (President and VP swapped; two digits of a PIN swapped), order matters. If you get the *same* outcome (two members of a committee swapped), it does not.

The "who chooses whom" test for repetition. When n things are each assigned one of k options independently, the answer is k^n – the base is the number of *options*, the exponent is the number of *things making a choice*. Five letters into three post boxes: each letter chooses a box, so 3^5, not 5^3.

Four lenses that solve almost everything

1. **Direct construction:** build the object step by step and multiply.
2. **Complement:** count what you don't want and subtract.
3. **Overcount then divide:** count with fake labels, then divide by how many times each real object was counted (repeated letters, circles, unlabelled groups).
4. **Bijection:** translate into a problem you already know (stars and bars, lattice paths, binary strings).

2. Fundamental Principles

2.1 Addition principle (OR)

If a task can be done in *one of* several mutually exclusive ways – m ways of type A **or** n ways of type B – the total is $m + n$.

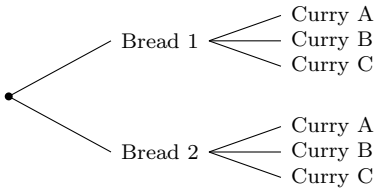
Real life: You travel Chennai→Bengaluru by one of 4 trains or one of 7 buses. Total options $= 4 + 7 = 11$.

The key word is **mutually exclusive**: no option is counted in both groups. If some trains are also listed as buses (impossible here, but think of overlapping categories like "students in finance club" and "students in consulting club"), you need inclusion–exclusion (Chapter 7).

2.2 Multiplication principle (AND)

If a task is a sequence of stages – m ways for stage 1 **and then** n ways for stage 2 – the total is $m \times n$, *provided the number of options at stage 2 is the same whatever was chosen at stage 1.* (The options themselves may differ; only their count must not.)

Real life: A food-delivery combo: 3 breads, 5 curries, 2 drinks. Combos $= 3 \cdot 5 \cdot 2 = 30$.



2 branches, each growing 3: $2 \times 3 = 6$ leaves.
Every leaf is one complete outcome.

Why it works

Draw a tree. Stage 1 has m branches. From *each* of them grow n branches for stage 2. The leaves (complete outcomes) number m groups of n, i.e. mn. If the second-stage count varied (say 5 branches after one choice and 3 after another), the leaves would be $5+3$, not a product – and you must add the cases separately. This is exactly why "split into cases" is needed when digits like 0 behave differently.

2.3 Complement principle

(Wanted) = (Total) − (Unwanted). Use it whenever you see "**at least one**" – the unwanted case ("none") is usually one clean term.

Speed shortcut

Fill the *most restricted* position first. For a 4-digit even number with distinct digits, the units digit and the thousands digit are both restricted – and they interact through 0. Split into cases on the digit that causes the interaction.

Example 2.1

From city A to B there are 3 roads, from B to C there are 4 roads, and there are 2 direct roads from A to C. How many ways to go from A to C?
Solution. Via B: $3 \times 4 = 12$ (AND). Direct: 2. These are alternatives (OR): $12 + 2 = 14$.

Example 2.2

How many 4-digit even numbers have all digits distinct?
Solution. The units digit is restricted (even) and the thousands digit can't be 0.

- Units $= 0$: thousands 9 choices, then $8 \cdot 7$ for the middle: $9 \cdot 8 \cdot 7 = 504$.
- Units $\in \{2, 4, 6, 8\}$: 4 ways; thousands: not 0, not the units digit $\Rightarrow$ 8; middle two: $8 \cdot 7$. Total $4 \cdot 8 \cdot 8 \cdot 7 = 1792$.

Total $= 504 + 1792 =$ **2296**.

Example 2.3

How many 4-digit numbers have at least one repeated digit?
Solution. Total 4-digit numbers $= 9000$. All distinct $= 9 \cdot 9 \cdot 8 \cdot 7 = 4536$. Wanted $= 9000 - 4536 =$ **4464**.

Common trap

In the even-number example, if you fill the thousands digit first (9 ways) and then say "units: 5 even digits", you go wrong, because whether 0 is still available for the units place depends on the first choice. When one choice changes the *number* of options for another, **split into cases**.

Exercises 2

2.1 A restaurant has 4 starters, 6 mains and 3 desserts. A meal must have exactly one main, and may optionally include one starter and optionally one dessert. How many different meals are possible?

2.2 How many 3-digit numbers contain at least one digit 7?

2.3 How many odd numbers between 3000 and 8000 have all four digits distinct?

2.4 In how many ways can 5 letters be posted in 3 letter boxes?

2.5 How many of the integers from 1 to 999 contain no digit 0?

3. Permutations (Arrangements)

3.1 Core formulas

- n distinct objects in a row: $n!$
- r out of n distinct objects in a row: ${}^nP_r = \dfrac{n!}{(n-r)!} = n(n-1)\cdots(n-r+1)$
- With repetition allowed, r positions from n symbols: n^r
- n objects where p are alike of one kind, q alike of another…: $\dfrac{n!}{p!\,q!\cdots}$

Why it works

The slot method. For ${}^nP_r$, think of r empty slots. Slot 1 can take any of n objects, slot 2 any of the remaining $n-1$, …, slot r any of $n-r+1$. Multiply.
Overcount then divide. For repeated letters, pretend the 3 S's in a word are different: S_1, S_2, S_3. Now every real arrangement appears $3! = 6$ times (once for each way of ordering the labels among the S-positions). Dividing by 3! removes the fake distinctions. With several repeated letters the divisions multiply: $\frac{n!}{p!q!\cdots}$.

3.2 Number formation and divisibility

Many questions ask for numbers formed from given digits that are divisible by something. Know the tests:

Divisor	Test	Divisor	Test
2	last digit even	8	last three digits divisible by 8
3	digit sum divisible by 3	9	digit sum divisible by 9
4	last two digits divisible by 4	11	(alternate sum) divisible by 11
5	last digit 0 or 5	6	divisible by 2 and 3

For "divisible by 3 or 9" questions the order of digits doesn't matter – first choose the *set* of digits with the right sum, then arrange.

Example 3.1 (divisibility by 3)

How many 3-digit numbers can be formed from 1, 2, 3, 4, 5 without repetition that are divisible by 3?
Solution. Find 3-element subsets with digit sum divisible by 3: $\{1,2,3\}$ (6), $\{1,3,5\}$ (9), $\{2,3,4\}$ (9), $\{3,4,5\}$ (12). Four subsets, each arranged in 3! ways: $4 \times 6 = \mathbf{24}$.

3.3 Restriction toolkit

Three standard restrictions

1. "Together" → glue. Treat the group as one block; arrange blocks; multiply by the internal arrangements of the block.
2. "No two together" → gap method. Arrange the *unrestricted* items first, then place restricted items in the gaps between them (including both ends). If k items are arranged in a row, there are $k+1$ gaps.
3. "A before B" (not necessarily adjacent) → symmetry. Exactly half of all arrangements have A before B. For A before B before C: 1/3! of all arrangements.

Why it works

Gap method: once the consonants are placed, two vowels are adjacent only if they go into the *same* gap. Putting each vowel into a *different* gap guarantees separation, and every valid arrangement arises this way exactly once.
Symmetry: pair every arrangement with the one obtained by swapping A and B. This pairing is a perfect matching between "A before B" and "B before A" arrangements, so the two classes are equal in size.

Common trap

"Not all together" $\neq$ "no two together". For vowels in a word: "not all together" = total − all together (complement). "No two together" needs the gap method. For three vowels, "AE together but O apart" is neither – it needs its own construction.

Example 3.2 (glue)

In how many arrangements of SOFTWARE are the vowels together?
Solution. Vowels O, A, E form one block. Units: [OAE], S, F, T, W, R = 6 units $\Rightarrow$ 6!. Block internally: 3!. Total = $720 \times 6 =$ **4320**.

Example 3.3 (gap method)

In how many arrangements of SOFTWARE are no two vowels adjacent?
Solution. Arrange 5 consonants: $5! = 120$. They create 6 gaps (_S_F_T_W_R_). Place 3 distinct vowels in 3 different gaps: ${}^{6}P_{3} = 120$. Total = $120 \times 120 =$ **14400**.

Example 3.4 (repeated letters)

How many arrangements of MISSISSIPPI are there?
Solution. 11 letters: I×4, S×4, P×2, M×1. $\dfrac{11!}{4!\,4!\,2!} =$ **34650**.

3.4 Rank of a word (dictionary order)

Very common in TCS/Infosys-style papers. Method: go letter by letter. At each position, count the words that agree with the target so far but have a *smaller* letter at this position. Each such block is "arrangements of the remaining letters". Add all blocks, then add 1 for the word itself.

Example 3.5

Find the rank of SURITI if all its arrangements are listed in dictionary order.

Solution. Letters sorted: I, I, R, S, T, U (I repeated twice).

- Position 1 is S. Words starting with I: remaining $\{I, R, S, T, U\}$, all distinct $\Rightarrow 5! = 120$. Starting with R: remaining $\{I, I, S, T, U\} \Rightarrow 5!/2! = 60$. Running total 180.
- Fix S. Position 2 is U; remaining $\{I, I, R, T, U\}$. Smaller than U: I, R, T. S-I-: $\{I, R, T, U\} \to 4! = 24$; S-R-: $\{I, I, T, U\} \to 12$; S-T-: $\{I, I, R, U\} \to 12$. Total so far $180 + 48 = 228$.
- Fix SU. Position 3 is R; remaining $\{I, I, R, T\}$. Smaller: I. SUI-: $\{I, R, T\} \to 3! = 6$. Total 234.
- Fix SUR. Position 4 is I; remaining $\{I, I, T\}$: nothing smaller. Position 5 is T; remaining $\{I, T\}$. Smaller: I $\to$ SURIIT: 1 word. Total 235.
- SURITI itself is next: rank = **236**.

Speed shortcut

Always use the *distinct* smaller letters. If the remaining pool has two I's, "starting with I" is one block, not two.

Exercises 3

3.1 How many arrangements of BANANA are there? In how many of them are the two N's not adjacent?

3.2 Find the rank of MOTHER in the dictionary listing of its arrangements.

3.3 How many 5-digit numbers formed from 0, 1, 2, 3, 4 (no repetition) are divisible by 4?

3.4 How many arrangements of the word ENGINEERING are possible?

3.5 How many arrangements of ALLAHABAD are possible?

3.6 Seven people stand in a row. In how many arrangements is A somewhere to the left of B?

4. Combinations (Selections)

4.1 Core facts

- ${}^nC_r = \dfrac{n!}{r!(n-r)!} = \dfrac{{}^nP_r}{r!}$
- ${}^nC_r = {}^nC_{n-r}$ (choosing who's in = choosing who's out).
- ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$ (each item: in or out).
- Pascal: ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ (a specific item is either chosen or not).

Why it works

${}^nP_r$ counts ordered lists of r items. Each unordered selection of r items can be listed in $r!$ orders, so it was counted $r!$ times. Hence ${}^nC_r = {}^nP_r/r!$ – another instance of "overcount then divide".

Real life: A placement committee of 5 from 10 students: ${}^{10}C_5 = 252$. A President and VP from 10: ${}^{10}P_2 = 90$ – order matters because the roles differ.

"Select then arrange". When you choose a group and then give them distinct roles, multiply: e.g. choose 5 from 10 and then a chairperson among them: $\binom{10}{5} \times 5 = 1260$. (Equivalently: choose the chair first, 10 ways, then 4 others from 9: $10 \times \binom{9}{4} = 1260$. Two ways of counting the same thing must agree – this "double counting" idea returns in Chapter 16.)

4.2 "At least" selections – split into exact cases

For "at least k of type X", either add the exact cases $k, k+1, \ldots$, or use the complement – pick whichever has fewer terms.

Common trap

The classic wrong method for "at least 2 women": "choose 2 women first (${}^4C_2$), then fill the remaining 3 seats from the other 8 people (${}^8C_3$)". This **overcounts**, because a committee with 3 women is counted once for each choice of which 2 were the "first" two. Always split into exact cases.

Example 4.1

A committee of 5 is formed from 6 men and 4 women. How many committees contain at least 2 women?
Solution. $\binom{4}{2}\binom{6}{3} + \binom{4}{3}\binom{6}{2} + \binom{4}{4}\binom{6}{1} = 6 \cdot 20 + 4 \cdot 15 + 1 \cdot 6 = \mathbf{186}$. (The wrong method in the trap box gives $6 \times 56 = 336$.)

4.3 Selecting from identical objects

If there are p identical items of one type, you can take $0, 1, \ldots, p$ of them: $(p+1)$ choices. For distinct items, each is in or out: a factor 2 each.

Number of selections (at least one item) from p alike, q alike, r alike and n distinct items:

$$(p+1)(q+1)(r+1)\cdot 2^n - 1.$$

Example 4.2

A fruit basket has 3 identical apples, 4 identical oranges and 5 different exotic fruits. How many ways to select at least one fruit?
Solution. $(3+1)(4+1)\cdot 2^5 - 1 = 4\cdot 5\cdot 32 - 1 = \mathbf{639}$.

4.4 Number of divisors

If $N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, a divisor picks an exponent $0..a_i$ for each prime – exactly "selection from identical objects" with the primes as the identical items.

$$\text{Number of divisors} = (a_1+1)(a_2+1)\cdots(a_k+1).$$

Variants: *even* divisors force the exponent of 2 to be ≥ 1; *odd* divisors force it to be 0; *perfect-square* divisors need every exponent even; ways to write N as a product of two factors $= \frac{d(N)}{2}$ (round up if N is a perfect square).

Example 4.3

How many divisors does 1800 have, and how many are even?
Solution. $1800 = 2^3\cdot 3^2\cdot 5^2$. Divisors $= 4\cdot 3\cdot 3 = \mathbf{36}$. Even divisors need exponent of 2 at least 1: $3\cdot 3\cdot 3 = \mathbf{27}$.

4.5 Selection of letters from a word with repeats

Split by *pattern*. For selecting 4 letters: all distinct / one pair + 2 distinct / two pairs / a triple + 1 / four alike. Count each pattern separately; if the question asks for *arrangements* rather than selections, multiply each pattern by $\frac{4!}{\text{repeats!}}$ before adding.

Exercises 4

4.1 At a networking event of 12 people, everyone shakes hands once with everyone else. How many handshakes?

4.2 A student must answer 7 of 10 questions, with at least 3 from the first 5. How many choices?

4.3 How many diagonals does a 12-sided polygon have?

4.4 In how many ways can you choose at least one item from 5 different fruits and 3 identical mangoes?

4.5 In how many ways can 3 numbers be chosen from $\{1, 2, \ldots, 20\}$ so that their sum is even?

4.6 How many selections of 4 letters can be made from the letters of EXAMINATION?

5. Circular Arrangements

5.1 Why $(n-1)!$

In a row there are $n!$ arrangements. Around a round table, rotating everyone one seat gives the *same* seating (only relative positions matter). Each circular arrangement corresponds to n row arrangements, so $n!/n = (n-1)!$.

Why it works

Take A, B, C, D. The row arrangements ABCD, BCDA, CDAB, DABC all describe the *same* circle (everyone has the same left and right neighbour). So the 24 rows collapse in groups of 4 into $24/4 = 6 = 3!$ circles. Practical method: **fix one person's seat** – this kills rotations – then arrange the remaining $n-1$ people as if in a row.

5.2 Necklaces and garlands

If the arrangement can also be *flipped* (beads on a necklace, flowers on a garland), clockwise and anticlockwise readings are the same object: $\dfrac{(n-1)!}{2}$ for $n \geq 3$. When beads are *not* all distinct, this simple formula fails – use Burnside's lemma (Chapter 15).

Common trap

People around a table: $(n-1)!$, **not** divided by 2 – a person's left neighbour and right neighbour are different, unless the question says clockwise and anticlockwise are the same. Seats that are numbered/distinguishable (a table with labelled chairs) $\Rightarrow$ it's just a row: $n!$.

Example 5.1

8 people sit around a round table. In how many ways can they sit if two particular people must sit together?
Solution. Glue them: 7 units around a circle $\Rightarrow 6!$. Internal order: 2. Total $= 720 \times 2 =$ **1440**.

Example 5.2

How many different necklaces can be made with 7 distinct beads?
Solution. $\dfrac{(7-1)!}{2} =$ **360**.

Example 5.3 (alternating)

5 men and 5 women sit around a table, alternating. How many ways?
Solution. Seat the men around the circle: $4!$. This creates 5 *distinguishable* gaps between men (each gap is identified by its neighbours), so the women fill them like a row: $5!$. Total $= 24 \times 120 =$ **2880**.

Exercises 5

5.1 10 people sit at a round table. In how many ways can they sit if 3 specific people must sit together?

5.2 How many garlands can be made from 8 different flowers?

5.3 6 people sit at a round table. In how many ways can they sit if two specific people must *not* sit next to each other?

5.4 Five boys and five girls sit at a round table such that boys and girls alternate. How many arrangements? (Re-derive without looking at the example.)

5.5 Four married couples sit at a round table. In how many ways can they sit if (a) every couple sits together; (b) men and women alternate?

6. Distributions: Stars & Bars and Grouping

6.1 Identical objects into distinct boxes – stars and bars

Distribute n identical items among r distinct people. Write n stars and $r-1$ bars; each arrangement is one distribution.

$$\text{Empty allowed: } \binom{n+r-1}{r-1} \qquad \text{Each gets at least 1: } \binom{n-1}{r-1}$$

Equivalent: number of non-negative integer solutions of $x_1 + \cdots + x_r = n$.

Why it works

Seven identical sweets for three children: the string

$$\star\star \mid \star\star\star\star \mid \star \quad \text{means} \quad (2,4,1), \qquad \star\star\star\star\star\star\star \mid\mid \quad \text{means} \quad (7,0,0).$$

Every distribution corresponds to exactly one string of 7 stars and 2 bars, and vice versa. The number of such strings is the number of ways to choose the 2 bar positions among 9 symbols: $\binom{9}{2} = 36$.
At least one each: line the 7 stars up; there are 6 internal gaps. Choosing 2 different gaps for the bars guarantees no empty group: $\binom{6}{2} = 15$. (Equivalently, hand one sweet to each child first, then distribute the remaining 4 freely: $\binom{4+2}{2} = 15$.)

Real life: 10 identical Rs. 100 gift vouchers among 4 interns. If someone may get none: $\binom{13}{3} = 286$.

Lower bounds ($x_1 \geq 2$): pre-give 2 to that person, reduce n. **Upper bounds** ($x_1 \leq 3$): use complement / inclusion–exclusion (Chapter 7) or generating functions (Chapter 14).

Example 6.1

How many non-negative integer solutions does $x + y + z = 10$ have?
Solution. $\binom{12}{2} = \mathbf{66}$.

Example 6.2

How many positive integer solutions does $a + b + c + d = 12$ have?
Solution. $\binom{11}{3} = \mathbf{165}$.

6.2 Distinct objects into groups

- Into groups of sizes p, q, r (all different sizes): $\dfrac{n!}{p!\,q!\,r!}$.
- Into k groups of *equal* size m (groups unlabelled): $\dfrac{(km)!}{(m!)^k\,k!}$.
- If those equal groups are then *given to k different people*: multiply by $k!$, which cancels the $k!$.

Why it works

Line all n objects up in $n!$ ways and cut the line into consecutive chunks of the required sizes. Shuffling inside a chunk doesn't change the group, so divide by $p!\,q!\,r!$. If several chunks have the same size and the groups are unlabelled, swapping whole chunks also doesn't change anything – divide again by $k!$ for k equal-sized groups.

Common trap

Dividing 8 people into two teams of 4 for a match: $\binom{8}{4}/2 = 35$, not 70. Choosing team A automatically makes team B, and "{A,B,C,D} vs rest" is the same split as "rest vs {A,B,C,D}". If the teams have names (Red team / Blue team), it is 70.

Example 6.3

In how many ways can 12 different books be divided into 3 bundles of 4? Into 3 bundles given to 3 students?

Solution. $\dfrac{12!}{(4!)^3\,3!} = \mathbf{5775}$. Given to 3 students: $5775 \times 3! = \mathbf{34650}$.

6.3 The four distribution cases (summary)

Objects	Boxes	Count (n objects, r boxes, empty allowed)
Distinct	Distinct	r^n
Identical	Distinct	$\binom{n+r-1}{r-1}$
Distinct	Identical	$S(n,1) + \cdots + S(n,r)$ (Stirling numbers, Ch. 16)
Identical	Identical	Partitions of n into at most r parts (Ch. 16)

For the last two rows, aptitude questions keep numbers small: **list the cases systematically**. Add "no box empty" by using $S(n,r)$ alone, or partitions into *exactly* r parts.

Example 6.4

In how many ways can 6 identical balls be put into 3 identical boxes (empty allowed)?

Solution. List partitions of 6 into at most 3 parts: 6; 5+1; 4+2; 3+3; 4+1+1; 3+2+1; 2+2+2. Total **7**.

Exercises 6

6.1 10 identical chocolates are distributed among 4 children so that each gets at least one. How many ways?

6.2 How many non-negative integer solutions of $a + b + c + d = 15$ have $a \geq 2$ and $b \geq 3$?

6.3 In how many ways can 9 different people be divided into 3 groups of 3?

6.4 In how many ways can 4 different toys be put into 2 identical boxes (a box may be empty)?

6.5 How many distinct terms are there in the expansion of $(a + b + c + d)^8$?

6.6 How many 3-digit numbers have digit sum 10?

6.7 How many non-negative integer solutions of $x + y + z = 6$ have each variable at most 3?

7. Inclusion–Exclusion

7.1 The formula

For two sets: $|A \cup B| = |A| + |B| - |A \cap B|$.
For three: $|A \cup B \cup C| = \sum |A| - \sum |A \cap B| + |A \cap B \cap C|$.
General pattern: add singles, subtract pairs, add triples, subtract quadruples, ...

Why it works

Take an element that lies in exactly $k \geq 1$ of the sets. The singles count it $\binom{k}{1}$ times, the pairs $\binom{k}{2}$ times, the triples $\binom{k}{3}$ times, and so on. Net count:

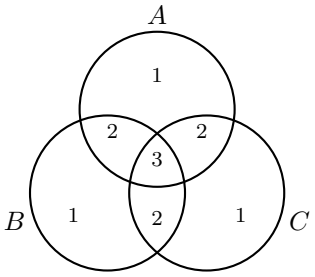
$$\binom{k}{1} - \binom{k}{2} + \binom{k}{3} - \cdots = 1 - (1-1)^k = 1.$$

So every element of the union is counted exactly once, and elements in none of the sets are counted zero times.

Real life: In a batch, 60 students opted for a finance elective, 45 for marketing, and 25 for both. Students with at least one of these: $60 + 45 - 25 = 80$.

Useful forms:

- "None of the properties" $=$ Total $-|A \cup B \cup \cdots|$.
- **Exactly one** of three sets: $\sum |A| - 2\sum |A \cap B| + 3|A \cap B \cap C|$.
- **Exactly two** of three sets: $\sum |A \cap B| - 3|A \cap B \cap C|$.



How many times $\sum |A|$ counts each region. Subtracting the pairs removes one count from each "2" region and three from the centre. Adding the triple restores the centre to 1.

7.2 Onto (surjective) functions

Distribute n distinct items to r distinct people, **everyone gets at least one**:

$$r^n - \binom{r}{1}(r-1)^n + \binom{r}{2}(r-2)^n - \cdots$$

(Subtract assignments where a particular person is left out, add back where two are left out, ...)

Example 7.1

How many integers from 1 to 1000 are divisible by 2, 3 or 5?
Solution. $500 + 333 + 200 - (166 + 100 + 66) + 33 = 1033 - 332 + 33 = \mathbf{734}$.

Example 7.2

In how many ways can 5 different projects be assigned to 3 analysts so that each analyst gets at least one?
Solution. $3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 = 243 - 96 + 3 = \mathbf{150}$.

Common trap

Answering "each gets at least one" by "give one each first, then distribute the rest freely" (${}^5P_3 \cdot 3^2$) is the same overcounting error as in Chapter 4: the project given "first" is an artificial label. Use inclusion–exclusion.

Exercises 7

7.1 Of 100 students, 60 study finance, 45 study marketing and 25 study both. How many study neither?

7.2 How many integers from 1 to 500 are divisible by neither 3 nor 7?

7.3 In how many ways can 6 different tasks be assigned to 3 analysts so that each gets at least one?

7.4 In how many arrangements of A, B, C, D, E is A not in position 1 and B not in position 2?

7.5 How many non-negative integer solutions of $x + y + z = 15$ have each variable at most 6?

8. Derangements

A derangement is an arrangement where **nobody is in their own place** – letters in wrong envelopes, a Secret Santa where nobody draws their own name.

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right), \qquad D_n = (n-1)(D_{n-1} + D_{n-2}).$$

Why it works

Formula: inclusion–exclusion over the events "person i gets their own item". Fixing a specific set of k people in place leaves $(n-k)!$ arrangements, and there are $\binom{n}{k}$ such sets: $D_n = \sum_k (-1)^k \binom{n}{k}(n-k)! = n!\sum_k \frac{(-1)^k}{k!}$.
Recurrence: letter 1 goes into some wrong envelope k ($n-1$ choices). Now either letter k goes into envelope 1 (a swap; the other $n-2$ letters must be deranged: D_{n-2}), or it doesn't (then treat envelope 1 as letter k's "forbidden" envelope, and the $n-1$ letters other than 1 form a derangement: D_{n-1}).

n	1	2	3	4	5	6	7
D_n	0	1	2	9	44	265	1854

Memorise this table up to $n = 6$ – it is faster than the formula in an exam. A sanity check: $D_n \approx n!/e$, rounded to the nearest integer. So the probability that a random permutation is a derangement is $\approx 1/e \approx 0.368$, for any reasonably large n.

Exactly k in the right place: choose which k are correct, derange the rest: $\binom{n}{k} D_{n-k}$.

Example 8.1

4 letters are put into 4 addressed envelopes at random. In how many ways do all go into wrong envelopes?
Solution. $D_4 = \mathbf{9}$.

Example 8.2

In how many ways can 5 letters be placed so that exactly 2 are in the correct envelopes?
Solution. $\binom{5}{2} D_3 = 10 \times 2 = \mathbf{20}$.

Exercises 8

8.1 In a 5-person Secret Santa, in how many ways can nobody draw their own name?

8.2 6 letters, 6 envelopes. In how many ways are exactly 3 letters correctly placed?

8.3 6 people leave their bags at a counter and get them back at random. In how many ways does at least one person get their own bag?

8.4 7 students' answer sheets are randomly redistributed for peer grading. In how many ways does no student get their own sheet?

9. Geometry and Grid Counting

9.1 Points, lines, triangles

From n points with no three collinear: lines $= \binom{n}{2}$, triangles $= \binom{n}{3}$.
If m of them are collinear: lines $= \binom{n}{2} - \binom{m}{2} + 1$; triangles $= \binom{n}{3} - \binom{m}{3}$.
Diagonals of an n-gon: $\binom{n}{2} - n = \dfrac{n(n-3)}{2}$.

Why it works

m collinear points would give $\binom{m}{2}$ "lines" if they were in general position, but they all lie on one line – so remove $\binom{m}{2}$ and add back 1. Any 3 of them form a degenerate (flat) triangle, so remove $\binom{m}{3}$.

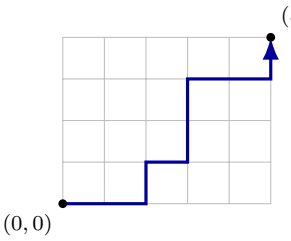
9.2 Rectangles and squares

A grid of $m \times n$ unit squares has $(m+1)$ vertical and $(n+1)$ horizontal lines. A rectangle is fixed by 2 vertical lines and 2 horizontal lines: $\binom{m+1}{2}\binom{n+1}{2}$.
Squares on an $n \times n$ board: $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ (there are $(n-k+1)^2$ squares of side k).

9.3 Lattice paths

Moving only Right or Up from $(0,0)$ to (m,n): a path is a word with m R's and n U's: $\binom{m+n}{m}$.
Through point P: (paths to P)×(paths from P). Avoiding P: total − through P.

Real life: In a grid of city blocks, the number of shortest walking routes between two corners is a lattice-path count.



This path spells RRURUURRU.
A path is any word with 5 R's and 4 U's,
so the count is the number of ways of choosing
where the U's go: $\binom{9}{4} = 126$.

Example 9.1

10 points in a plane, of which exactly 4 are collinear. How many lines and how many triangles?
Solution. Lines: $45 - 6 + 1 = \mathbf{40}$. Triangles: $120 - 4 = \mathbf{116}$.

Example 9.2

How many rectangles and how many squares are on a chessboard?
Solution. Rectangles: $\binom{9}{2}^2 = 36^2 = \mathbf{1296}$. Squares: $1 + 4 + 9 + \cdots + 64 = \mathbf{204}$.

Example 9.3

Shortest paths from $(0,0)$ to $(5,4)$: how many in total, and how many pass through $(2,2)$?
Solution. Total $\binom{9}{4} = \mathbf{126}$. Through $(2,2)$: $\binom{4}{2} \cdot \binom{5}{2} = 6 \cdot 10 = \mathbf{60}$.

Exercises 9

9.1 12 points lie in a plane, 5 of which are collinear (no other 3 collinear). How many triangles can be formed?

9.2 How many rectangles are there in a 5×4 grid of unit squares?

9.3 How many triangles can be formed using vertices of a regular decagon such that no side of the triangle is a side of the decagon?

9.4 A set of 4 parallel lines intersects another set of 5 parallel lines. How many parallelograms are formed?

9.5 How many shortest paths from $(0,0)$ to $(6,6)$ avoid the point $(3,3)$?

10. Catalan Numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n-1}: \qquad 1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132,\ \ldots \quad (n = 0, 1, 2, \ldots)$$

Why it works: the reflection principle

Count paths from $(0,0)$ to (n,n) (R/U steps) that never go *above* the diagonal $y = x$. Total paths: $\binom{2n}{n}$. A bad path touches the line $y = x+1$ at some first point. Reflect the part of the path *after* that first touch across $y = x+1$ (swap R and U). The reflected path ends at $(n-1, n+1)$. This reflection is a one-to-one correspondence between bad paths and *all* paths to $(n-1, n+1)$, of which there are $\binom{2n}{n-1}$. So good paths $= \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}$.

The same number appears in many disguises – the skill is **recognising** them:

- Balanced arrangements of n pairs of brackets.
- Lattice paths from $(0,0)$ to (n,n) that never go above the diagonal.
- A queue of $2n$ people, n holding Rs. 100 and n holding Rs. 200 notes, ticket Rs. 100, cashier starts with no change: orders in which the cashier never runs out of change.
- Ways to triangulate a convex polygon with $n+2$ sides.
- $2n$ people round a table shaking hands in pairs with no arms crossing.

Common thread: a sequence of n "up" and n "down" moves where the running count must never go negative.

Example 10.1

How many balanced strings of 3 pairs of parentheses are there?
Solution. $C_3 = 5$: ((())), (()()), (())(), ()(()), ()()().

Example 10.2

8 people queue for a Rs. 100 ticket; 4 have Rs. 100 notes and 4 have Rs. 200 notes. The cashier starts with no change. Treating people with the same note as identical, how many orderings let the cashier always give change?
Solution. At every point, Rs. 100-holders so far $\geq$ Rs. 200-holders so far. This is $C_4 = \mathbf{14}$. (If people are distinct, multiply by $4! \cdot 4!$.)

Exercises 10

10.1 In how many ways can a convex hexagon be divided into triangles by non-intersecting diagonals?

10.2 10 people queue with 5 holding Rs. 100 and 5 holding Rs. 200 notes (otherwise as in the example). How many valid orderings (same-note people identical)?

10.3 How many shortest paths from $(0,0)$ to $(5,5)$ never go above the line $y = x$?

10.4 6 people sit around a round table and simultaneously shake hands in 3 pairs so that no two handshakes cross. How many ways?

11. Pigeonhole Principle

If more than $n \cdot k$ items go into n boxes, some box has at least $k+1$ items. General form: if N items go into n boxes, some box has at least $\lceil N/n \rceil$ items. In aptitude tests this appears as **"minimum number to guarantee"** questions.

Method: build the worst case. Imagine the most unlucky sequence of draws that still fails, then add 1. In proofs, the art is choosing the right "boxes".

Example 11.1

A drawer has 10 black and 10 white socks. Minimum socks drawn (in the dark) to guarantee a matching pair?
Solution. Worst case: 1 black, 1 white. The next one must match. **3**. To guarantee a *black* pair: worst case all 10 white, then 2 black $\Rightarrow$ **12**.

Example 11.2

How many people are needed to guarantee that two share a birth month?
Solution. 12 people could all differ; **13** guarantees a repeat.

Exercises 11

11.1 How many people must be in a room to guarantee that at least 3 share a birth month?

11.2 Show that if 6 numbers are chosen from $\{1, 2, \ldots, 10\}$, two of them must sum to 11.

11.3 A bag has 5 red, 7 blue and 9 green balls. (a) Minimum draws to guarantee 4 of the same colour? (b) Minimum draws to guarantee at least one of each colour?

12. Number-Based Counting

12.1 Sum of all numbers formed by permuting digits

With n distinct non-zero digits used once each, each digit appears $(n-1)!$ times in every place:

$$\text{Sum} = (n-1)! \times (\text{sum of digits}) \times \underbrace{111\ldots1}_{n}.$$

If 0 is among the digits, subtract the "numbers" that start with 0 (compute them the same way with $n-1$ digits).

Why it works

Fix a place, say the units place. By symmetry each of the n digits sits there in exactly $(n-1)!$ of the $n!$ arrangements. So the units column adds up to $(n-1)!\times$(digit sum). The tens column is the same times 10, and so on, giving the factor $1+10+100+\cdots = 111\ldots1$.

12.2 Highest power of a prime in $n!$ (Legendre)

$$\text{Power of } p \text{ in } n! = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

$\lfloor n/p \rfloor$ counts the multiples of p up to n (one factor each); multiples of p^2 contribute an extra factor, counted by $\lfloor n/p^2 \rfloor$, and so on. Trailing zeros of $n!$ = power of 5 (there are always more 2's). For a composite like $12 = 2^2 \cdot 3$: find powers of 2 and 3, then take $\min(\lfloor e_2/2 \rfloor,\ e_3)$.

Example 12.1

Find the sum of all 4-digit numbers formed using 1, 2, 3, 4 without repetition. Repeat with 0, 1, 2, 3.

Solution. $3! \times 10 \times 1111 = \mathbf{66660}$.

With 0, 1, 2, 3: all 24 arrangements give $6 \times 6 \times 1111 = 39996$. Arrangements starting with 0 are 3-digit arrangements of 1, 2, 3: $2! \times 6 \times 111 = 1332$. Answer $39996 - 1332 = \mathbf{38664}$.

Example 12.2

Number of trailing zeros in 100!, and the highest power of 12 dividing 40!.

Solution. $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = \mathbf{24}$.

In 40!: power of $2 = 20+10+5+2+1 = 38$; power of $3 = 13+4+1 = 18$. 12^k needs $2^{2k}3^k$: $\min(19, 18) = \mathbf{18}$.

Exercises 12

12.1 Find the sum of all 3-digit numbers formed using 2, 5, 7 without repetition.

12.2 How many trailing zeros does 250! have?

12.3 What is the highest power of 2 that divides 30!?

12.4 What is the highest power of 3 that divides 50!?

12.5 How many times is the digit 3 written when listing the integers from 1 to 100?

12.6 How many digits are used to number the pages of a 250-page report?

Part II

Advanced Counting

13. Recurrence Relations

When a direct formula is hard to see, count **small cases** and find how the answer for size n depends on answers for smaller sizes. This is the single most powerful technique in combinatorics, and it is also how dynamic programming works in coding rounds.

The recipe

1. Let a_n be the answer for size n.
2. Look at the **first (or last) step** and split into cases by what happens there.
3. In each case, recognise the rest as a smaller instance of the same problem.
4. Write down base cases (a_0, a_1, ...) by hand, then iterate. In an exam, a table of 8–10 values takes under a minute.

13.1 The Fibonacci family

Example 13.1 (staircase)

You climb a 10-step staircase taking 1 or 2 steps at a time. How many ways?
Solution. Look at the *last* move. If it was a 1-step, before it you climbed $n-1$ steps (a_{n-1} ways); if it was a 2-step, you climbed $n-2$ (a_{n-2} ways). So $a_n = a_{n-1} + a_{n-2}$, with $a_1 = 1$, $a_2 = 2$:

$$1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ 55,\ 89. \qquad a_{10} = \mathbf{89}.$$

Example 13.2 (binary strings)

How many binary strings of length 8 have no two consecutive 1s?
Solution. Look at the first character. If it is 0, the rest is any valid string of length $n-1$. If it is 1, the next must be 0, and the rest is any valid string of length $n-2$. So $b_n = b_{n-1} + b_{n-2}$, $b_1 = 2$ (0, 1), $b_2 = 3$ (00, 01, 10):

$$2,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ 55. \qquad b_8 = \mathbf{55}.$$

Example 13.3 (domino tiling)

In how many ways can a 2×8 board be tiled with 1×2 dominoes?
Solution. Look at the leftmost column. Either a vertical domino covers it (leaving $2\times(n-1)$), or two horizontal dominoes stacked cover the first two columns (leaving $2 \times (n-2)$). A single horizontal domino can't be alone – the cell below it would be stranded. So $t_n = t_{n-1} + t_{n-2}$, $t_1 = 1$, $t_2 = 2$, and $t_8 = \mathbf{34}$.

Seeing the common structure

All three problems are "sequences built from pieces of size 1 and size 2". Whenever the objects are sequences of pieces, the recurrence is $a_n = \sum_{\text{piece sizes } s} a_{n-s}$. With sizes $\{1, 2, 3\}$ you get the "Tribonacci" recurrence $a_n = a_{n-1} + a_{n-2} + a_{n-3}$.

13.2 Other classic recurrences

- **Tower of Hanoi:** $h_n = 2h_{n-1} + 1$ (move $n-1$ disks off, move the largest, move $n-1$ back on) $\Rightarrow h_n = 2^n - 1$.
- **Regions made by n lines** (no two parallel, no three concurrent): the n-th line crosses the previous $n-1$ lines at $n-1$ points, is cut into n pieces, and each piece splits a region in two: $r_n = r_{n-1} + n \Rightarrow r_n = 1 + \frac{n(n+1)}{2}$.
- **Regions made by n circles** (each pair meets in 2 points, no three through a point): the n-th circle meets the others in $2(n-1)$ points, is cut into $2(n-1)$ arcs: $c_n = c_{n-1} + 2(n-1)$, $c_1 = 2 \Rightarrow c_n = n^2 - n + 2$.
- **Derangements:** $D_n = (n-1)(D_{n-1} + D_{n-2})$ (Chapter 8).
- **Catalan:** $C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}$ (split at the bracket that matches the first one).

13.3 Recurrences with states

Sometimes one sequence isn't enough; track several "states".

Example 13.4 (strings with a forbidden pattern)

How many strings of length n over $\{A, B, C\}$ have no two consecutive A's? Find the answer for $n = 4$.

Solution. Let x_n = valid strings ending in A, y_n = valid strings ending in B or C.

$$x_n = y_{n-1} \text{ (append A only after a non-A)}, \qquad y_n = 2(x_{n-1} + y_{n-1}).$$

Start: $x_1 = 1$, $y_1 = 2$. Then $(x_2, y_2) = (2, 6)$, $(x_3, y_3) = (6, 16)$, $(x_4, y_4) = (16, 44)$. Answer for $n = 4$: $16 + 44 = \mathbf{60}$. Eliminating x gives a single recurrence $a_n = 2a_{n-1} + 2a_{n-2}$ (check: $3, 8, 22, 60$).

13.4 Solving linear recurrences (closed forms)

For $a_n = pa_{n-1} + qa_{n-2}$, solve the **characteristic equation** $x^2 = px + q$. If the roots $\alpha \neq \beta$, then $a_n = A\alpha^n + B\beta^n$, with A, B fixed by the base cases. (Repeated root α: $a_n = (A + Bn)\alpha^n$.) For Fibonacci, $x^2 = x + 1$ gives $\alpha, \beta = \frac{1 \pm \sqrt{5}}{2}$ – the golden ratio appears. In aptitude tests you almost never need the closed form; iterate the table instead. In quant interviews, knowing that solutions grow like α^n (the largest root) is often enough.

Common trap

Get the base cases right – most errors are there. Check by brute-forcing $n = 1, 2, 3$ by hand. For the staircase, $a_0 = 1$ (one way to climb zero steps: do nothing) makes the recurrence work from $n = 2$.

Exercises 13

13.1 In how many ways can you climb 8 steps if each move is 1, 2 or 3 steps?

13.2 How many binary strings of length 10 have no three consecutive 1s? (Hint: condition on how the string starts: 0, 10, 110.)

13.3 How many strings of length 7 over $\{A, B, C\}$ have no two consecutive A's?

13.4 What is the maximum number of regions into which 8 straight lines can divide a plane?

13.5 What is the minimum number of moves to solve the Tower of Hanoi with 7 disks?

13.6 What is the maximum number of regions into which 5 circles can divide a plane?

13.7 (Challenge) In how many ways can a 3×6 board be tiled with 1×2 dominoes? (Hint: $f(2) = 3$, $f(4) = 11$ and $f(n) = 4f(n-2) - f(n-4)$ with $f(0) = 1$.)

14. Generating Functions

A generating function turns a counting question into polynomial multiplication. It shines when variables have **irregular constraints** (upper bounds, only even values, coin denominations, dice).

The dictionary

For each variable, write a polynomial whose exponents are the *allowed values*:

Any value ≥ 0	$1 + x + x^2 + \cdots = \dfrac{1}{1-x}$
Between 0 and m	$1 + x + \cdots + x^m = \dfrac{1-x^{m+1}}{1-x}$
At least 2	$x^2 + x^3 + \cdots = \dfrac{x^2}{1-x}$
Even only	$1 + x^2 + x^4 + \cdots = \dfrac{1}{1-x^2}$
A die face	$x + x^2 + \cdots + x^6$

Multiply the polynomials. The number of solutions with total n is the **coefficient of** x^n.

Why it works

When you expand $(1+x+x^2)(1+x)$, each term of the product is $x^a \cdot x^b = x^{a+b}$, one term for every choice (a, b). Collecting like powers groups together exactly the choices with the same total $a + b$. So the coefficient of x^n *counts* the choices whose values add to n.

Two expansions you need:

$$\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n \quad \text{(this } \textit{is} \text{ stars and bars)}, \qquad (1-x^m)^k = \sum_j (-1)^j \binom{k}{j} x^{mj}.$$

The second one, multiplied with the first, is how inclusion–exclusion for upper bounds falls out automatically.

Example 14.1 (three dice)

In how many ways can three dice show a total of 10?
Solution. We need the coefficient of x^{10} in $(x + \cdots + x^6)^3 = x^3(1-x^6)^3(1-x)^{-3}$, i.e. the coefficient of x^7 in $(1 - 3x^6 + \cdots)(1-x)^{-3}$:

$$\binom{7+2}{2} - 3\binom{1+2}{2} = 36 - 9 = \mathbf{27}.$$

(The famous Galileo puzzle: 10 comes up 27 ways, 9 only 25 ways, although both have six "partitions".)

Example 14.2 (coins)

In how many ways can you make Rs. 10 using Rs. 1, Rs. 2 and Rs. 5 coins?
Solution. Coefficient of x^{10} in $\frac{1}{(1-x)(1-x^2)(1-x^5)}$. By hand, case on the Rs. 5 coins: 0 fives $\Rightarrow$ make 10 from 1s and 2s: twos $= 0..5$, 6 ways; 1 five $\Rightarrow$ make 5: twos $= 0..2$, 3 ways; 2 fives: 1 way. Total **10**.

Speed shortcut

For coin problems, a quick table (the coding-round "coin change" DP) beats algebra: start with an array $[1, 0, 0, \dots]$ and for each coin c do $a[i] \mathrel{+}= a[i-c]$ for i increasing.

Example 14.3 (mixed constraints)

A fruit seller packs 10 fruits from apples, bananas and mangoes. Apples come in pairs (so an even number), at most 3 bananas, and at least 2 mangoes. How many packs?
Solution. Coefficient of x^{10} in $(1+x^2+x^4+\cdots)(1+x+x^2+x^3)(x^2+x^3+\cdots)$. Equivalently, list by apples a: we need $b \le 3$ and $a+b \le 8$. $a = 0, 2, 4$: 4 choices of b each; $a = 6$: $b \le 2$, 3 choices; $a = 8$: $b = 0$, 1. Total $4+4+4+3+1 = \mathbf{16}$.

14.1 Partitions and Euler's theorem

The number of ways to write n as a sum of positive integers *ignoring order* has the generating function $\prod_{k\ge1} \frac{1}{1-x^k}$. Partitions into *distinct* parts: $\prod(1+x^k)$. Partitions into *odd* parts: $\prod_{k \text{ odd}} \frac{1}{1-x^k}$. Since $1 + x^k = \frac{1-x^{2k}}{1-x^k}$, the two products are equal (the even factors cancel) – **Euler's theorem**: distinct-part partitions and odd-part partitions are equinumerous.

Example 14.4

Check Euler's theorem for $n = 7$.
Solution. Distinct parts: 7, 6+1, 5+2, 4+3, 4+2+1 → 5. Odd parts: 7, 5+1+1, 3+3+1, 3+1+1+1+1, 1×7 → 5. ✓

Exercises 14

14.1 In how many ways can four dice show a total of 15?

14.2 In how many ways can you pay Rs. 20 using Rs. 1, Rs. 2, Rs. 5 and Rs. 10 coins?

14.3 How many solutions does $a+b+c+d = 20$ have with each variable an integer from 0 to 8?

14.4 With three dice, in how many ways can the total be 9? Compare with the total 10.

14.5 List and count the partitions of 8 into distinct parts and into odd parts.

14.6 Four friends each take 0, 1 or 2 laddoos from a plate, 5 laddoos in all. In how many ways? (Find the coefficient of x^5 in $(1+x+x^2)^4$.)

15. Symmetry: Burnside's Lemma

"How many different necklaces can be made with 6 beads, each red or blue?" The answer is **not** $2^6/6$ (that isn't even an integer). Some colourings look the same after fewer rotations than others, so simple division fails. Burnside's lemma fixes this.

Burnside's lemma

$$\text{Number of distinct objects} = \frac{1}{|G|}\sum_{g\in G}\text{Fix}(g),$$

where G is the set of symmetries (rotations, and reflections if allowed) and $\text{Fix}(g)$ is the number of colourings that look *unchanged* under symmetry g. In words: **average the number of colourings fixed by each symmetry**.

Why it works

Count pairs (symmetry, colouring it fixes) in two ways. Grouped by symmetry, the count is $\sum_g \text{Fix}(g)$. Grouped by colouring: a colouring with a "large" family of look-alikes (its orbit) is fixed by few symmetries, and a very symmetric colouring has few look-alikes but is fixed by many symmetries. Precisely, (orbit size)$\times$(number of symmetries fixing it) $= |G|$. So each orbit contributes exactly $|G|$ pairs in total, and dividing by $|G|$ counts the orbits.

How to compute Fix(g): break the positions into **cycles** under g (positions that g moves into each other). A colouring is fixed iff each cycle is a single colour. With k colours, $\text{Fix}(g) = k^{\text{number of cycles}}$.

Object	Symmetries	Cycle facts
n-bead necklace	n rotations	rotation by j has $\gcd(n,j)$ cycles
n-bead bracelet	n rotations + n reflections	n odd: each reflection has $\frac{n+1}{2}$ cycles n even: half have $\frac{n}{2}+1$, half have $\frac{n}{2}$
Cube faces	24 rotations	see example

Necklace formula: with k colours and n beads, $\frac{1}{n}\sum_{d\mid n}\varphi(d)\,k^{n/d}$, where φ is Euler's totient.

Example 15.1 (square corners)

Colour the 4 corners of a square, each black or white. How many distinct colourings up to rotation?

Solution. Rotations: 0° (4 cycles, $2^4 = 16$ fixed), 90° and 270° (1 cycle each, 2 fixed), 180° (2 cycles, 4 fixed). Average: $\frac{16+2+4+2}{4} = \mathbf{6}$. (Check: all white, all black, one black, two adjacent black, two opposite black, three black.)

Example 15.2 (necklace and bracelet)

6 beads, each red or blue. (a) Necklaces up to rotation? (b) Bracelets (rotation and flipping)?

Solution. (a) Rotation by $j = 0,1,2,3,4,5$ has $\gcd(6,j) = 6,1,2,3,2,1$ cycles:

$$\tfrac{1}{6}(2^6+2+2^2+2^3+2^2+2) = \tfrac{84}{6} = \mathbf{14}.$$

(b) Add 6 reflections: 3 pass through two opposite beads (4 cycles: $2^4 = 16$ each), 3 pass between beads (3 cycles: 8 each):

$$\tfrac{1}{12}(84+48+24) = \tfrac{156}{12} = \mathbf{13}.$$

Example 15.3 (cube)

How many ways to colour the faces of a cube with 2 colours, up to rotation?
Solution. The cube has 24 rotations:

Rotation type	How many	Face cycles	Fixed ($k = 2$)
Identity	1	6	64
90°/270° about face axes	6	3 (top, bottom, a 4-cycle)	8
180° about face axes	3	4	16
120°/240° about vertex diagonals	8	2	4
180° about edge axes	6	3	8

$\frac{64+48+48+32+48}{24} = \frac{240}{24} = \mathbf{10}$.

Common trap

Burnside counts colourings where colours *may* repeat or be unused. For "exactly 3 red and 3 blue", $\mathrm{Fix}(g)$ must count only fixed colourings with that exact colour count – do it by hand for each g.

Exercises 15

15.1 How many necklaces (rotation only) of 5 beads can be made using 3 colours?

15.2 How many bracelets (rotation and reflection) of 5 beads can be made using 3 colours?

15.3 How many ways can the faces of a cube be coloured with 3 colours (repetition allowed), up to rotation?

15.4 Each cell of a 3×3 grid is coloured black or white. How many distinct patterns are there up to rotation of the grid?

15.5 How many necklaces (rotation only) of 6 beads have exactly 3 red and 3 blue beads?

15.6 How many bracelets of 4 beads can be made from 3 colours?

16. Bijections, Identities, Stirling Numbers and Partitions

16.1 Double counting

If you count the same set in two different ways, the answers are equal. This proves identities and, more usefully in exams, lets you swap a hard sum for a single binomial coefficient.

Identity	Story
$\sum_k \binom{n}{k} = 2^n$	choose any subset
$\sum_k k\binom{n}{k} = n2^{n-1}$	committee with a chairperson: choose chair, then any subset of the rest
$\sum_k \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$	(Vandermonde) choose r from m men and n women
$\sum_k \binom{n}{k}^2 = \binom{2n}{n}$	Vandermonde with $m = n = r$, using $\binom{n}{k} = \binom{n}{n-k}$
$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$	(hockey stick) choose $r+1$ from $\{0, \ldots, n\}$; condition on the largest

Example 16.1

Evaluate $\sum_{k=0}^{n} k\binom{n}{k}$ for $n = 6$ without adding terms.
Solution. Committees with a chair from 6 people: $6 \cdot 2^5 = \mathbf{192}$. (Check: $6 + 30 + 60 + 60 + 30 + 6 = 192$.)

16.2 The non-consecutive selection bijection

Choose k numbers from $\{1, \ldots, n\}$ with no two consecutive: $\binom{n-k+1}{k}$.

Why it works

If $a_1 < a_2 < \cdots < a_k$ have gaps of at least 2, then $b_i = a_i - (i-1)$ are strictly increasing numbers from $\{1, \ldots, n-k+1\}$ with no restriction. The map is reversible ($a_i = b_i + i - 1$), so the two sets have the same size.

Around a circle of n seats, choosing k with no two adjacent: $\dfrac{n}{n-k}\binom{n-k}{k}$.

Example 16.2

In how many ways can 3 numbers be chosen from 1 to 10 so that no two are consecutive?
Solution. $\binom{10-3+1}{3} = \binom{8}{3} = \mathbf{56}$.

16.3 Stirling numbers of the second kind: distinct objects into identical boxes

$S(n, k)$ = number of ways to split n distinct objects into k non-empty *unlabelled* groups.

$$S(n, k) = k\,S(n-1, k) + S(n-1, k-1).$$

(Object n either joins one of the k groups formed by the others, or forms a group on its own.)

$n\backslash k$	1	2	3	4	5	6	7
1	1						
2	1	1					
3	1	3	1				
4	1	7	6	1			
5	1	15	25	10	1		
6	1	31	90	65	15	1	
7	1	63	301	350	140	21	1

Connections: $S(n,2) = 2^{n-1} - 1$; onto functions to k labelled people $= k!\,S(n,k)$ (Chapter 7); the **Bell number** $B_n = \sum_k S(n,k)$ counts all ways to partition a set: $1, 1, 2, 5, 15, 52, 203, \ldots$

16.4 Stirling numbers of the first kind: round tables

$c(n,k)$ = ways to seat n people at k identical round tables, none empty (equivalently, permutations of n with k cycles). $c(n,k) = (n-1)\,c(n-1,k) + c(n-1,k-1)$: person n either sits to the left of one of the other $n-1$ people, or sits alone. Small values: $c(n,1) = (n-1)!$, $c(4,2) = 11$, $c(5,2) = 50$, $c(5,3) = 35$.

16.5 Integer partitions

$p(n)$ = ways to write n as an unordered sum of positive integers (identical objects into identical boxes).

n	1	2	3	4	5	6	7	8	9	10
$p(n)$	1	2	3	5	7	11	15	22	30	42

Conjugation bijection: draw a partition as rows of dots (a Ferrers diagram); reading columns instead of rows gives another partition. This shows "partitions into at most k parts" = "partitions with largest part at most k".

Example 16.3

In how many ways can 6 different gifts be packed into 3 identical bags with no bag empty?
Solution. $S(6,3) = \mathbf{90}$. (If the bags are given to 3 different people: $3! \times 90 = 540$, matching the onto-function count $3^6 - 3 \cdot 2^6 + 3 = 540$.)

Exercises 16

16.1 Evaluate $\sum_{k=0}^{10} k\binom{10}{k}$.

16.2 Evaluate $\binom{8}{0}^2 + \binom{8}{1}^2 + \cdots + \binom{8}{8}^2$.

16.3 An analyst must pick 4 of the next 15 days for site visits, with no two visits on consecutive days. How many ways?

16.4 In how many ways can 7 different jobs be split among 3 identical teams, each team getting at least one job?

16.5 In how many ways can the set $\{1,2,3,4,5\}$ be partitioned into non-empty subsets?

16.6 In how many ways can 5 people be seated at 2 identical round tables, neither table empty?

16.7 In how many ways can 10 be written as a sum of positive integers, ignoring order?

16.8 Evaluate $\binom{4}{4} + \binom{5}{4} + \binom{6}{4} + \cdots + \binom{12}{4}$.

16.9 10 chairs stand around a round table. In how many ways can 3 people choose chairs so that no two sit next to each other? (Count sets of chairs, ignoring who sits where.)

17. Mixed Counting Practice (Exam Simulation)

These are not labelled by chapter – identifying the model is part of the test. Aim for under 90 seconds each.

Exercises 17

17.1 In how many arrangements of ALLAHABAD are all the A's together?

17.2 How many 6-digit numbers have their digits strictly increasing from left to right?

17.3 How many 6-digit numbers have digits that never decrease from left to right?

17.4 4 identical red flags and 3 identical green flags are placed in a row so that no two green flags are adjacent. How many arrangements?

17.5 How many 4-letter codes (letters A–Z, repetition allowed) contain at least one vowel?

17.6 In how many ways can 8 people be split into two teams of 4 for a match?

17.7 In how many ways can two squares be chosen on a chessboard so that they are neither in the same row nor in the same column?

17.8 A lift with 5 people stops at 7 floors. In how many ways can they get off if no two get off at the same floor?

17.9 A 4-digit PIN uses digits 0–9 (leading zeros allowed). How many PINs use exactly two distinct digits?

17.10 How many binary strings of length 12 have no two consecutive 0s?

Part III

Number Sense and Speed

18. Exam Strategy: Playing the Paper, Not Just the Questions

An aptitude screen is a time-limited sorting test. Most papers give you around a minute per question, and the cut-off is usually well below full marks. So the skill being tested is partly maths and partly *choosing what to do with the next sixty seconds.*

18.1 The two-pass method

1. **First pass (about 60% of the time):** attempt every question you can see a route through within 15 seconds of reading it. Leave the rest.
2. **Second pass:** return to the questions you left and pick the ones closest to done.
3. **Last two minutes:** apply the guessing rule below to whatever remains.

Check first whether the paper allows you to go back. Some online tests lock each section or each question. If it does lock, replace the two passes with a hard limit: when a question passes 90 seconds, make your best elimination and move on.

Why skipping raises your score

Suppose easy questions take 40 s and hard ones 150 s, each worth one mark. In ten minutes you could do 15 easy questions or 4 hard ones. A hard question costs you nearly four easy marks *unless you run out of easy ones.* So the order you attempt things in matters even if your accuracy does not change.

18.2 When should you guess?

With k options, $+1$ for correct and $-m$ for wrong, a random guess is worth

$$\mathbb{E}[\text{guess}] = \frac{1}{k} - \frac{k-1}{k}\,m.$$

Marking scheme (4 options)	Blind guess	After eliminating one option
No negative marking	$+0.25$	$+0.33$
$-\frac{1}{4}$ per wrong	$+0.0625$	$+0.17$
$-\frac{1}{3}$ per wrong	0	$+0.11$
$-\frac{1}{2}$ per wrong	-0.125	0

Rule: with no negative marking, never leave a blank. With a $-\frac{1}{3}$ penalty, guess as soon as you can rule out one option.

18.3 Using the options as a tool

Four ways to answer without solving

1. **Backsolve:** plug options into the question. Start with a middle option, since being too big or too small tells you which way to go.
2. **Eliminate by last digit or remainder:** a product's unit digit, or its remainder mod 9, often kills most of the options.
3. **Eliminate by size:** estimate roughly. If the options are far apart, a rough estimate is

enough.

4. **Test a simple case:** for algebraic options, put in $x = 1$ or $n = 2$ and see which options survive.

Example 18.1 (backsolving)

A number's square exceeds five times the number by 84. Is the number 10, 11, 12 or 14?
Solution. Try 12: $144 - 60 = 84$. ✓ Done in seconds, without writing a quadratic.

Example 18.2 (testing a simple case)

The sum $1 + 3 + 5 + \cdots + (2n - 1)$ equals which of: $2n - 1$, n^2, $n(n + 1)$, $n^2 - 1$?
Solution. Put $n = 2$: the sum is $1 + 3 = 4$. The options give $3, 4, 6, 3$. Only n^2 survives.

Common trap

Options are often built from the mistakes people actually make. If your first answer appears straight away in the options, that is not evidence that it is right. Common decoys: the answer for the wrong unit, the intermediate value, the percentage on the wrong base, and the count that forgot the leading zero.

18.4 A four-week preparation plan

Week	Study	Practice target
1	Speed maths, numbers, percentages, profit, interest, ratio, averages	Tables and fractions from memory; 20 timed questions a day
2	Mixtures, work, speed–distance, algebra, geometry	Finish the exercises, then redo all the errors two days later
3	Counting, probability foundations, data interpretation, reasoning	Timed sets of 15 mixed questions; keep a log of errors
4	Mock tests, the error log, formula sheet	One full paper every two days, with a review as long as the paper

The error log matters most. For every wrong answer, write down the *type* of error: wrong model, slow method, careless slip, or misread question. Careless slips need slower checks. Slow methods need a shortcut from this booklet. Wrong models need the chapter again.

19. Speed Maths and Estimation

Calculators are usually banned, and arithmetic is where the minutes leak away. A small set of memorised numbers and about ten tricks covers most of what you will need.

19.1 Numbers worth memorising

n	11	12	13	14	15	16	17	18	19	20
n^2	121	144	169	196	225	256	289	324	361	400
n	21	22	23	24	25	26	27	28	29	30
n^2	441	484	529	576	625	676	729	784	841	900

n	2	3	4	5	6	7	8	9	10	11
n^3	8	27	64	125	216	343	512	729	1000	1331
2^n	4	8	16	32	64	128	256	512	1024	2048

Also: $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$, $\pi \approx \frac{22}{7} \approx 3.1416$, $e \approx 2.718$, $\ln 2 \approx 0.693$, $\log_{10} 2 \approx 0.301$, $\log_{10} 3 \approx 0.477$.

19.2 Multiplication tricks

Situation	Method	Example
$\times 5$, $\times 25$, $\times 125$	$\times \frac{10}{2}$, $\times \frac{100}{4}$, $\times \frac{1000}{8}$	$68 \times 25 = \frac{6800}{4} = 1700$
$\times 11$	Write the digits apart; insert each neighbouring sum	$63 \times 11 = 6\,(6{+}3)\,3 = 693$
$\times 9$, $\times 99$	$\times 10 - 1$, $\times 100 - 1$	$47 \times 99 = 4700 - 47 = 4653$
Square ending in 5	$(n5)^2$ is $n(n+1)$ followed by 25	85^2: $8 \cdot 9 = 72 \to 7225$
Squares near 50	$(50 + a)^2 = 2500 + 100a + a^2$	$53^2 = 2500 + 300 + 9 = 2809$
Numbers either side of a round number	$(m + a)(m - a) = m^2 - a^2$	$47 \times 53 = 2500 - 9 = 2491$
Both near 100	Base method (below)	$96 \times 97 = 9312$
Percent swap	$a\%$ of $b = b\%$ of a	16% of $25 = 25\%$ of $16 = 4$

Why the base method works

For numbers near 100, let the shortfalls be a and b:

$$(100 - a)(100 - b) = 100\,(100 - a - b) + ab.$$

So cross-subtract to get the leading digits, then multiply the shortfalls for the last two digits. For 96×97: the shortfalls are 4 and 3; $96 - 3 = 93$; $4 \times 3 = 12$; the answer is 9312. If the shortfall product has three digits, carry into the hundreds. Numbers above 100 have negative shortfalls: 104×103: 107 then 12, giving 10712.

19.3 Checking with remainders (casting out nines)

A number has the same remainder mod 9 as its digit sum. So you can check any product or sum in a few seconds.

Example 19.1

Is $4327 \times 568 = 2457836$ correct?
Solution. $4327 \to 16 \to 7$. $568 \to 19 \to 1$. The product must be $\equiv 7 \pmod 9$. But $2457836 \to 35 \to 8$. **The answer is wrong.** (A passing check does not prove it is right, since swapped digits slip through. A failing check always proves it is wrong.)

19.4 Comparing fractions

- **Cross-multiply:** $\frac{7}{9}$ vs $\frac{11}{14}$: $7 \cdot 14 = 98$ and $9 \cdot 11 = 99$, so $\frac{11}{14}$ is larger.
- **Complement:** $\frac{11}{14} = 1 - \frac{3}{14}$ and $\frac{15}{19} = 1 - \frac{4}{19}$. Compare the gaps from 1. This is often much easier for fractions close to 1.
- **Percentage change:** going from $\frac{a}{b}$ to $\frac{c}{d}$, the fraction grows exactly when the top grows by a bigger percentage than the bottom. This is the fast way to compare ratios in data interpretation.

19.5 Estimation

Round, then compensate

Round each number to something friendly, and note the direction of the error. Rounding a numerator up and a denominator down both push the result up, so the true value must be smaller.
Use $\frac{1}{3} \approx 33\%$, $\frac{1}{8} = 12.5\%$, $\frac{1}{7} \approx 14.3\%$ to turn awkward percentages into fractions. Use $2^{10} \approx 10^3$ to size large powers.

Example 19.2 (estimating a compound expression)

Estimate $\dfrac{498 \times 61}{19.7}$.
Solution. $\approx \frac{500 \times 60}{20} = 1500$. The true value is $\frac{30378}{19.7} \approx 1542$. One part in thirty is good enough when the options are 800, 1200, 1500, 2000.

Example 19.3 (sizing a power)

How many digits does 2^{50} have?
Solution. $2^{50} = (2^{10})^5 \approx (10^3)^5 = 10^{15}$. Since $2^{10} = 1024$ is a bit more than 10^3, the true value is 1.13×10^{15}, which has **16** digits.

Exercises 19

19.1 Compute mentally: 98×97.

19.2 Compute mentally: 65^2 and 115^2.

19.3 Compute mentally: 43×37.

19.4 Find 16.67% of 738.

19.5 Compute $25 \times 48 \times 125$ in your head.

19.6 Which is largest: $\frac{7}{9}$, $\frac{11}{14}$, $\frac{15}{19}$?

19.7 How many digits does 2^{30} have? Estimate first, then check against $2^{10} = 1024$.

19.8 Estimate 49.8% of 1202 plus 24.9% of 804 to the nearest hundred.

19.9 Compute 999×999.

19.10 Use casting out nines to test whether $738 \times 256 = 188928$ could be correct.

20. Number System: Divisibility, Factors, HCF and LCM

20.1 Vocabulary and parity

Natural numbers start at 1; whole numbers add 0; integers add negatives. A **prime** has exactly two divisors, so 1 is not prime, and 2 is the only even prime. Two numbers are **coprime** if their HCF is 1.

Parity: odd $\pm$ odd is even; even $\pm$ anything keeps the other number's parity; a product is odd only if every factor is odd. Parity alone answers a surprising number of data-sufficiency questions.

Primality test: check divisibility by primes up to $\sqrt{n}$. For 437: $\sqrt{437} < 21$, so test 2, 3, 5, 7, 11, 13, 17, 19. It fails at 19 ($437 = 19 \times 23$), so it is not prime. There are 25 primes below 100.

20.2 More divisibility tests

Chapter 3 lists the tests for 2–6, 8, 9 and 11. Here are two extensions:

- **Composite divisors:** split into coprime parts. Divisibility by 12 means by 3 and by 4; by 72 means by 8 and by 9. (Splitting 12 as 2×6 is useless: 2 and 6 are not coprime, and 18 passes both.)
- **7, 11, 13 via 1001:** since $1001 = 7 \times 11 \times 13$, split the number into blocks of three from the right and take the alternating sum. The result has the same remainder mod 7, 11 and 13 as the original.

Example 20.1

Is 286286 divisible by 7? Is 123456 divisible by 13?
Solution. $286286 = 286 \times 1001$, so it is divisible by 7, 11 and 13. For 123456: $456 - 123 = 333$, and $333 = 13 \times 25 + 8$, so **no**, the remainder is 8.

20.3 Counting and adding factors

For $N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$:

Quantity	Formula
Number of divisors	$d(N) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$
Sum of divisors	$\sigma(N) = \prod_i (1 + p_i + p_i^2 + \cdots + p_i^{a_i})$
Product of divisors	$N^{d(N)/2}$
Numbers below N coprime to N (Euler's totient)	$\varphi(N) = N \prod_i \left(1 - \frac{1}{p_i}\right)$

Why it works

Sum: expanding the product $\prod_i (1 + p_i + \cdots + p_i^{a_i})$ produces each divisor exactly once, because a divisor is one choice of power from each bracket.
Product: divisors pair up as $(x, N/x)$, each pair multiplying to N. There are $d(N)/2$ pairs.
Totient: a fraction $\frac{1}{p}$ of numbers is divisible by p, and these events are independent across

different primes, so the fraction avoiding all of them is $\prod(1-\frac{1}{p_i})$.

Example 20.2

For $N = 360 = 2^3 \cdot 3^2 \cdot 5$, find the number of divisors, the sum of divisors and $\varphi(N)$.
Solution. $d = 4 \cdot 3 \cdot 2 = \mathbf{24}$. $\sigma = (1+2+4+8)(1+3+9)(1+5) = 15 \cdot 13 \cdot 6 = \mathbf{1170}$. $\varphi = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = \mathbf{96}$.

20.4 HCF and LCM

Three facts

1. In prime factors: HCF takes the *lowest* power of each prime, and LCM takes the *highest*.
2. For two numbers, HCF $\times$ LCM $= a \times b$. (This fails for three numbers.)
3. **Euclid:** $\mathrm{HCF}(a, b) = \mathrm{HCF}(b,\ a \bmod b)$. For example, $\mathrm{HCF}(1651, 2032)$: $2032 \to 381$, $1651 \to 127$, $381 \to 0$. The answer is 127.

For fractions: HCF of fractions $= \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$; LCM of fractions $= \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$.

Why HCF $\times$ LCM $= ab$

For each prime, one of the two numbers has the lower power and the other has the higher. HCF collects the lower powers and LCM the higher, so between them they collect every prime power in a and b exactly once.

20.5 Standard word problems

The question says…	Use
Greatest number dividing a, b, c exactly	$\mathrm{HCF}(a, b, c)$
Greatest number leaving remainders r_1, r_2, r_3	$\mathrm{HCF}(a - r_1,\ b - r_2,\ c - r_3)$
Greatest number leaving the *same* remainder	HCF of the differences $\lvert a-b\rvert, \lvert b-c\rvert$
Least number divisible by a, b, c	$\mathrm{LCM}(a, b, c)$
Least number leaving the same remainder r	LCM$+r$
Least number with a constant gap $(a - r_1 = b - r_2 = \cdots = g)$	LCM$-g$
Bells, runners, traffic lights together again	LCM of the intervals
Largest tile, largest measuring jug	HCF

Why the differences work

If a and b leave the same remainder when divided by d, then $a - b$ leaves remainder 0. So d must divide every difference, and the largest such d is the HCF of the differences.

Example 20.3 (same remainder)

Find the greatest number that divides 43, 91 and 183 leaving the same remainder.
Solution. Differences: 48, 92, 140. Their HCF is **4**. (Check: all three leave remainder 3.)

Example 20.4 (constant gap)

Find the least number which, divided by 5, 6 and 7, leaves remainders 3, 4 and 5.
Solution. In each case the remainder is 2 short of the divisor, so $n+2$ is a multiple of all three. LCM $= 210$, so $n =$ **208**.

Example 20.5 (pairs from HCF and product)

Two numbers have HCF 12 and product 2880. How many such pairs exist?
Solution. Write them as $12x$, $12y$ with x, y coprime. $144xy = 2880 \Rightarrow xy = 20$. Coprime factorisations: $(1, 20)$, $(4, 5)$. (Not $(2, 10)$, which share a factor of 2.) **Two pairs:** $(12, 240)$ and $(48, 60)$.

20.6 Recurring decimals

The rule

Pure recurring: the block over as many 9s as digits. $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$.
Mixed recurring: (all digits − non-recurring digits) over (one 9 per recurring digit, one 0 per non-recurring digit). $0.1\overline{6} = \frac{16-1}{90} = \frac{1}{6}$.

Why it works

If $x = 0.\overline{27}$ then $100x = 27.\overline{27}$. Subtracting, $99x = 27$. The recurring tails cancel, and this is the entire idea.

Exercises 20

20.1 Find the number of divisors and the sum of divisors of 720.

20.2 How many integers from 1 to 100 are coprime to 30?

20.3 Find the greatest number that divides 1657 and 2037 leaving remainders 6 and 5 respectively.

20.4 Find the least number which leaves remainder 8 when divided by each of 12, 15, 20 and 54.

20.5 Find the least number which, divided by 6, 7 and 8, leaves remainders 4, 5 and 6.

20.6 Two numbers have HCF 15 and LCM 180. Find all such pairs.

20.7 Evaluate $0.\overline{36} + 0.\overline{63}$.

20.8 Five bells toll at intervals of 2, 4, 6, 8 and 10 seconds, starting together. How many times do they toll together in 30 minutes, including the start?

20.9 Is 437 prime? Is 221?

20.10 Find the product of all the divisors of 36, as a power of 6.

20.11 Find the smallest positive integer by which 1800 must be multiplied to get (a) a perfect square; (b) a perfect cube.

20.12 How many integers from 1 to 100 have exactly 3 divisors?

20.13 A hall measuring 6.48 m by 5.04 m is to be paved with identical square tiles, with no cutting. What is the largest tile, and how many are needed?

21. Remainders, Cyclicity and Modular Arithmetic

21.1 Working mod n

$a \equiv b \pmod{n}$ means a and b leave the same remainder on division by n. The key property: you can **replace any number by its remainder** before adding, subtracting, multiplying or raising to a power.

$$a \equiv a',\ b \equiv b' \ \Rightarrow\ a+b \equiv a'+b', \quad ab \equiv a'b', \quad a^k \equiv (a')^k.$$

Speed shortcut: negative remainders

Remainders may be negative while you work. $29 \equiv -1 \pmod{10}$ makes $29^{51} \equiv (-1)^{51} = -1 \equiv 9$ immediately. Look for a base that is ± 1 mod n.

Example 21.1

Find the remainder when $17 \times 23 \times 29$ is divided by 7.
Solution. $17 \equiv 3$, $23 \equiv 2$, $29 \equiv 1$. Product $\equiv 6$. Remainder **6**.

21.2 Powers: find the cycle

Method

Compute $a^1, a^2, a^3, \ldots \bmod n$ until you reach 1 (or a repeat). If $a^k \equiv 1$, then $a^{qk+r} \equiv a^r$. So reduce the exponent modulo the cycle length.

Example 21.2

Find the remainder when 2^{100} is divided by 7.
Solution. $2^3 = 8 \equiv 1$. $100 = 3 \cdot 33 + 1$. So $2^{100} \equiv 2^1 = \mathbf{2}$.

Fermat's little theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Euler's theorem: if a and n are coprime, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

These guarantee a cycle length that divides $p-1$ (or $\varphi(n)$), even when you cannot find the shortest one quickly.

Example 21.3 (Fermat)

Find the remainder when 2^{1000} is divided by 13.
Solution. $2^{12} \equiv 1$. $1000 = 12 \cdot 83 + 4$. $2^4 = 16 \equiv \mathbf{3}$.

21.3 Unit digits

The unit digit of a power depends only on the last digit of the base, and it cycles with period dividing 4.

Last digit	Cycle of unit digits	Period
0, 1, 5, 6	constant	1
4	4, 6	2
9	9, 1	2
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

Method: reduce the exponent mod 4, treating a remainder of 0 as 4. For 7^{2026}: $2026 \equiv 2 \pmod 4$, so the unit digit is **9**.

21.4 Last two digits

- Base ending in 1: $(10a+1)^n \equiv 1 + 10an \pmod{100}$. Example: $11^{23} \equiv 1 + 230 \equiv \mathbf{31}$.
- Otherwise look for a small power that is ± 1 mod 100: $7^4 = 2401 \equiv 1$, so $7^{2026} \equiv 7^2 = \mathbf{49}$. And $3^{20} \equiv 1$, while $9^{10} \equiv 1$.
- Bases ending in 5: 25^n always ends in 25 for $n \geq 1$.

21.5 Algebraic divisibility

$a^n - b^n$	always divisible by $a - b$; divisible by $a + b$ when n is even
$a^n + b^n$	divisible by $a + b$ when n is odd

Why it works

Mod $(a-b)$ we have $a \equiv b$, so $a^n - b^n \equiv 0$. Mod $(a+b)$ we have $a \equiv -b$, so $a^n \equiv (-1)^n b^n$, which gives the even and odd cases.

Example 21.4

Show $17^{25} + 13^{25}$ is divisible by 30.
Solution. The exponent is odd, so $a + b = 30$ divides the sum.

21.6 Simultaneous conditions

Chinese remainder idea: when divisors are coprime, conditions combine into a single condition mod their product. Solve by sieving: list numbers satisfying the largest modulus and test the others.

Example 21.5 (Sun Tzu, about 1,600 years old)

Find the least n with remainder 2 on division by 3, remainder 3 by 5 and remainder 2 by 7.
Solution. Numbers $\equiv 2 \pmod 7$: 2, 9, 16, 23, 30, ... Of these, $\equiv 3 \pmod 5$: 23. And $23 \equiv 2 \pmod 3$ ✓. So $n = \mathbf{23}$, and all solutions are $23 + 105k$.

21.7 Number bases

To convert to base b, repeatedly divide by b and read the remainders from the last to the first. Going back, $(d_k \dots d_1 d_0)_b = \sum d_i b^i$.

Example 21.6

Write 45 in binary, and evaluate $(234)_8$ in decimal.
Solution. $45 = 32 + 8 + 4 + 1 = (101101)_2$. And $(234)_8 = 2 \cdot 64 + 3 \cdot 8 + 4 = \mathbf{156}$.

Binary fact for coding rounds: a positive n is a power of 2 exactly when n has a single 1 bit. Every number has a unique binary form, which is why weights $1, 2, 4, 8, \dots$ can weigh any integer (Chapter 54).

Exercises 21

21.1 Find the remainder when 1234×5678 is divided by 9.

21.2 Find the remainder when 5^{100} is divided by 7.

21.3 Find the unit digit of $3^{2023} \times 7^{2024}$.

21.4 Find the last two digits of 21^{37}.

21.5 Is $2^{60} - 1$ divisible by 3? By 5? By 7? Explain using cycles.

21.6 Find the remainder when $1! + 2! + 3! + \cdots + 50!$ is divided by 8.

21.7 Find the smallest positive n leaving remainders 1, 2, 3, 4, 5 on division by 2, 3, 4, 5, 6.

21.8 Write 156 in binary and in base 8.

21.9 Find the remainder when 7^{99} is divided by 4.

21.10 Find the remainder when 2^{2026} is divided by 5.

21.11 What is the sum of the digits of $10^{25} - 7$?

21.12 Find the unit digit of 3^{3^3}.

21.13 Find the unit digit of $1^1 + 2^2 + 3^3 + 4^4 + 5^5$.

21.14 A number leaves remainder 7 when divided by 12. What remainder does it leave when divided by 4? Can you find its remainder when divided by 8?

Part IV

Arithmetic Aptitude

22. Percentages

22.1 The multiplier mindset

Never think "add 20%". Think "multiply by 1.2". A $r\%$ increase is a multiplier of $(1 + \frac{r}{100})$; a $r\%$ decrease is $(1 - \frac{r}{100})$. Successive changes *multiply*:

$$\text{Net multiplier} = (1+a)(1+b)(1+c)\cdots \qquad \Rightarrow \qquad \text{two changes: } a + b + \frac{ab}{100}\ \%.$$

Why it works

$(1 + \frac{a}{100})(1 + \frac{b}{100}) = 1 + \frac{a+b}{100} + \frac{ab}{100^2}$. The cross term $\frac{ab}{100}\%$ is "the change on the change". With $a = +20$, $b = -20$: $20 - 20 - 4 = -4\%$, which is why a 20% rise followed by a 20% fall leaves you 4% poorer.

22.2 Fraction–percent table (memorise)

$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	$\frac{1}{8}$	$\frac{1}{9}$	$\frac{1}{11}$	$\frac{1}{12}$
50%	33.33%	25%	20%	16.67%	14.29%	12.5%	11.11%	9.09%	8.33%

These turn "what is 37.5% of 640" into "$\frac{3}{8} \times 640 = 240$" in two seconds.

22.3 Base effect and the "reverse" percentage

If A is $r\%$ more than B, then B is $\dfrac{r}{100 + r} \times 100\%$ *less* than A. The base changes.

Constant expenditure rule. If price rises by $r\%$, consumption must fall by $\frac{r}{100+r} \times 100\%$ to keep spending constant (since price $\times$ quantity is fixed, the multipliers must cancel: $(1 + r)(1 - s) = 1$).

Percent vs percentage points. If the repo rate goes from 6.5% to 6%, it fell by 0.5 *percentage points*, which is a $\frac{0.5}{6.5} = 7.69\%$ fall. Finance interviews love this distinction.

22.4 The fraction method for percent change

Many percentages are simple fractions, and a change of $\frac{1}{n}$ has a neat reverse.

Change	As a fraction	Multiplier	To undo it, change by
$+25\%$	$+\frac{1}{4}$	$\frac{5}{4}$	$-\frac{1}{5} = -20\%$
$+20\%$	$+\frac{1}{5}$	$\frac{6}{5}$	$-\frac{1}{6} = -16.67\%$
$+33.33\%$	$+\frac{1}{3}$	$\frac{4}{3}$	$-\frac{1}{4} = -25\%$
$+50\%$	$+\frac{1}{2}$	$\frac{3}{2}$	$-\frac{1}{3} = -33.33\%$
-12.5%	$-\frac{1}{8}$	$\frac{7}{8}$	$+\frac{1}{7} = +14.29\%$

Pattern: to undo $+\frac{1}{n}$, apply $-\frac{1}{n+1}$. To undo $-\frac{1}{n}$, apply $+\frac{1}{n-1}$.

Why it works

$\left(1+\frac{1}{n}\right)\left(1-\frac{1}{n+1}\right)=\frac{n+1}{n}\cdot\frac{n}{n+1}=1$. The two multipliers are reciprocals. Treat a percentage as a fraction and "undo" becomes "flip".

22.5 Growth and decay

A constant percentage change is compound interest under another name:

$$\text{Population after } n \text{ years} = P\left(1+\frac{r}{100}\right)^n, \qquad \text{Value after depreciation} = V\left(1-\frac{r}{100}\right)^n.$$

Going back in time just divides by the same multiplier.

Example 22.1 (real life: the price-fall gain)

A price falls 25%, so a buyer gets 2 kg more for Rs. 120. Find the original price.
Solution. A 25% price fall leaves the money buying $\frac{4}{3}$ of the old quantity, i.e. $\frac{1}{3}$ more. So 2 kg is $\frac{1}{3}$ of the old quantity: old quantity 6 kg, old price $\frac{120}{6}=$ **Rs. 20**/kg. (New price Rs. 15: $\frac{120}{15}=8$ kg. ✓)

Example 22.2 (overlapping percentages)

In a batch, 60% know Python and 50% know SQL; 10% know neither. What percentage know both?
Solution. At least one: 90%. So both $=60+50-90=$ **20**%.

Example 22.3 (real life: fuel prices)

Petrol price rises 25%. By what percentage should a household cut consumption so its fuel bill is unchanged?
Solution. $\frac{25}{125}=\frac{1}{5}=$ **20**%. (Multipliers: $1.25\times0.8=1$.)

Example 22.4

A town's population grows 10% in year 1, 20% in year 2, and falls 10% in year 3. Net change?
Solution. $1.1\times1.2\times0.9=1.188\Rightarrow$ **+18.8%**.

Example 22.5 (elections)

In a two-candidate election, 10% of votes cast were invalid. The winner got 58% of the valid votes and won by 2880 votes. How many votes were cast?
Solution. Margin $=58\%-42\%=16\%$ of valid votes $=2880\Rightarrow$ valid $=18000$. Valid is 90% of total: total $=$ **20000**.

Common trap

Percentages of different bases can't be added. "Sales rose 10% in Q1 and 10% in Q2" is a 21% rise, not 20%. And an average of percentages is not the percentage of the totals unless the bases are equal.

Exercises 22

22.1 A's salary is 25% more than B's. By what percentage is B's salary less than A's?

22.2 The length of a rectangle increases by 20% and its breadth decreases by 10%. Percentage change in area?

22.3 What single discount is equivalent to successive discounts of 20% and 10%?

22.4 Rent is 30% of a person's income. Income rises 20% and rent rises 10%. What percentage of the new income is the new rent?

22.5 A number is increased by 40% and then decreased by 40%. The result is 168 less than the original. Find the number.

22.6 The pass mark is 40%. A student scores 178 and fails by 22 marks. What are the maximum marks?

22.7 A bank's lending rate falls from 6.5% to 6%. State the fall in percentage points and in percent.

22.8 A's income is 20% more than B's, and B's is 25% less than C's. A's income is what percentage of C's?

22.9 In an exam, 35% fail in Maths, 25% fail in English, and 15% fail in both. What percentage pass in both?

22.10 A town of 20,000 grows at 10% a year. What is its population after 3 years?

22.11 A car bought for Rs. 8,00,000 depreciates 15% a year. What is its value after 2 years?

22.12 The price of sugar falls 20%, so a buyer gets 5 kg more for Rs. 400. Find the new price per kg and the original price.

22.13 A shop's revenue is price $\times$ units. The price rises 10% and units sold fall 10%. What happens to revenue? What fall in units would keep revenue exactly unchanged?

23. Profit, Loss and Discount

23.1 Vocabulary and the master equation

Cost Price (CP), Selling Price (SP), Marked Price (MP, the list price). Profit% is **always on CP**; discount% is **always on MP**.

$$SP = CP\,(1+p) = MP\,(1-d) \qquad \Rightarrow \qquad \frac{MP}{CP} = \frac{1+p}{1-d}.$$

This single equation answers most "markup, discount, profit" questions.

23.2 Standard results

- **Same SP, x% profit on one and x% loss on the other** $\Rightarrow$ always a net *loss* of $\frac{x^2}{100}$% (because the loss-making item had a higher CP).
- **CP of a items = SP of b items** $\Rightarrow$ profit% $= \frac{a-b}{b} \times 100$.
- **False weights:** a dealer who gives w grams instead of 1000 g while charging for 1 kg at CP makes $\frac{1000-w}{w} \times 100$% profit (profit is measured against what he actually gave).
- **Successive discounts** d_1, d_2: single equivalent $= d_1 + d_2 - \frac{d_1 d_2}{100}$.

Why the false-weight formula divides by w

He receives the price of 1000 g but his cost is the cost of w g. Profit = cost of $(1000 - w)$ g; profit% is profit over *his* cost, i.e. over w. Dividing by 1000 is the classic slip.

Example 23.1 (retail markup)

An electronics store marks goods 40% above cost and offers a 15% festive discount. Profit%?
Solution. $1.4 \times 0.85 = 1.19 \Rightarrow$ **19**% profit.

Example 23.2 (same selling price trap)

Two phones are sold at Rs. 24,000 each, one at 20% profit and the other at 20% loss. Net result?
Solution. CPs: $24000/1.2 = 20000$ and $24000/0.8 = 30000$. Total CP 50000, total SP 48000: loss Rs. 2000 = **4**% (matches $\frac{20^2}{100}$).

Example 23.3 (dishonest dealer)

A grocer claims to sell at cost price but uses a 900 g weight for 1 kg. Profit%?
Solution. $\frac{100}{900} =$ **11.11**%.

Example 23.4

The CP of 15 pens equals the SP of 12 pens. Profit%?
Solution. $\frac{15-12}{12} =$ **25**%.

23.3 Margin versus markup

These two are confused constantly, including in business conversations.

$$\text{Markup} = \frac{\text{Profit}}{\text{Cost}}, \qquad \text{Margin} = \frac{\text{Profit}}{\text{Selling price}}, \qquad \text{Margin} = \frac{\text{Markup}}{1 + \text{Markup}}.$$

A 25% markup is a 20% margin; a 100% markup is a 50% margin.

Example 23.5 (real life: break-even analysis)

A startup's product has a fixed monthly cost of Rs. 3,00,000, a variable cost of Rs. 400 per unit and a price of Rs. 1,000. How many units per month are needed to break even? How many to make Rs. 1,20,000 profit?

Solution. Each unit contributes $1000 - 400 = Rs.\,600$ towards fixed costs. Break-even: $\frac{300000}{600} = \mathbf{500}$ units. For the target: $\frac{420000}{600} = \mathbf{700}$ units.

Example 23.6 (buy-one-get-one offers)

A store offers "buy 2, get 1 free" and still makes a 20% profit. By what percentage above cost is the marked price?

Solution. Selling 3 units brings in $2M$, and the cost is $3C$. So $2M = 1.2 \times 3C \Rightarrow M = 1.8C$. Marked **80**% above cost.

Exercises 23

23.1 An article is sold at a 12% loss. Had it been sold for Rs. 180 more, there would have been an 8% profit. Find the CP.

23.2 A trader wants a 26% profit after allowing a 10% discount. By what percentage above CP should he mark the goods?

23.3 A dealer claims to sell at cost price but gives 20% less weight. Profit%?

23.4 A dealer marks goods 50% above CP, uses an 800 g weight for 1 kg, and allows a 20% discount. Profit%?

23.5 A vendor buys bananas at 11 for Rs. 10 and sells them at 10 for Rs. 11. Profit%?

23.6 What single discount is equivalent to successive discounts of 20%, 15% and 10%?

23.7 Selling an item for Rs. 840 gives a 20% profit. At what price should it be sold for a 35% profit?

23.8 In a "buy 4, get 1 free" offer, what is the effective discount on the marked price?

23.9 A shopkeeper buys 20 kg of rice for Rs. 800. 2 kg is spoiled. At what price per kg should the remainder be sold for a 26% overall profit?

23.10 An article's list price is Rs. 1,000. With 18% GST on the discounted price, a customer pays Rs. 944. What discount was given?

23.11 A sells to B at 10% profit, B sells to C at 20% profit, and C pays Rs. 660. What did A pay?

23.12 A trader buys at 20% below the marked price and sells at 10% above the marked price. Profit%?

23.13 A retailer's markup is 25% on cost. What is this as a percentage of the selling price (the gross margin)?

24. Simple and Compound Interest

24.1 Formulas

$$\text{SI} = \frac{P\,R\,T}{100}, \qquad A_{\text{CI}} = P\left(1 + \frac{R}{100\,m}\right)^{mT} \quad (m = \text{compounding periods per year}).$$

Effective annual rate for nominal rate R compounded m times a year: $\left(1 + \frac{R}{100m}\right)^{m} - 1$.

Real life: A bank FD quoted at 7% "compounded quarterly" actually earns $(1.0175)^4 - 1 = 7.19\%$ a year. That is why banks quote both a rate and an "annualised yield".

24.2 The CI–SI difference

On the same P and R:

$$\text{2 years: } \text{CI} - \text{SI} = P\left(\frac{R}{100}\right)^2, \qquad \text{3 years: } \text{CI} - \text{SI} = P\left(\frac{R}{100}\right)^2\left(3 + \frac{R}{100}\right).$$

Why it works

In year 2, compound interest also earns interest on year 1's interest: $P \cdot r \cdot r$. In 3 years, the interest-on-interest terms are $3Pr^2$ (from years 1–2 feeding years 2–3) plus Pr^3 (interest on interest on interest).

24.3 Doubling times

- SI: money doubles when total interest = 100%, i.e. in $\frac{100}{R}$ years; it becomes k times in $\frac{100(k-1)}{R}$ years.
- CI: if money doubles in t years, it becomes 4 times in $2t$, 8 times in $3t$ (multipliers compound).
- **Rule of 72:** at $R\%$ compounded annually, doubling time $\approx \frac{72}{R}$ years.

24.4 Equal instalments

A loan P at r per year repaid by n equal annual instalments x (first one a year from now): the present value of the instalments equals the loan:

$$P = \frac{x}{1+r} + \frac{x}{(1+r)^2} + \cdots + \frac{x}{(1+r)^n}.$$

This is exactly the annuity/EMI logic from corporate finance.

Example 24.1

Rs. 10,000 at 8% for 2 years: compare CI and SI.
Solution. CI amount $= 10000 \times 1.08^2 = 11664$; SI amount $= 11600$. Difference **Rs. 64** $= 10000 \times 0.08^2$. ✓

Example 24.2

A sum doubles in 5 years at simple interest. In how many years will it become 4 times? What if the interest were compound?
Solution. SI: doubling means 100% in 5 years, so 20% a year; 4 times needs 300%: **15 years**. CI: doubling every 5 years, so $\times 4$ in **10 years**.

Example 24.3 (EMI-style instalments)

A loan of Rs. 21,000 is repaid in 2 equal annual instalments at 10% compound interest. Find the instalment.
Solution. $\frac{x}{1.1} + \frac{x}{1.21} = 21000 \Rightarrow x \cdot \frac{1.1+1}{1.21} = 21000 \Rightarrow x = \frac{21000 \times 1.21}{2.1} = \mathbf{Rs.\,12,100}$.

Example 24.4 (different rates in different years)

Rs. 5,000 earns 10% in year 1 and 20% in year 2, compounded. Find the amount.
Solution. Multipliers multiply: $5000 \times 1.1 \times 1.2 = \mathbf{Rs.\,6,600}$.

Example 24.5 (half-yearly compounding)

Rs. 8,000 at 10% per annum compounded half-yearly for $1\frac{1}{2}$ years?
Solution. Rate 5% per half-year, 3 periods: $8000 \times 1.05^3 = \mathbf{Rs.\,9,261}$.

24.5 Financial maths: the same idea, used by professionals

Time value of money

A rupee today beats a rupee next year, since today's rupee can earn interest. To compare cash flows at different dates, move them all to one date:

$$\text{Present value of } C \text{ received in } n \text{ years} = \frac{C}{(1+r)^n}.$$

Tool	Formula	What it answers
NPV	$\sum_t \frac{C_t}{(1+r)^t}$	Is this project worth more than it costs? Take it if NPV > 0.
IRR	The r that makes NPV $= 0$	What rate of return is built into these cash flows?
CAGR	$\left(\frac{\text{End}}{\text{Start}}\right)^{1/n} - 1$	What constant yearly growth gives the same result?
Perpetuity	$\frac{C}{r}$	Worth of C a year, forever
Growing perpetuity	$\frac{C}{r-g}$	Worth of C next year, growing at g (Gordon growth model)
Annuity	$\frac{C}{r}\left(1 - \frac{1}{(1+r)^n}\right)$	Worth of C a year for n years
EMI	$\frac{Pr(1+r)^n}{(1+r)^n - 1}$	Equal payment on a loan P, per-period rate r

Why the annuity is a difference of perpetuities

An annuity for n years is a perpetuity starting now, minus a perpetuity starting n years from now. The first is worth $\frac{C}{r}$. The second is worth $\frac{C}{r}$ at year n, and so $\frac{C}{r} \cdot \frac{1}{(1+r)^n}$ today. Subtract, and you have the formula without summing a series. The EMI formula is the same equation solved for C.

Why bond prices fall when yields rise

A bond's price is the present value of its fixed coupons and face value. A higher discount rate shrinks every term. If the coupon rate equals the yield, the price equals face value exactly. Above that yield it trades at a discount, below at a premium.

Example 24.6 (how an EMI splits)

A Rs. 10,000 loan at 10% is repaid in 2 equal annual instalments. Find each one and draw up the schedule.

Solution. $x = \frac{10000 \times 0.1 \times 1.21}{0.21} = Rs.\,5,761.90.$

Year	Opening balance	Interest (10%)	Payment	Closing balance
1	10,000.00	1,000.00	5,761.90	5,238.10
2	5,238.10	523.81	5,761.90	0.00

Early payments are mostly interest, later ones mostly principal. That is why prepaying a loan early saves the most.

Common trap

A return of +50% then −50% is a net −25%, though the arithmetic average return is 0. Averages of returns overstate the growth you actually get; CAGR (the geometric mean) is what compounds. Similarly, a CAGR of 8% is not 8% every year. It only describes the start and end.

Exercises 24

24.1 The difference between CI and SI on a sum for 2 years at 10% per annum is Rs. 150. Find the sum.

24.2 Find the difference between CI and SI on Rs. 8,000 for 3 years at 10% per annum.

24.3 A sum amounts to Rs. 1,210 in 2 years and Rs. 1,331 in 3 years at compound interest. Find the rate and the principal.

24.4 At what rate of simple interest does a sum triple in 12.5 years?

24.5 What is the effective annual rate of 12% per annum compounded monthly?

24.6 A loan is repaid in 3 equal annual instalments of Rs. 1,331 at 10% compound interest. What was the loan amount?

24.7 Using the rule of 72, roughly how long does money take to double at 9% compounded annually? How close is this to the exact value?

24.8 Find the amount on Rs. 10,000 for 1 year at 12% compounded half-yearly, and compare with annual compounding.

24.9 Find the amount on Rs. 20,000 for 3 years if the rates are 5%, 10% and 20% in successive years.

24.10 An investment grows from Rs. 1,00,000 to Rs. 1,61,051 in 5 years. What is its CAGR?

24.11 A project costs Rs. 1,000 today and pays Rs. 600 at the end of each of the next 2 years. At a 10% discount rate, what is its NPV? Should you take it?

24.12 A perpetuity pays Rs. 5,000 a year forever, starting one year from now. What is it worth at 8%?

24.13 Find the monthly EMI on a Rs. 1,00,000 loan over 12 months at 12% a year (1% a month). (Use $1.01^{12} \approx 1.12683$.)

24.14 A 2-year bond with face value Rs. 1,000 pays a 10% annual coupon. What is its price if the market yield is 10%? If the yield is 12%?

25. Ratio, Proportion, Partnership and Ages

25.1 Working with ratios

- **The k-method:** if $A : B = 3 : 5$, write $A = 3k$, $B = 5k$. Every ratio problem becomes one equation in k.
- **Chaining:** $A : B = 2 : 3$ and $B : C = 4 : 5$. Make the common term equal (LCM of 3 and 4 is 12): $A : B : C = 8 : 12 : 15$.
- **Dividing in inverse ratio:** "in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$" = multiply by LCM $12 = 6 : 4 : 3$.
- **Componendo–dividendo:** if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

25.2 Partnership

Profit is shared in the ratio of **capital** $\times$ **time** (in a common unit, usually months). A working partner's salary is paid first; the rest is shared.

Real life: This is the same logic as weighted-average capital in a startup's cap table before priced rounds: money that stayed in longer earns a proportionally larger share.

25.3 Ages

Let present ages be x (or k-multiples of a ratio). Translate "n years ago" into $x - n$ for *every* person. The difference between two ages never changes – a quick consistency check.

25.4 Variation

Type	Equation	Example
Direct	$y = kx$	Cost and quantity at a fixed price
Inverse	$y = \frac{k}{x}$	Workers and days for a fixed job
Joint	$z = kxy$	Interest varies with principal and time
Mixed	$z = \frac{kx}{y^2}$	Gravity or light intensity

Method: write the equation with a constant k, use the given data to find k, then answer. For a ratio question, you can skip k by dividing the two situations.

Example 25.1 (real life: a diamond's value)

A diamond's price varies as the square of its weight. A diamond worth Rs. 1,00,000 breaks into pieces with weights in the ratio $3 : 2$. What is the loss in value?
Solution. Let the weight be 5 units, so $k \cdot 25 = 100000$, value per unit^2 $= 4000$. Pieces: $4000 \times 9 + 4000 \times 4 = 52000$. Loss = $\mathbf{Rs.\,48,000}$.

Example 25.2 (fourth and mean proportional)

Find the mean proportional of 9 and 16, and the fourth proportional to 4, 6, 10.
Solution. Mean proportional: $\sqrt{9 \times 16} = \mathbf{12}$. Fourth proportional: $\frac{6 \times 10}{4} = \mathbf{15}$.

Example 25.3 (partnership)

A invests Rs. 50,000 for 12 months, B invests Rs. 75,000 for 8 months, and C invests Rs. 60,000 for 6 months. The year's profit is Rs. 1,30,000. Shares?
Solution. Capital-months: $6,00,000 : 6,00,000 : 3,60,000 = 5 : 5 : 3$. Shares: Rs. 50,000, Rs. 50,000, Rs. 30,000.

Example 25.4 (ages)

The ratio of two people's present ages is $3 : 4$. Five years ago it was $5 : 7$. Find their ages.
Solution. $\frac{3k-5}{4k-5} = \frac{5}{7} \Rightarrow 21k - 35 = 20k - 25 \Rightarrow k = 10$: ages **30 and 40**.

Example 25.5 (coins)

A bag has Rs. 1, 50 paise and 25 paise coins in the ratio $5 : 6 : 8$ by number. The total value is Rs. 210. How many coins of each?
Solution. Value ratio $= 5 \times 1 : 6 \times 0.5 : 8 \times 0.25 = 5 : 3 : 2$, so values are Rs. 105, Rs. 63, Rs. 42. Coins: 105, 126, 168.

Exercises 25

25.1 If $A : B = 3 : 4$, $B : C = 6 : 7$ and $C : D = 14 : 15$, find $A : D$.

25.2 Divide Rs. 7,800 among A, B, C in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$.

25.3 Two numbers are in the ratio $3 : 5$. If 10 is added to each, the ratio becomes $5 : 7$. Find the numbers.

25.4 A starts a business with Rs. 40,000. After 4 months B joins with Rs. 60,000. The year-end profit is Rs. 34,000. Find each share.

25.5 A father is three times as old as his son. After 15 years he will be twice as old. Find their present ages.

25.6 The salaries of A and B are in the ratio $4 : 5$ and their expenses in the ratio $7 : 9$. Each saves Rs. 5,000 a month. Find their salaries.

25.7 A and B invest in the ratio $3 : 2$. A is a working partner who first takes 10% of the profit as a salary. The rest is shared in the investment ratio. From a profit of Rs. 60,000, what does each receive?

25.8 The speed of a train varies inversely as the square root of the number of wagons attached. With 4 wagons the speed is 60 km/h. What is the speed with 9 wagons?

25.9 5 years ago, A's age was 3 times B's. 5 years hence, A's age will be 2 times B's. Find their present ages.

25.10 Find the third proportional to 4 and 6.

25.11 A bag has Rs. 5, Rs. 2 and Rs. 1 coins in the ratio $2 : 3 : 5$ by number, worth Rs. 210 in all. How many Rs. 2 coins are there?

26. Averages

$$\text{Average} = \frac{\text{Sum of values}}{\text{Number of values}} \qquad \Longleftrightarrow \qquad \text{Sum} = \text{Average} \times \text{Count}.$$

Nearly every average question is solved by the right-hand form: **convert averages to totals, do the arithmetic on totals, and convert back.**

Tools

- **Deviation method:** pick a convenient guess, average the deviations, and add back. It also avoids big sums.
- **Evenly spaced numbers:** the average is the middle term, or $\frac{\text{first}+\text{last}}{2}$. The first n naturals average $\frac{n+1}{2}$.
- **One new member:** new value = old average + (new count) $\times$ (rise in average).
- **One replaced member:** new $-$ old = (count) $\times$ (rise in average).
- **Combining groups:** use the weighted average. For two groups, use alligation (Chapter 27).

The "share out" picture

Imagine levelling the values so that everyone holds the same amount; that level is the average. A new person with value x joining n people at average a first takes a for themselves. Any excess $x - a$ is then spread over $n + 1$ people, raising the average by $\frac{x-a}{n+1}$. Rearranged: $x = a + (n+1) \times$ rise. No formula to memorise, only the picture.

Example 26.1 (deviation method)

Find the average of 72, 75, 78, 81, 84, 79.
Solution. Guess 78. Deviations: $-6, -3, 0, +3, +6, +1$, summing to $+1$. Average $= 78 + \frac{1}{6} \approx$ **78.17**.

Example 26.2 (teacher joins)

The average age of 30 students is 20. When the teacher is included, the average becomes 21. Find the teacher's age.
Solution. $21 \times 31 - 20 \times 30 = 651 - 600 =$ **51**. Or by the picture: $20 + 31 \times 1 = 51$.

Example 26.3 (batting average)

A batter averages 40 after 10 innings. How many runs in the 11th innings would raise the average to 42?
Solution. $42 \times 11 - 40 \times 10 = 462 - 400 =$ **62**.

Example 26.4 (wrong entry)

The mean of 40 observations is 50. One value, 36, was misread as 56. Find the correct mean.
Solution. The total was too high by 20. Correct mean $= 50 - \frac{20}{40} =$ **49.5**.

Example 26.5 (overlapping groups)

The average of 11 numbers is 50. The first six average 49 and the last six average 52. Find the sixth number.

Solution. The sixth number is counted twice in the two groups: $6 \times 49 + 6 \times 52 - 11 \times 50 = 294 + 312 - 550 = \mathbf{56}$.

Common trap

The average of averages is not the overall average unless the groups have equal sizes. Section A (30 students, average 60) with section B (20 students, average 70) averages $\frac{1800+1400}{50} = 64$, not 65.

Exercises 26

26.1 Find the average of the first 20 multiples of 7.

26.2 The average of 5 consecutive odd numbers is 27. Find the largest.

26.3 The mean of 25 observations is 36. The first 13 average 32 and the last 13 average 40. Find the 13th observation.

26.4 A batter scores 87 in the 17th innings, raising the average by 3. Find the new average.

26.5 A class of 40 has an average age of 15. Eight new students with an average age of 18 join. Find the new average.

26.6 The average of A, B, C is 45. The average of A and B is 40, and of B and C is 43. Find B.

26.7 The average salary of 20 workers is Rs. 15,000. Including the manager, it rises by Rs. 1,000. Find the manager's salary.

26.8 Boys in a class average 70 and girls average 85. The class average is 76. Find the ratio of boys to girls.

26.9 The average of 8 numbers is 20. Two numbers, 16 and 24, are removed. Find the new average.

26.10 A person's average on 5 tests is 72. What score on a sixth test would make the average 75?

26.11 Three years ago, the average age of a family of 5 was 17. A baby has since been born, and the family's average age is the same today. How old is the baby?

27. Mixtures and Alligation

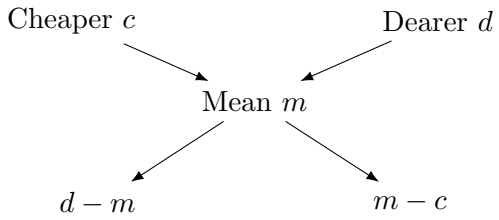
27.1 Weighted averages

If quantity q_1 has value v_1 and q_2 has v_2, the mixture's value is $\bar{v} = \frac{q_1 v_1 + q_2 v_2}{q_1 + q_2}$. "Value" can be price per kg, concentration, marks, speed, interest rate – anything averaged by quantity.

27.2 The alligation rule

To get a mean value m by mixing a cheaper ingredient (value c) and a dearer one (value d):

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{d-m}{m-c}.$$



Why it works

Balance the deviations: $q_c(m-c) = q_d(d-m)$. Whatever the cheaper part "loses" relative to the mean must be made up by what the dearer part "gains". Rearranging gives the rule. It's the see-saw principle: the mean sits closer to the larger quantity.

27.3 Repeated replacement

From a vessel of volume V of pure liquid, remove x litres and replace with water, n times. Pure liquid left:

$$V\left(1-\frac{x}{V}\right)^n.$$

Each operation keeps the fraction $1-\frac{x}{V}$ of whatever pure liquid is there – a compound *decay*, just like CI with a negative rate.

Example 27.1 (rice blend)

Rice at Rs. 40/kg and Rs. 55/kg is mixed to sell at a cost of Rs. 45/kg. Ratio?
Solution. Cheaper : dearer $= (55-45) : (45-40) = 10 : 5 = \mathbf{2 : 1}$.

Example 27.2 (replacement)

A 40 L can of milk: 4 L is removed and replaced by water, three times. How much milk remains?
Solution. $40 \times (0.9)^3 = \mathbf{29.16}$ L.

Example 27.3 (mixing two mixtures)

Vessel 1 has milk:water $= 3:1$, vessel 2 has $5:3$. In what ratio should they be mixed to get $2:1$?

Solution. Work with milk fractions: $\frac{3}{4}$, $\frac{5}{8}$, target $\frac{2}{3}$. Cheaper (vessel 2) : dearer (vessel 1) $= (\frac{3}{4} - \frac{2}{3}) : (\frac{2}{3} - \frac{5}{8}) = \frac{1}{12} : \frac{1}{24} = 2:1$. So vessel 1 : vessel 2 $=$ **1 : 2**.

Example 27.4 (milkman's profit)

A milkman buys milk at Rs. 60/L, adds water, and sells the mixture at Rs. 60/L, making a 20% profit. Ratio of milk to water?

Solution. Effective CP of the mixture $= 60/1.2 = 50$ per litre. Water costs 0. Alligation: water : milk $= (60-50) : (50-0) = 1:5$, so milk : water $=$ **5 : 1**.

Example 27.5 (alligation as a tool for anything averaged)

A class of 50 has an average mark of 64. Boys average 60 and girls average 70. How many girls are there?

Solution. Boys : girls $= (70-64) : (64-60) = 3:2$, so there are **20** girls. The "mean" can be marks, a speed, an interest rate or a profit percentage: any quantity-weighted average.

Example 27.6 (profit on parts)

A trader has 100 kg of sugar. Part is sold at 8% profit and the rest at 18% profit. The overall profit is 14%. How much was sold at 18%?

Solution. 8% part : 18% part $= (18-14) : (14-8) = 2:3$. So **60** kg at 18%.

Example 27.7 (three ingredients)

Rice at Rs. 30, Rs. 35 and Rs. 50 per kg is blended to cost Rs. 40 per kg. Give one possible ratio.

Solution. Pair each cheaper type with a dearer one. Pairing 30 with 50: $10:10$ for the pair. Pairing 35 with 50: $10:5$. Totals: 30-rice : 35-rice : 50-rice $= 10:10:(10+5) =$ **2 : 2 : 3**. Check: $\frac{60+70+150}{7} = 40$. ✓ Infinitely many ratios work, because there is one equation and two free ratios.

Common trap

In mixture-of-mixtures problems, always alligate on *one component's fraction* (e.g. milk fraction), never on the ratios themselves.

Exercises 27

27.1 Tea at Rs. 320/kg is mixed with tea at Rs. 420/kg to give a blend worth Rs. 350/kg. In what ratio?

27.2 60 L of a mixture contains 20% water. How much water must be added to make water 25% of the mixture?

27.3 An 80 L cask of wine: 8 L is drawn off and replaced with water, twice. Final ratio of wine to water?

27.4 Two alloys have gold and copper in ratios $7:2$ and $7:11$. Equal weights are melted together. Ratio of gold to copper in the new alloy?

27.5 The average weight of 8 people rises by 2.5 kg when one of them, weighing 65 kg, is replaced. Weight of the new person?

27.6 How much water must be added to 48 L of milk costing Rs. 60/L so that the mixture costs Rs. 48/L?

27.7 Rs. 10,000 is invested partly at 6% and partly at 9% simple interest. The total annual interest is Rs. 780. How much is at each rate?

27.8 A 60 L mixture has milk and water in the ratio $2:1$. How much water must be added to make the ratio $1:2$?

27.9 From a 100 L tank of pure acid, 10 L is replaced by water, and this is done 3 times. What percentage of acid remains?

27.10 A 30% acid solution and a 50% acid solution are mixed to get 40 L of 35% solution. How much of each is needed?

27.11 In what ratio must water be mixed with a liquid so that selling the mixture at cost price gives $16\frac{2}{3}\%$ profit?

27.12 A vessel holds 20 L of a $3:1$ milk–water mixture. How much mixture should be replaced with water to make it $1:1$?

28. Time and Work, Pipes and Cisterns

28.1 The LCM (work-units) method

If A finishes a job in 12 days and B in 18, **let the job be LCM**$(12, 18) = 36$ **units**. Then A does 3 units/day and B does 2. Everything becomes integer arithmetic. Together: $36/5 = 7.2$ days.

Core relations

- Rate $= \frac{1}{\text{time}}$; rates add when people work together.
- Efficiency is inversely proportional to time: if A is twice as efficient as B, A takes half the time.
- Together time for two: $\frac{xy}{x+y}$.
- **Man-days formula:** $\dfrac{M_1 D_1 H_1}{W_1} = \dfrac{M_2 D_2 H_2}{W_2}$ (men, days, hours/day, work done).
- **Wages** are shared in the ratio of *work done*, not time spent.
- **Pipes:** inlets have positive rates, outlets/leaks negative rates. Same algebra.

28.2 Efficiency and equivalence

Efficiency is rate. If A is 40% more efficient than B, then A's rate is $1.4\times$ B's, and A's time is $\frac{1}{1.4}$ of B's.

Equivalence. If 3 men do the work of 5 women, then 1 man $= \frac{5}{3}$ women. Convert everyone to one kind of worker, then use man-days.

Working backwards from a deadline. When someone leaves early, count each person's working days from the *finish*: in "A leaves 5 days before the end", A works $T - 5$ days and the others work T.

Why it works

"Work" is just quantity = rate × time, like distance = speed × time. Choosing the total as the LCM just picks a convenient unit – like measuring in paise instead of rupees to avoid fractions.

Example 28.1

A, B and C together finish a job in 10 days; A and B together in 15 days. How long would C take alone?
Solution. C's rate $= \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$: **30 days**.

Example 28.2 (man-days, real life: a project deadline)

12 people working 8 hours a day complete $\frac{2}{5}$ of a project in 10 days. How many *additional* people are needed to finish the rest in 10 more days at 6 hours a day?
Solution. $\frac{12 \cdot 10 \cdot 8}{2/5} = \frac{M \cdot 10 \cdot 6}{3/5} \Rightarrow 2400 = 100M \Rightarrow M = 24$. Additional: **12**.

Example 28.3 (alternate days)

A can do a job in 10 days, B in 15. They work on alternate days, A starting. When is it done?

Solution. Total 30 units; A 3/day, B 2/day; each 2-day cycle does 5 units. 6 cycles = 12 days do exactly 30 units: **12 days**. (Watch out: if the total is not a multiple of the cycle, finish the leftover with whoever's turn it is.)

Example 28.4 (pipes)

Pipe A fills a tank in 20 min, B in 30 min, and C empties it in 15 min. All three are opened. Time to fill?

Solution. $\frac{1}{20}+\frac{1}{30}-\frac{1}{15}=\frac{3+2-4}{60}=\frac{1}{60}$: **60 minutes**.

Example 28.5 (wages)

A can do a job in 6 days and B in 8 days. With C's help they finish it in 3 days and are paid Rs. 3,200. How is it split?

Solution. In 3 days A does $\frac{3}{6}=\frac{1}{2}$, B does $\frac{3}{8}$, so C does $\frac{1}{8}$. Split 4 : 3 : 1: A Rs. 1,600, B Rs. 1,200, C Rs. 400.

Example 28.6 (converting workers)

2 men and 3 women finish a job in 10 days; 3 men and 2 women finish it in 8 days. How long would 1 man take alone?

Solution. Let the rates be m and w of the job per day. $2m+3w=\frac{1}{10}$ and $3m+2w=\frac{1}{8}$. Add: $5(m+w)=\frac{9}{40}$, so $m+w=\frac{9}{200}$. Subtract: $m-w=\frac{1}{8}-\frac{1}{10}=\frac{1}{40}=\frac{5}{200}$. Thus $m=\frac{7}{200}$, so a man alone takes $\frac{200}{7}\approx$ **28.6** days.

Example 28.7 (real life: a sprint falling behind)

A team of 8 planned to finish 240 story points in 30 days. After 10 days it has done only 60 points. How many extra people are needed to finish on time?

Solution. The rate so far is $\frac{60}{8\times10}=0.75$ points per person-day. Remaining: 180 points in 20 days, or $\frac{180}{20\times0.75}=12$ people. So **4** more, assuming new people work at the same rate immediately. (They rarely do. Brooks's law says adding people to a late software project makes it later.)

Exercises 28

28.1 A is twice as efficient as B. Together they finish a job in 14 days. How long would A take alone?

28.2 A can do a job in 20 days and B in 30 days. They start together, but A leaves 5 days before the job is finished. How many days does the job take?

28.3 A, B and C can do a job in 12, 15 and 20 days. All three start, but C leaves after 2 days. How long does the job take in total?

28.4 A pipe fills a tank in 6 hours, but because of a leak it takes 8 hours. How long would the leak take to empty a full tank?

28.5 15 workers build a wall in 12 days working 8 hours a day. How many days would 20 workers take working 9 hours a day?

28.6 A and B can do a job in 12 days, B and C in 15 days, C and A in 20 days. How long would all three take together, and A alone?

28.7 Pipes A and B fill a tank in 12 and 15 minutes. They are opened alternately for 1 minute each, starting with A. When is the tank full?

28.8 A can do a job in 10 days. A works for 4 days and then B finishes the rest in 9 days. How long would B take alone?

28.9 3 men or 5 women can do a job in 12 days. How long will 6 men and 5 women take?

28.10 A does 40% more work than B per day. If B takes 21 days alone, how long does A take?

28.11 A, B and C can do a job in 20, 30 and 60 days. A works every day, and is assisted by B and C together on every third day. How long does the job take?

28.12 A tank has an inlet which fills at 4 L/min and a leak. Full, it empties in 6 hours through the leak with the inlet closed, and in 8 hours with the inlet open. What is the tank's capacity?

28.13 20 people planned to finish a job in 30 days. After 10 days, only one quarter of the work is done. How many more people are needed to finish on time?

28.14 Two pipes fill a tank in 20 and 24 min. A third drains 3 L/min. With all three open, the tank fills in 15 min. Find its capacity.

29. Time, Speed and Distance

$$\text{Distance} = \text{Speed} \times \text{Time}.$$

Everything else here is a way of holding one of the three fixed and reasoning about the other two.

The three proportionalities

Held fixed	Relationship
Time	Distance $\propto$ speed. In the same time, speeds $3:2$ give distances $3:2$.
Speed	Distance $\propto$ time.
Distance	Time $\propto \frac{1}{\text{speed}}$. Speeds $3:2$ give times $2:3$.

Unit conversion: km/h $\times \frac{5}{18}$ = m/s. So 36 km/h = 10 m/s, 54 km/h = 15 m/s, 72 km/h = 20 m/s.

29.1 Average speed

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}.$$

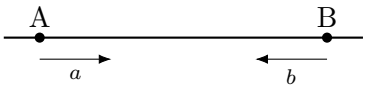
- Equal **distances** at speeds a, b: the harmonic mean, $\dfrac{2ab}{a+b}$.
- Equal **times** at speeds a, b: the arithmetic mean, $\dfrac{a+b}{2}$.
- n equal distances: $\dfrac{n}{\frac{1}{a_1} + \cdots + \frac{1}{a_n}}$.

Why not simply the average?

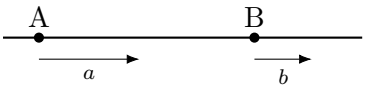
You spend *more time* at the slower speed, so it gets more weight. Go 60 km at 30 km/h (2 hours) and return at 60 km/h (1 hour): 120 km in 3 hours is 40 km/h, not 45. In the extreme case, if you go there at 1 km/h and back at 1000 km/h, your average is about 2 km/h, because you spent nearly all your time crawling.

29.2 Relative speed

Two moving things: **add speeds if they move towards or away from each other, subtract if they move the same way**. Then treat one as stationary.



towards each other: gap closes at $a + b$



same way: gap closes at $a - b$

Example 29.1 (the chase)

A thief runs at 8 km/h and is spotted 100 m ahead of a police officer, who runs at 10 km/h. How far does the thief get before being caught?

Solution. The gap closes at 2 km/h. Time $= \frac{0.1}{2} = 0.05$ h. The thief runs $8 \times 0.05 = 0.4$ km $=$ **400** m.

29.3 Standard patterns

Late and early. If a speed change converts "t_1 late" into "t_2 early", the two journey times differ by $t_1 + t_2$.

Fraction of usual speed. At $\frac{a}{b}$ of usual speed, time is $\frac{b}{a}$ of usual. So the extra time is $\left(\frac{b}{a} - 1\right) \times$ usual.

Stoppages. If a train averages v with stops and u without, it stops $\frac{u-v}{u}$ of the time: that many hours per hour.

After meeting. Two travellers start together from opposite ends, meet, and then take t_1 and t_2 more to reach the far ends. Then $\dfrac{S_1}{S_2} = \sqrt{\dfrac{t_2}{t_1}}$.

Why the square root

Let the meeting take time T. Traveller 1's remaining distance is what traveller 2 covered: $S_1 t_1 = S_2 T$. Likewise $S_2 t_2 = S_1 T$. Dividing, $\frac{S_1 t_1}{S_2 t_2} = \frac{S_2}{S_1}$, and so $\left(\frac{S_1}{S_2}\right)^2 = \frac{t_2}{t_1}$.

Repeated meetings. Two people start at opposite ends of a line of length D and turn back at the ends. By the first meeting they have jointly covered D; by the second, $3D$; by the nth, $(2n-1)D$. So each has covered $(2n-1)$ times their distance at the first meeting.

Example 29.2 (late and early)

At 40 km/h a person arrives 10 minutes late; at 50 km/h, 5 minutes early. Find the distance and the on-time journey duration.
Solution. The times differ by 15 min: $\frac{D}{40} - \frac{D}{50} = \frac{1}{4} \Rightarrow \frac{D}{200} = \frac{1}{4} \Rightarrow D = \mathbf{50}$ km. At 40 km/h the trip takes 75 min, which is 10 late, so on time is **65** minutes.

Example 29.3 (fraction of usual speed)

Walking at $\frac{3}{4}$ of the usual speed, a person is 20 minutes late. Find the usual time.
Solution. Time is $\frac{4}{3}$ of usual. Extra $= \frac{1}{3}$ of usual $= 20$ min. Usual time $=$ **60** min.

29.4 Circular tracks

On a track of length L with speeds $a > b$:

- They meet every $\dfrac{L}{a-b}$ in the same direction, and every $\dfrac{L}{a+b}$ in opposite directions.
- They are both back at the start together after $\mathrm{LCM}\left(\frac{L}{a}, \frac{L}{b}\right)$.

Exercises 29

29.1 Convert 72 km/h to m/s, and 25 m/s to km/h.

29.2 A person cycles to the office at 12 km/h and returns at 18 km/h. Find the average speed.

29.3 Walking at $\frac{3}{4}$ of the usual speed, a person is 16 minutes late. Find the usual time.

29.4 Two cyclists 72 km apart ride towards each other at 10 and 8 km/h. A bird starts with the first and flies at 15 km/h between them until they meet. How far does it fly?

29.5 Walking at 5 km/h a student misses a train by 7 minutes. Walking at 6 km/h the student is 5 minutes early. How far is the station?

29.6 An officer starts 100 m behind a thief. Their speeds are 10 and 8 km/h. How far does the thief run before being caught? How long does it take?

29.7 A train averages 80 km/h without stoppages and 60 km/h with them. How many minutes per hour does it stop?

29.8 Two trains start at the same moment from opposite ends of a route and meet. They then take 16 hours and 9 hours to complete their trips. Find the ratio of their speeds.

29.9 On a 600 m circular track, A runs at 5 m/s and B at 3 m/s from the same point. How often do they meet running the same way? Running opposite ways? When are they first both at the start?

29.10 Three equal stretches are covered at 10, 20 and 30 km/h. Find the average speed.

29.11 A and B start at opposite ends of a 100 km road and shuttle back and forth at constant speeds. Their first meeting is 60 km from A's end. How far from B's end is their second meeting?

29.12 A car covers half a journey's *time* at 40 km/h and the other half at 60 km/h. Find its average speed. How does this compare to covering half the *distance* at each speed?

30. Trains, Boats, Races and Escalators

30.1 Trains

A train is not a point, so crossing something means covering the **extra length**.

A train of length L, speed v, crossing…	Distance to cover	Speed to use
A pole or a person standing still	L	v
A platform or bridge of length P	$L+P$	v
A person walking the same way at w	L	$v-w$
A person walking the opposite way at w	L	$v+w$
A train of length M, speed u, same way	$L+M$	$v-u$
A train of length M, speed u, opposite way	$L+M$	$v+u$

Example 30.1 (platform)

A 150 m train at 54 km/h crosses a 250 m platform. How long does it take?
Solution. 54 km/h = 15 m/s. Distance 400 m. Time = $\frac{400}{15} \approx$ **26.7** s.

Example 30.2 (two trains)

Trains of 120 m and 180 m run at 50 and 40 km/h. How long do they take to pass each other in opposite and in the same directions?
Solution. Distance 300 m. Opposite: 90 km/h = 25 m/s, so **12** s. Same way: 10 km/h = $\frac{25}{9}$ m/s, so **108** s.

Example 30.3 (length from two crossings)

A train passes a pole in 15 s and a 100 m platform in 25 s. Find its length and speed.
Solution. The extra 10 s is spent covering the platform: speed = 10 m/s. Length = 10×15 = **150** m. Speed = **36** km/h.

30.2 Boats and streams

With boat speed u in still water and stream speed v:

$$\text{Downstream } d=u+v, \quad \text{Upstream } e=u-v \quad\Longrightarrow\quad u=\frac{d+e}{2}, \quad v=\frac{d-e}{2}.$$

A round trip's average speed is $\dfrac{2de}{d+e}=\dfrac{u^2-v^2}{u}$. So a current always hurts a round trip: the upstream leg takes longer and so counts more.

Example 30.4

A boat goes 24 km downstream in 2 hours and returns in 4 hours. Find the boat and stream speeds.
Solution. $d=12$ and $e=6$. So $u=$ **9** km/h and $v=$ **3** km/h.

A frame-of-reference trick

A rower drops a hat in the river and notices after 10 minutes. How long to get back to the hat? In the water's frame, the hat is stationary and the rower moves at the same speed each way. So the return also takes 10 minutes, *whatever the current*. The right frame of reference can remove a whole variable.

30.3 Races

In a race, the speed ratio equals the distance ratio over the same time.

- "A beats B by x m in a D m race": when A runs D, B runs $D - x$. So $\frac{A}{B} = \frac{D}{D-x}$.
- "A gives B a start of x m": the same equation.
- "Dead heat": both finish together.
- Chain the speed ratios: $\frac{A}{C} = \frac{A}{B} \times \frac{B}{C}$.

Example 30.5

In a 200 m race, A beats B by 20 m. In a 300 m race, B beats C by 30 m. By how much does A beat C in a 500 m race?
Solution. $\frac{B}{A} = \frac{180}{200} = \frac{9}{10}$, and $\frac{C}{B} = \frac{270}{300} = \frac{9}{10}$. So $\frac{C}{A} = \frac{81}{100}$. When A runs 500, C runs 405. A wins by **95** m.

Common trap

Margins do not add. If A beats B by 10 m and B beats C by 10 m in a 100 m race, A beats C by 19 m and not by 20. When A finishes, B has 10 m to go; C is 10% of *B's 90 m* behind B, not 10 m behind.

30.4 Escalators

Example 30.6 (moving staircase)

Walking up a moving escalator at 1 step/s, a person takes 20 steps to reach the top. At 2 steps/s, they take 32 steps. How many steps are visible on the escalator?
Solution. Let it move e steps/s and have N steps. Times: 20 s and 16 s. $N = 20 + 20e = 32 + 16e$, so $e = 3$ and $N = \mathbf{80}$.

Exercises 30

30.1 A 180 m train at 72 km/h passes a pole. How long does it take?

30.2 A 240 m train crosses a 360 m bridge in 30 s. Find its speed in km/h.

30.3 Trains of 100 m and 150 m run in opposite directions at 36 and 54 km/h. How long to pass?

30.4 A train passes a person walking the same way at 3 km/h in 10 s, and a person walking the opposite way at 6 km/h in 9 s. Find its speed and length.

30.5 A boat with still-water speed 15 km/h in a stream of 3 km/h goes 36 km downstream and back. How long does it take?

30.6 A boat takes twice as long upstream as downstream over the same distance. Find the ratio of boat speed to stream speed.

30.7 A person rows at 6 km/h in still water in a stream of 2 km/h. A round trip takes 3 hours. How far away is the turning point?

30.8 In a 100 m race, A beats B by 10 m and B beats C by 10 m. By how much does A beat C?

30.9 A runs $\frac{5}{3}$ as fast as B. In an 80 m race, how much of a start should A give B for a dead heat?

30.10 An escalator carries a standing person up in 60 s. Walking on it while switched off takes 90 s. How long does walking up it take while it moves?

30.11 A rower drops a hat and notices 20 minutes later. The rower turns round at once and retrieves the hat 1 km downstream of where it fell. Find the stream's speed.

30.12 Two boats 60 km apart head towards each other, one downstream at a still-water speed of 12 km/h and the other upstream at 18 km/h. The stream runs at 3 km/h. When do they meet?

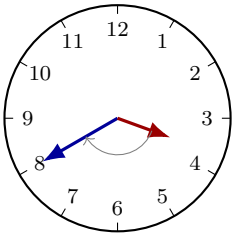
31. Clocks and Calendars

31.1 Clock angles

Minute hand	6° per minute
Hour hand	0.5° per minute (30° per hour)
Minute hand gains	5.5° per minute on the hour hand

At H hours and M minutes:

$$\text{Angle} = |30H - 5.5M| \quad (\text{take } 360^\circ - \text{this if it exceeds } 180^\circ).$$



3:40: the minute hand is at 240° from 12, the hour hand at 110°; the gap is 130°.

Example 31.1

What is the angle between the hands at 3:40?
Solution. $|90 - 220| = \mathbf{130}^\circ$.

31.2 Meetings of the hands

Relative to the hour hand, the minute hand gains a full 360° every $\frac{360}{5.5} = \frac{720}{11} = 65\frac{5}{11}$ minutes.

- Hands coincide 11 times in 12 hours, or **22 times a day**.
- Hands are straight (opposite) **22 times a day**.
- Hands are at right angles **44 times a day**.

Why not 24 and 48? Because the minute hand laps the hour hand only 11 times in 12 hours, as the hour hand itself makes one full turn.

To find when: set $30H - 5.5M = \pm\theta$ and solve for M.

Example 31.2

When do the hands coincide between 4 and 5?
Solution. $120 - 5.5M = 0 \Rightarrow M = \frac{240}{11} = \mathbf{21\frac{9}{11}}$ minutes past 4.

31.3 Fast and slow clocks

A clock that gains or loses is simply running at a different rate. True time elapsed = shown time elapsed $\times \frac{\text{true minutes}}{\text{shown minutes}}$.

Example 31.3

A clock gains 5 minutes every hour and is set right at noon. What is the true time when it shows 6:30 pm?
Solution. It shows 65 minutes per true 60. Shown elapsed: 390 min. True elapsed: $390\times\frac{60}{65} = 360$ min. True time: **6:00** pm.

Mirror image: the time seen in a vertical mirror is 11:60− time. For 3:40, the mirror shows 8:20. **Water image** and similar variations are best done by sketching.

31.4 Calendars: the odd-days method

Only the remainder mod 7 matters. An **odd day** is a day left over after complete weeks.

Ordinary year	365 days = 1 odd day
Leap year	366 days = 2 odd days
100 years	5 odd days (76 ordinary + 24 leap $\Rightarrow 76+48=124\equiv 5$)
200 years; 300 years; 400 years	3; 1; 0 odd days
Months (Jan–Dec)	3, 0/1, 3, 2, 3, 2, 3, 3, 2, 3, 2, 3

Leap years: divisible by 4, but century years must be divisible by 400. So 2000 was a leap year and 1900 was not. Number the days with 0 = Sunday, 1 = Monday, …, 6 = Saturday.

Method

For a date: add the odd days from complete centuries, then complete years in the century, then complete months in the year, then the day of the month. The total mod 7 gives the weekday.

Example 31.4

What day was 15 August 1947?
Solution. 1600 years: 0. Next 300 years: 1. Then 46 years (1901–1946), with 11 leap years: $46+11=57\equiv 1$. Jan–Jul 1947: $3+0+3+2+3+2+3=16\equiv 2$. Day: $15\equiv 1$. Total $0+1+1+2+1=5$: **Friday**.

Same calendar: two years share a calendar when they are the same type (leap or ordinary) and the odd days between them total a multiple of 7. Between ordinary years the gap is usually 6 or 11 years, and a leap year's calendar repeats after 28 years within a century that follows the usual rule.

Exercises 31

31.1 Find the angle between the hands at 7:20.

31.2 Find the angle between the hands at 5:30.

31.3 Find both times between 7 and 8 when the hands are at right angles.

31.4 The hands of a faulty clock coincide every 65 minutes of true time. Does it run fast or slow, and by how much per day?

31.5 What does a clock showing 8:25 look like in a mirror?

31.6 A clock loses 10 minutes every hour and is set right at 6 am. What is the true time when it shows 10 am?

31.7 What day of the week was 1 January 2000?

31.8 What day of the week is 15 March 2028?

31.9 If 1 January 2026 is a Thursday, what day is 1 January 2027?

31.10 Which is the next year after 2026 with exactly the same calendar?

31.11 What is the probability that a randomly chosen leap year has 53 Sundays?

31.12 What is the largest possible number of Fridays-the-13th in one year? (Check which months' 13ths fall on the same weekday.)

Part V

Algebra

32. Linear Equations and Word Problems

Aptitude algebra is mostly **translation**: turning a sentence into an equation. The solving is the easy part.

The words	The equation
A is 5 more than twice B	$A = 2B + 5$
A exceeds B by 20%	$A = 1.2B$
Two consecutive integers; odd integers	$n,\ n+1;\ \ n,\ n+2$
Two-digit number with digits a, b; reversed	$10a + b;\ \ 10b + a$
Sum of three numbers in the ratio $2 : 3 : 5$	$2k + 3k + 5k$
After 5 years, and 5 years ago	$x + 5;\ \ x - 5$ (for *every* person)

32.1 Two equations: when is there a solution?

For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$:

Condition	Lines	Solutions
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	intersect	exactly one
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	parallel	none
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	the same line	infinitely many

This matters for data sufficiency: two equations do not always pin down two unknowns.

32.2 Tricks that avoid solving

Add everything

With symmetric equations, add them first. If $x + y = 10$, $y + z = 12$, $z + x = 14$, then $2(x + y + z) = 36$, so $x + y + z = 18$. Now each variable is immediate: $z = 8$, $x = 6$, $y = 4$. Ask what the question needs. If you want $x+y+z$, you may not need the individual values at all. For example, if $3a+2b+c = 20$ and $a+2b+3c = 12$, then adding gives $4(a+b+c) = 32$, so $a + b + c = 8$ with neither a nor b known.

Example 32.1 (digits)

The digits of a two-digit number sum to 10. Reversing the digits lowers the number by 36. Find it.

Solution. $(10a+b)-(10b+a) = 9(a-b) = 36$, so $a-b = 4$. With $a+b = 10$: $a = 7$, $b = 3$. The number is **73**.

Example 32.2 (heads and legs)

A farm has 35 heads and 94 legs among chickens and rabbits. How many of each?

Solution. If all were chickens, there would be 70 legs. Each rabbit adds 2 legs: $\frac{94-70}{2} = 12$ rabbits, 23 chickens. This "assume all of one kind" method works for any two-type problem, such as coins, tickets, or marks with negative scoring.

Example 32.3 (negative marking)

A test gives $+4$ for correct and -1 for wrong. A student attempts all 60 questions and scores 130. How many were correct?
Solution. All correct would be 240. Each wrong answer costs 5. $\frac{240-130}{5} = 22$ wrong, so **38** correct.

32.3 Integer solutions (Diophantine equations)

When the unknowns must be whole numbers (coins, tickets, people), one equation can be enough.

Example 32.4

How many positive integer solutions does $3x + 5y = 47$ have?
Solution. $47 - 5y$ must be divisible by 3, and this happens for $y \equiv 1 \pmod 3$. So $y = 1, 4, 7$, giving $x = 14, 9, 4$. Beyond that x turns negative. **3 solutions.** Once you have one solution, the rest follow by adding 5 to x and subtracting 3 from y (the coefficients, swapped).

Exercises 32

32.1 The digits of a two-digit number sum to 9. Adding 27 reverses the digits. Find the number.

32.2 Solve $2x + 3y = 12$, $3x - y = 7$.

32.3 For what k does $3x + ky = 7$, $6x + 8y = 5$ have no solution?

32.4 A piggy bank has 40 coins of Rs. 2 and Rs. 5, worth Rs. 110 in all. How many of each?

32.5 How many positive integer solutions does $7x + 4y = 100$ have?

32.6 200 tickets are sold at Rs. 50 (adult) and Rs. 30 (child) for a total of Rs. 8,000. How many adult tickets are sold?

32.7 A fraction becomes $\frac{1}{2}$ when 1 is added to its numerator, and $\frac{1}{3}$ when 1 is subtracted from its denominator. Find the fraction.

32.8 Solve $x + y + z = 6$, $x + 2y + 3z = 14$, $x + y - z = 0$.

32.9 In a test with $+3$ for correct and -1 for wrong, a student attempts 50 questions and scores 90. How many are correct?

32.10 $x + y = 10$ and $x^2 - y^2 = 40$. Find x and y.

32.11 If $5a + 3b + c = 30$ and $a + 3b + 5c = 18$, find $a + b + c$.

32.12 A person buys some Rs. 8 pens and Rs. 5 pencils for exactly Rs. 100, at least one of each. How many combinations are possible?

33. Quadratics and Polynomials

33.1 Quadratic essentials

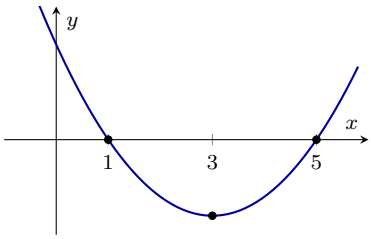
For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac; \qquad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

$D > 0$	two real roots (rational if D is a perfect square and the coefficients are rational)
$D = 0$	one repeated root; the parabola touches the axis
$D < 0$	no real root; the parabola stays on one side of the axis

Where the formula comes from

Complete the square: $x^2 + \frac{b}{a}x = -\frac{c}{a}$ becomes $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2-4ac}{4a^2}$. The vertex of the parabola is at $x = -\frac{b}{2a}$, and the roots sit symmetrically around it at a distance of $\frac{\sqrt{D}}{2|a|}$. That is why their *average* is $-\frac{b}{2a}$, so their sum is $-\frac{b}{a}$.



$y = (x-1)(x-5) = x^2 - 6x + 5$.
Roots 1 and 5 average to 3.
Vertex at $x = -\frac{b}{2a} = 3$, where $y = -4$ is the **minimum**.
For $a < 0$ the parabola opens downward and the vertex is the **maximum**.

33.2 Symmetric functions of the roots

Do not solve for the roots. Express what you need in terms of $s = \alpha + \beta$ and $p = \alpha\beta$:

$\alpha^2 + \beta^2$	$s^2 - 2p$
$\frac{1}{\alpha} + \frac{1}{\beta}$	$\frac{s}{p}$
$(\alpha - \beta)^2$	$s^2 - 4p$
$\alpha^3 + \beta^3$	$s^3 - 3ps$

New equation from old: an equation with roots α, β is $x^2 - sx + p = 0$. Roots reciprocal: reverse the coefficients. Roots doubled: replace x by $\frac{x}{2}$.

33.3 Identities worth knowing

$$(a \pm b)^2 = a^2 \pm 2ab + b^2, \quad a^2 - b^2 = (a-b)(a+b), \quad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2),$$
$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

So **if** $a+b+c = 0$**, then** $a^3+b^3+c^3 = 3abc$. And $x + \frac{1}{x} = t$ gives $x^2 + \frac{1}{x^2} = t^2 - 2$ and $x^3 + \frac{1}{x^3} = t^3 - 3t$.

33.4 Remainder and factor theorems

Dividing a polynomial $f(x)$ by $x-a$ leaves the remainder $f(a)$. So $x-a$ is a factor exactly when $f(a)=0$.

Cubics: for $ax^3+bx^2+cx+d=0$, the roots satisfy: sum $=-\frac{b}{a}$; sum of pairs $=\frac{c}{a}$; product $=-\frac{d}{a}$.

Example 33.1

If α,β are the roots of $x^2-7x+10=0$, find $\alpha^2+\beta^2$ and $\frac{1}{\alpha}+\frac{1}{\beta}$.
Solution. $s=7$ and $p=10$. $\alpha^2+\beta^2=49-20=\mathbf{29}$; $\frac{s}{p}=\mathbf{\frac{7}{10}}$.

Example 33.2 (maximum of a quadratic)

A shop sells $100-2p$ units a day at price p. Which price maximises revenue?
Solution. Revenue $=p(100-2p)$, with roots at 0 and 50. The peak lies midway at $p=\mathbf{25}$, with revenue $25\times 50=1250$. You need no calculus.

Example 33.3 (reciprocal identity)

If $x+\frac{1}{x}=3$, find $x^3+\frac{1}{x^3}$.
Solution. $27-9=\mathbf{18}$.

Common trap

Never cancel a variable without considering zero. From $x^2=5x$ you get $x=5$ *or* $x=0$. Also, "both roots positive" needs $D\geq 0$, $s>0$ and $p>0$ all together, not just one of them.

Exercises 33

33.1 Solve $2x^2-7x+3=0$.

33.2 If α,β are the roots of $x^2-6x+4=0$, find $\alpha^2+\beta^2$ and $\alpha^3+\beta^3$.

33.3 For what k does $x^2-4x+k=0$ have equal roots?

33.4 Write the equation whose roots are the reciprocals of the roots of $x^2-5x+6=0$.

33.5 The product of two consecutive positive integers is 182. Find them.

33.6 A rectangle has perimeter 34 and area 60. Find its sides.

33.7 If $x+\frac{1}{x}=4$, find $x^2+\frac{1}{x^2}$ and $x^3+\frac{1}{x^3}$.

33.8 Find the remainder when $2x^3-3x^2+x-5$ is divided by $x+1$.

33.9 Find the minimum of $x^2-8x+20$, and where it occurs.

33.10 Evaluate $13^3-8^3-5^3$ without cubing.

33.11 A train would take 1 hour less for a 180 km trip if its speed were 15 km/h higher. Find its speed.

33.12 Find the sum and product of the roots of $x^3-6x^2+11x-6=0$, and then find the roots.

34. Inequalities, Functions and Optimisation

34.1 Rules for inequalities

- Adding a number or multiplying by a positive number keeps the direction.
- **Multiplying or dividing by a negative flips the direction.** If you do not know a variable's sign, you cannot multiply by it.
- Squaring is safe only when both sides are known to be non-negative. ($-3 < 2$ but $9 > 4$.)

34.2 Sign analysis (the wavy curve)

To solve a product or fraction inequality:

1. Factorise, and mark the zeros on a number line.
2. Go right to left, alternating $+, -, +, \ldots$ (when each factor has a positive coefficient of x).
3. Read off the regions you need. Points where the denominator vanishes are always excluded.



$(x-1)(x-4) < 0 \iff 1 < x < 4$

Example 34.1

Solve $\dfrac{x-2}{x+3} \geq 0$.

Solution. Zeros at -3 and 2. Signs from the right: $+, -, +$. So $x < -3$ or $x \geq 2$. We exclude $x = -3$ since the fraction is undefined there, and we include $x = 2$ since the fraction is 0.

34.3 Absolute value is distance

$|x - a|$ is the distance from x to a on the number line.

- $|x-a| < r \iff a - r < x < a + r$; $\quad |x-a| > r \iff x < a - r$ or $x > a + r$.
- $\sum_i |x - a_i|$ is smallest when x is the **median** of the a_i.

Why the median

Move x right by a small step. It gets closer to every point on its right and further from every point on its left. So the sum falls as long as more points are on the right. It stops falling when the two sides balance, which is at the median. This is why median-based estimates ignore extreme outliers, and why the best meeting point on a road is at the median house.

Example 34.2

Minimise $|x-1| + |x-5| + |x-9|$.

Solution. At the median, $x = 5$: $4 + 0 + 4 = \mathbf{8}$.

34.4 AM–GM: the optimisation workhorse

For non-negative numbers,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}, \quad \text{with equality exactly when all are equal.}$$

Two consequences

1. **Fixed sum ⇒ the product is largest when the parts are equal.** Perimeter 40: the best rectangle is the 10×10 square, area 100.
2. **Fixed product ⇒ the sum is smallest when the parts are equal.** $x + \frac{16}{x} \geq 2\sqrt{16} = 8$, at $x = 4$.

With weights: to maximise xy given $2x + 3y = 12$, make $2x = 3y$ and not $x = y$. The equal parts must be the ones that add up to the constant.

Why it works, for two numbers

$(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a + b \geq 2\sqrt{ab}$, with equality exactly when $a = b$. Geometrically: a right triangle in a semicircle of diameter $a + b$ has altitude $\sqrt{ab}$, which can never exceed the radius $\frac{a+b}{2}$.

34.5 Extreme-value puzzles with integers

Push the others to their limits

To *maximise* one value under a fixed total, make all the others as small as allowed. To *minimise the maximum*, make all the values as equal as possible.

Example 34.3

Five different positive integers sum to 100. What are the largest possible maximum and the smallest possible maximum?

Solution. Largest maximum: $1, 2, 3, 4$ and **90**. Smallest maximum: as close together as possible, $18, 19, 20, 21, 22$, so **22**.

34.6 Functions and the floor function

A function maps each input to one output. Composition: $f(g(x))$ means apply g first. Usually $f(g(x)) \neq g(f(x))$. The **floor** $\lfloor x \rfloor$ is the largest integer $\leq x$; the **ceiling** $\lceil x \rceil$ is the smallest integer $\geq x$.

Example 34.4 (functional equation)

If $f(x + y) = f(x) + f(y)$ for all x, y and $f(1) = 3$, find $f(10)$.

Solution. $f(2) = f(1) + f(1) = 6$, and repeating gives $f(n) = 3n$. So $f(10) = \mathbf{30}$. For such questions, compute small cases until a pattern shows.

Exercises 34

34.1 Solve $x^2 - 5x + 6 > 0$.

34.2 How many integers satisfy both $3x - 7 < 11$ and $2x + 5 > 9$?

34.3 How many integers satisfy $|x - 3| \leq 4$?

34.4 For positive x, y with $2x + 3y = 12$, find the maximum of xy.

34.5 For $x > 0$, find the minimum of $4x + \frac{9}{x}$.

34.6 Find the minimum of $|x - 2| + |x - 6| + |x - 7| + |x - 15|$.

34.7 Five different positive integers have an average of 20. What is the largest possible value of one of them?

34.8 Positive integers x, y satisfy $x + y = 20$. Find the maximum of xy. Find the maximum when $x \neq y$.

34.9 If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$ and $g(f(3))$.

34.10 Find the minimum of $x^2 + y^2$ given $x + y = 10$.

34.11 Evaluate $\lfloor\sqrt{1}\rfloor + \lfloor\sqrt{2}\rfloor + \cdots + \lfloor\sqrt{16}\rfloor$.

34.12 For $0 < x < 1$, put x, x^2, $\sqrt{x}$ and $\frac{1}{x}$ in increasing order. Is the order the same for $x > 1$?

34.13 Solve $\frac{(x-1)(x+2)}{x-3} \leq 0$.

34.14 Three numbers sum to 12. What is the maximum of their product if they must be non-negative?

35. Indices, Surds and Logarithms

35.1 Laws of indices

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n,$$

$$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}, \quad a^{p/q} = \sqrt[q]{a^p}.$$

Method: write everything in terms of the smallest possible base. $27^x = 9^{x+1}$ becomes $3^{3x} = 3^{2x+2}$, so $x = 2$.

Comparing powers: equalise the exponents. For 2^{300} vs 3^{200}: these are $(2^3)^{100} = 8^{100}$ and $(3^2)^{100} = 9^{100}$. So 3^{200} is bigger.

35.2 Surds

- **Rationalise with the conjugate:** $\dfrac{1}{\sqrt{3}+\sqrt{2}} = \dfrac{\sqrt{3}-\sqrt{2}}{3-2} = \sqrt{3}-\sqrt{2}$.
- **Nested radicals:** $\sqrt{a+2\sqrt{b}} = \sqrt{x}+\sqrt{y}$, where $x+y=a$ and $xy=b$. So $\sqrt{7+4\sqrt{3}} = \sqrt{7+2\sqrt{12}} = \sqrt{4}+\sqrt{3} = 2+\sqrt{3}$.
- **Infinite radicals:** if $x = \sqrt{6+\sqrt{6+\cdots}}$ then $x^2 = 6+x$, so $x = 3$. And $\sqrt{6\sqrt{6\sqrt{\cdots}}} = 6$.
- **Comparing:** $\sqrt{11}-\sqrt{10}$ vs $\sqrt{10}-\sqrt{9}$: multiply each by its conjugate. They equal $\frac{1}{\sqrt{11}+\sqrt{10}}$ and $\frac{1}{\sqrt{10}+\sqrt{9}}$, so the first is smaller.

35.3 Logarithms

$\log_a x = y$ means $a^y = x$. A logarithm is an exponent, and all its laws are the laws of indices in disguise.

Product and quotient	$\log(xy) = \log x + \log y$; $\log \frac{x}{y} = \log x - \log y$
Power	$\log x^n = n \log x$
Change of base	$\log_a x = \dfrac{\log_b x}{\log_b a}$; $\log_a b \cdot \log_b a = 1$
Chain	$\log_a b \cdot \log_b c \cdot \log_c d = \log_a d$
Base and argument powers	$\log_{a^p} x^q = \frac{q}{p} \log_a x$
Inverse	$a^{\log_a x} = x$

Values to memorise: $\log_{10} 2 \approx 0.301$, $\log_{10} 3 \approx 0.477$, $\log_{10} 7 \approx 0.845$. Then $\log_{10} 5 = 1 - \log_{10} 2 \approx 0.699$.

Domain: $\log_a x$ needs $x > 0$, $a > 0$ and $a \neq 1$. **Always check your roots in the original equation.**

35.4 Number of digits

A positive integer N has $\lfloor \log_{10} N \rfloor + 1$ digits.

Example 35.1

How many digits does 2^{64} have?
Solution. $64 \times 0.30103 = 19.27$. So **20** digits. Famously, $2^{64} \approx 1.8 \times 10^{19}$ grains is the chessboard legend.

Example 35.2 (real life: how long to multiply money)

At 10% annual growth, how long until money multiplies 10 times?
Solution. $1.1^n = 10 \Rightarrow n = \frac{1}{\log_{10} 1.1} = \frac{1}{0.0414} \approx \mathbf{24.2}$ years.

Example 35.3 (log equation with a trap)

Solve $\log_2 x + \log_2(x-2) = 3$.
Solution. $x(x-2) = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow x = 4$ or $x = -2$. The domain needs $x > 2$, so only $x = \mathbf{4}$.

Exercises 35

35.1 Evaluate $16^{3/4}$.

35.2 Solve $27^x = 9^{x+1}$.

35.3 Which is the largest: 2^{30}, 3^{20} or 5^{10}?

35.4 Rationalise $\frac{1}{\sqrt{5}-2}$.

35.5 Simplify $\sqrt{11 + 2\sqrt{30}}$.

35.6 Evaluate $\sqrt{12 + \sqrt{12 + \sqrt{12 + \cdots}}}$.

35.7 Evaluate $\log_2 64 + \log_5 125$.

35.8 Evaluate $\log_{10} 8 + \log_{10} 125$.

35.9 Solve $\log_3(x+1) + \log_3(x-1) = 1$.

35.10 How many digits does 3^{100} have?

35.11 Evaluate $\log_9 27$.

35.12 At 8% annual growth, about how long until money triples? (Use $\log_{10} 1.08 \approx 0.0334$.)

35.13 If $2^x = 3$ and $3^y = 4$, find xy.

35.14 Evaluate $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

36. Sequences and Series

36.1 Arithmetic progressions

Constant difference d:

$$a_n = a + (n-1)d, \qquad S_n = \frac{n}{2}\big(2a + (n-1)d\big) = \frac{n}{2}(\text{first} + \text{last}).$$

Gauss's pairing

Write the sum forwards and backwards, and add. Each of the n columns totals first + last. So twice the sum is $n(\text{first} + \text{last})$. Average of the terms = average of the ends.

Choosing terms: three terms in AP: $a-d,\ a,\ a+d$. Four terms: $a-3d,\ a-d,\ a+d,\ a+3d$. The sum then cancels the d's.

36.2 Geometric progressions

Constant ratio r:

$$a_n = ar^{n-1}, \qquad S_n = a\frac{r^n - 1}{r - 1}, \qquad S_\infty = \frac{a}{1-r}\ (|r| < 1).$$

Why it works

Write $S = a + ar + \cdots + ar^{n-1}$. Then $rS = ar + \cdots + ar^n$. Subtracting, all but two terms cancel: $(r-1)S = ar^n - a$. The same trick handles arithmetic–geometric sums (below).

36.3 Means

For positive a, b: AM $= \frac{a+b}{2}$, GM $= \sqrt{ab}$, HM $= \frac{2ab}{a+b}$.

$$\text{AM} \geq \text{GM} \geq \text{HM}, \qquad \text{GM}^2 = \text{AM} \times \text{HM}.$$

A **harmonic progression** is one whose reciprocals are in AP. HM is what average speed over equal distances uses.

36.4 Standard sums

$1 + 2 + \cdots + n$	$\frac{n(n+1)}{2}$
$1^2 + 2^2 + \cdots + n^2$	$\frac{n(n+1)(2n+1)}{6}$
$1^3 + 2^3 + \cdots + n^3$	$\left(\frac{n(n+1)}{2}\right)^2$
$1 + 3 + 5 + \cdots + (2n-1)$	n^2
$\sum_{k=1}^{n} \frac{1}{k(k+1)}$	$1 - \frac{1}{n+1}$ (telescoping)
$\sum_{k\geq 0}(k+1)x^k$	$\frac{1}{(1-x)^2}$ for $\lvert x\rvert < 1$

Telescoping

Rewrite each term as a difference so that neighbours cancel: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. Likewise $\frac{1}{\sqrt{k+1}+\sqrt{k}} = \sqrt{k+1} - \sqrt{k}$. Quant interviews love this.

Example 36.1 (multiples)

Find the number and sum of the multiples of 7 between 100 and 500.
Solution. First 105, last 497. Count $= \frac{497-105}{7} + 1 = \mathbf{57}$. Sum $= 57 \times \frac{105+497}{2} = 57 \times 301 = \mathbf{17157}$.

Example 36.2 (bouncing ball)

A ball dropped from 10 m rebounds to $\frac{3}{5}$ of its previous height each time. Find the total distance travelled.
Solution. Down 10, then each rebound is travelled up and down: $10 + 2\,(6 + 3.6 + \cdots) = 10 + 2 \cdot \frac{6}{1-3/5} = 10 + 30 = \mathbf{40}$ m.

Example 36.3 (arithmetic vs geometric raises)

A job pays Rs. 30,000 a month in year 1. Compare 10-year totals for a raise of Rs. 3,000 a year and a raise of 10% a year.
Solution. AP: $\frac{10}{2}(60000 + 27000) = 4,35,000$. GP: $30000 \cdot \frac{1.1^{10}-1}{0.1} \approx 4,78,125$. Both begin with the same first raise, and the gap comes entirely from compounding. Over 30 years the difference would be enormous, which is the main lesson of geometric growth.

Example 36.4 (the arithmetic–geometric trick)

Evaluate $S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \cdots$.
Solution. $\frac{1}{2}S = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \cdots$. Subtracting, $\frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$. So $S = \mathbf{2}$. This is the expected number of tosses to the first head, in disguise.

Example 36.5 (digit-repeat sums)

Find $5 + 55 + 555 + \ldots$ to 4 terms using the trick that $\underbrace{55\ldots5}_{k} = \frac{5}{9}(10^k - 1)$.

Solution. $\frac{5}{9}\big((10+100+1000+10000)-4\big) = \frac{5}{9} \times 11106 = \mathbf{6170}$. (Check: $5+55+555+5555 = 6170$.)

Exercises 36

36.1 For the AP $3, 7, 11, \ldots$, find the 20th term and the sum of 20 terms.

36.2 Find the sum of all two-digit numbers divisible by 3.

36.3 For the GP $2, 6, 18, \ldots$, find the 8th term and the sum of 8 terms.

36.4 Evaluate $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$.

36.5 Three numbers in GP have product 216 and sum 19. Find them.

36.6 Evaluate $1^2 + 2^2 + \cdots + 20^2$ and $1^3 + 2^3 + \cdots + 10^3$.

36.7 Evaluate $\frac{1}{2\cdot3} + \frac{1}{3\cdot4} + \cdots + \frac{1}{9\cdot10}$.

36.8 A person saves Rs. 200 in the first month, and Rs. 50 more each month than in the previous one. How much is saved in 2 years?

36.9 Two numbers have AM 10 and GM 8. Find their HM, and the numbers.

36.10 Evaluate $1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \cdots$.

36.11 Find $5 + 55 + 555 + \cdots$ to 10 terms.

36.12 Logs are stacked with 20 in the bottom row, 19 in the next, down to 10 in the top row. How many logs are there?

36.13 Evaluate $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \frac{1}{\sqrt{99}+\sqrt{100}}$.

36.14 In an AP, the 5th term is 20 and the 12th term is 48. Find the first term, the difference, and the sum of the first 15 terms.

Part VI

Geometry and Mensuration

37. Lines, Angles and Triangles

37.1 Angle facts

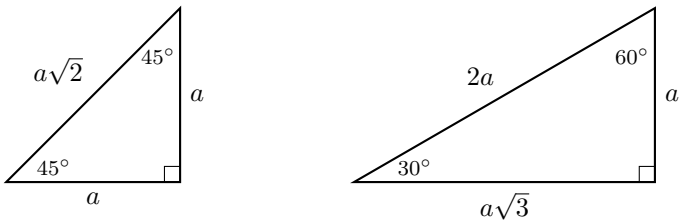
- Angles on a straight line add to 180°; vertically opposite angles are equal.
- A transversal across parallel lines: corresponding and alternate angles are equal, and co-interior angles add to 180°.
- Triangle angles add to 180°. An **exterior angle** equals the sum of the two opposite interior angles.

37.2 Triangle sides

Triangle inequality: each side is less than the sum of the other two, so the third side satisfies $|a-b| < c < a+b$.

Pythagoras: $a^2+b^2=c^2$ in a right triangle. Compare c^2 with a^2+b^2 for the longest side c: if smaller the triangle is acute, if larger it is obtuse.

Triples to memorise: $(3,4,5)$, $(5,12,13)$, $(8,15,17)$, $(7,24,25)$, $(9,40,41)$, $(20,21,29)$ and their multiples.



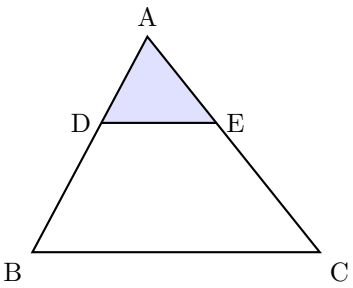
37.3 Area and the special segments

Item	Fact
Area	$\frac{1}{2}bh = \frac{1}{2}ab\sin C = \sqrt{s(s-a)(s-b)(s-c)}$ (Heron), $s=\frac{a+b+c}{2}$
Equilateral, side a	Height $\frac{\sqrt{3}}{2}a$, area $\frac{\sqrt{3}}{4}a^2$, inradius $\frac{a}{2\sqrt{3}}$, circumradius $\frac{a}{\sqrt{3}}$
Median (vertex to midpoint)	Medians meet at the **centroid**, which splits each $2:1$ from the vertex. They cut the triangle into 6 equal areas.
Angle bisector	Meets the opposite side in the ratio of the adjacent sides: $\frac{BD}{DC}=\frac{AB}{AC}$. Bisectors meet at the **incentre**.
Inradius	$r=\frac{\text{Area}}{s}$. For a right triangle, $r=\frac{a+b-c}{2}$.
Circumradius	$R=\frac{abc}{4\cdot\text{Area}}$. For a right triangle, $R=\frac{c}{2}$.
Altitude on the hypotenuse	$h=\frac{ab}{c}$
Apollonius	$AB^2+AC^2=2(AD^2+BD^2)$, where D is the midpoint of BC

37.4 Similarity

Scaling

Similar triangles have equal angles and proportional sides, with ratio k. Then: **lengths** (including heights, medians and perimeters) scale by k, and **areas** scale by k^2. In 3D, **volumes** scale by k^3.
A line parallel to one side makes a smaller similar triangle (the basic proportionality theorem): if $DE \parallel BC$, then $\frac{AD}{DB} = \frac{AE}{EC}$.



$DE \parallel BC$ with $AD : DB = 2 : 3$.
So $AD : AB = 2 : 5$.
Areas: $ADE : ABC = 4 : 25$.
Trapezium $DECB$ is the remaining 21 parts.

Example 37.1 (the 13-14-15 triangle)

Find the area, inradius and circumradius of the triangle with sides 13, 14, 15.
Solution. $s = 21$. Area $= \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = \mathbf{84}$. $r = \frac{84}{21} = \mathbf{4}$. $R = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{2730}{336} = \mathbf{8.125}$.

Example 37.2 (angle bisector)

In triangle ABC, $AB = 6$, $AC = 9$ and $BC = 10$. The bisector of A meets BC at D. Find BD.
Solution. $BD : DC = 6 : 9 = 2 : 3$, so $BD = \frac{2}{5} \times 10 = \mathbf{4}$.

Example 37.3 (counting possible triangles)

Two sides are 5 and 9. How many integer values can the third take?
Solution. $4 < c < 14$: from 5 to 13, so **9** values.

Common trap

A figure in an exam is often not to scale. Do not read a right angle or equal lengths from the picture unless they are given or implied. Also, the area ratio is the *square* of the side ratio: a triangle with sides doubled has 4 times the area, not 2.

Exercises 37

37.1 The angles of a triangle are in the ratio $2 : 3 : 4$. Find them.

37.2 A right triangle has legs 9 and 12. Find the hypotenuse, the area and the altitude on the hypotenuse.

37.3 Find the area of a triangle with sides 7, 8, 9.

37.4 Two sides of a triangle are 4 and 11. How many integer values can the third take?

37.5 An equilateral triangle has side 6. Find its area, height, inradius and circumradius.

37.6 Two similar triangles have areas in the ratio $25 : 49$. The smaller has perimeter 30. Find the larger's perimeter.

37.7 In triangle ABC, $DE \parallel BC$ with $AD : DB = 2 : 3$. Find the ratio of the area of ADE to that of trapezium DECB.

37.8 In triangle ABC, $AB = 8$, $AC = 12$, $BC = 15$. The bisector of angle A meets BC at D. Find BD and DC.

37.9 The area of triangle ABC is 60, and G is its centroid. Find the area of triangle GBC.

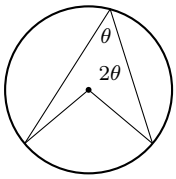
37.10 A 13 m ladder rests against a wall with its foot 5 m out. If the top slips 7 m down the wall, how far does the foot slide outwards?

37.11 How many triangles with integer sides have perimeter 12?

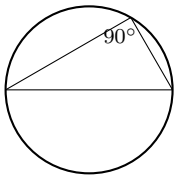
37.12 Find the inradius and circumradius of a triangle with sides 9, 40, 41.

38. Circles, Quadrilaterals and Polygons

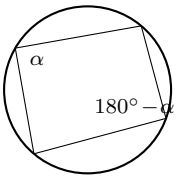
38.1 Circle theorems



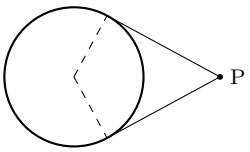
Angle at centre is twice angle at circle



Angle in a semicircle is 90°



Cyclic quadrilateral: opposite angles sum to 180°



Tangents from a point are equal; radius ⊥ tangent

- Angles in the same segment (on the same chord, same side) are equal.
- A perpendicular from the centre to a chord bisects it: $\text{radius}^2 = \text{distance}^2 + \left(\frac{\text{chord}}{2}\right)^2$.
- **Alternate segment:** the angle between a tangent and a chord equals the angle in the opposite segment.

38.2 Power of a point

Chords AB and CD crossing at P	$PA \cdot PB = PC \cdot PD$
Secant PAB and tangent PT from outside	$PT^2 = PA \cdot PB$

38.3 Two circles

With centres d apart and radii $r_1 \geq r_2$:

Direct common tangent	$\sqrt{d^2 - (r_1 - r_2)^2}$
Transverse common tangent	$\sqrt{d^2 - (r_1 + r_2)^2}$
Touching externally ($d = r_1 + r_2$): direct tangent	$2\sqrt{r_1 r_2}$

38.4 Quadrilaterals

Shape	Key properties	Area
Parallelogram	Opposite sides equal and parallel; diagonals bisect each other	$bh = ab\sin\theta$
Rectangle	Diagonals equal	lw
Rhombus	All sides equal; diagonals bisect at right angles: $4a^2 = d_1^2 + d_2^2$	$\frac{1}{2}d_1 d_2$
Square	All of the above; $d = a\sqrt{2}$	$a^2 = \frac{1}{2}d^2$
Trapezium	One pair of parallel sides	$\frac{1}{2}(a+b)h$
Circumscribed	Has an incircle: $AB + CD = BC + DA$	rs

38.5 Regular polygons

n sides: interior angle sum $(n-2) \times 180°$; each exterior angle $\frac{360°}{n}$; each interior angle $180° - \frac{360°}{n}$; diagonals $\frac{n(n-3)}{2}$. A regular hexagon splits into 6 equilateral triangles, so its area is $\frac{3\sqrt{3}}{2}a^2$.

Why exterior angles add to 360°

Walk round the polygon. At each corner you turn through the exterior angle, and by the time you are back at your start facing the same way, you have turned one full circle. This is the fastest way to get any regular polygon's angles.

Example 38.1 (chord)

A circle of radius 13 has a chord of length 24. How far is the chord from the centre?
Solution. $\sqrt{13^2 - 12^2} = \mathbf{5}$.

Example 38.2 (crossing chords)

Chords AB and CD meet inside a circle at P, with $PA = 3$, $PB = 8$, $PC = 4$. Find PD.
Solution. $3 \cdot 8 = 4 \cdot PD \Rightarrow PD = \mathbf{6}$.

Example 38.3 (common tangents)

Two circles have radii 8 and 3, with centres 13 apart. Find the common tangent lengths.
Solution. Direct: $\sqrt{169 - 25} = \mathbf{12}$. Transverse: $\sqrt{169 - 121} = \sqrt{48} \approx \mathbf{6.93}$.

Example 38.4 (rhombus)

A rhombus has diagonals 16 and 30. Find its side and area.
Solution. The half-diagonals 8 and 15 form a right triangle: side $= \mathbf{17}$. Area $= \frac{1}{2} \cdot 16 \cdot 30 = \mathbf{240}$.

Exercises 38

38.1 An arc subtends 110° at the centre. What does it subtend at a point on the major arc?

38.2 One angle of a cyclic quadrilateral is 75°. Find the opposite angle.

38.3 A chord of length 12 lies in a circle of radius 10. How far is it from the centre?

38.4 From an external point P, a secant meets a circle at A and B with $PA = 4$ and $PB = 9$. Find the tangent length from P.

38.5 The tangent from P to a circle of radius 5 has length 12. How far is P from the centre?

38.6 Find the area of a regular hexagon of side 4.

38.7 Each interior angle of a regular polygon is 140°. Find its number of sides and diagonals.

38.8 A rhombus has diagonals 10 and 24. Find its side and area.

38.9 A parallelogram has adjacent sides 5 and 8 with an included angle of 60°. Find its area.

38.10 A trapezium has parallel sides 10 and 16 and height 7. Find its area.

38.11 Two circles of radii 9 and 4 touch externally. Find the length of their direct common tangent.

38.12 A quadrilateral with an incircle has $AB = 7$, $BC = 6$, $CD = 9$. Find DA.

39. Mensuration: Areas and Volumes

39.1 Plane figures

Figure	Perimeter or length	Area
Circle, radius r	$2\pi r$	πr^2
Sector, angle θ	Arc $= \frac{\theta}{360} 2\pi r$	$\frac{\theta}{360}\pi r^2 = \frac{1}{2} r \times \text{arc}$
Segment (sector minus triangle)	Arc + chord	$\frac{\theta}{360}\pi r^2 - \frac{1}{2} r^2 \sin\theta$
Ring, radii $R > r$	$2\pi(R + r)$	$\pi(R^2 - r^2)$

39.2 Solids

Solid	Volume	Curved or lateral area	Total surface area
Cuboid l, b, h	lbh	$2h(l + b)$	$2(lb + bh + hl)$
Cube a	a^3	$4a^2$	$6a^2$
Cylinder r, h	$\pi r^2 h$	$2\pi rh$	$2\pi r(r + h)$
Cone r, h, slant l	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r + l)$
Sphere r	$\frac{4}{3}\pi r^3$	–	$4\pi r^2$
Hemisphere r	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
Frustum R, r, h, slant l	$\frac{1}{3}\pi h(R^2 + r^2 + Rr)$	$\pi l(R + r)$	$\pi l(R + r) + \pi(R^2 + r^2)$
Prism	Base area $\times h$	Base perimeter $\times h$	Lateral + 2 bases
Pyramid	$\frac{1}{3}\times$ base area $\times h$	$\frac{1}{2}\times$ perimeter $\times$ slant	Lateral + base

Cuboid diagonal: $\sqrt{l^2 + b^2 + h^2}$. Cone slant: $l^2 = r^2 + h^2$. For hollow shapes and shells, subtract.

Why $\frac{1}{3}$ for cones and pyramids

Three identical square pyramids (each with its apex at one corner of a cube) fit together to fill the cube, so each is $\frac{1}{3}$ of the box around it. Stacking thin slices shows the same holds for any apex position and base shape. A cone is just a pyramid with a circular base.

Three problem types

1. **Melting and recasting:** volume is conserved, and surface area is not.
2. **Scaling:** if lengths scale by k, areas scale by k^2 and volumes by k^3. So a 10% rise in all sides means a 21% rise in area and a 33.1% rise in volume.
3. **Unfolding:** a sector folds into a cone. The sector's radius becomes the slant height, and its arc becomes the base circumference. To find the shortest path on a surface, flatten it out.

Example 39.1 (path around a garden)

A 40×30 m garden has a 2 m wide path around the outside. Find the path's area.
Solution. Outer rectangle: $44 \times 34 = 1496$. Path $= 1496 - 1200 =$ **296** m^2.

Example 39.2 (cuboid)

For a $3 \times 4 \times 12$ cuboid, find the volume, surface area and diagonal.
Solution. $V =$ **144**; $S = 2(12 + 36 + 48) =$ **192**; diagonal $= \sqrt{9 + 16 + 144} =$ **13**.

Example 39.3 (melting)

A sphere of radius 6 is melted into cones of radius 3 and height 4. How many cones?
Solution. Sphere: $\frac{4}{3}\pi \cdot 216 = 288\pi$. Cone: $\frac{1}{3}\pi \cdot 9 \cdot 4 = 12\pi$. $\frac{288}{12} = \mathbf{24}$ cones.

Example 39.4 (sector into a cone)

A semicircle of radius 12 is folded into a cone. Find the cone's radius and height.
Solution. Slant $= 12$. The arc 12π becomes the base circumference: $2\pi r = 12\pi$, so $r = \mathbf{6}$. $h = \sqrt{144 - 36} = \mathbf{6\sqrt{3}}$.

Example 39.5 (shortest path on a cylinder)

An ant crawls from the bottom to the top of a cylinder of height 12 and base circumference 5 going once round. Find the shortest path.
Solution. Unroll the side to a 5×12 rectangle. The path is its diagonal: $\sqrt{25 + 144} = \mathbf{13}$.

Common trap

Keep units consistent: 1 $\text{m}^3 = 1000$ L, 1 hectare $= 10{,}000\ \text{m}^2$, and 1 km/h is $\frac{5}{18}$ m/s. Also, when a cone is cut, h is the height and l the slant. Using l in the volume formula is a classic slip.

Exercises 39

39.1 A circle has circumference 44. Find its radius and area ($\pi = \frac{22}{7}$).

39.2 A square has diagonal 12. Find its area.

39.3 Find the area of a ring with radii 7 and 14 ($\pi = \frac{22}{7}$).

39.4 A cylinder has radius 7 and height 10. Find its curved area, total area and volume ($\pi = \frac{22}{7}$).

39.5 A cone has radius 3 and height 4. Find its slant height, curved area and volume, in terms of π.

39.6 A cube has surface area 150. Find its volume.

39.7 Find the fraction of a cube's volume filled by the largest sphere that fits inside.

39.8 Eight equal spheres are melted into one. Find the ratio of the total surface area before to after.

39.9 5 cm of rain falls on 2 hectares. How many litres is this?

39.10 Water flows at 5 km/h through a pipe of radius 7 cm. How many cubic metres flow in an hour ($\pi = \frac{22}{7}$)?

39.11 Each edge of a cube rises by 20%. Find the percentage rise in volume and in surface area.

39.12 A wheel of diameter 70 cm rolls 1.1 km. How many revolutions does it make ($\pi = \frac{22}{7}$)?

39.13 Find the arc length and area of a sector with radius 21 and angle 60° ($\pi = \frac{22}{7}$).

39.14 A cone is cut parallel to its base halfway up its height. What fraction of the volume is in the top part?

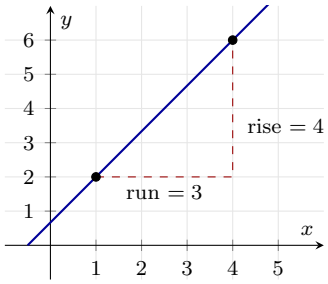
40. Coordinate Geometry

40.1 Toolkit

Item	Formula
Distance	$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
Section point $m:n$	$\left(\frac{mx_2+nx_1}{m+n},\ \frac{my_2+ny_1}{m+n}\right)$; midpoint is $m=n$
Centroid	Average of the three vertices
Slope	$m=\frac{y_2-y_1}{x_2-x_1}=\tan\theta$
Line	$y=mx+c$; $y-y_1=m(x-x_1)$; $\frac{x}{a}+\frac{y}{b}=1$ (intercepts a and b)
Parallel, perpendicular	$m_1=m_2$; $m_1m_2=-1$
Distance from a point to a line	$\frac{\lvert ax_0+by_0+c\rvert}{\sqrt{a^2+b^2}}$
Between parallel lines	$\frac{\lvert c_1-c_2\rvert}{\sqrt{a^2+b^2}}$
Triangle area (shoelace)	$\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$
Circle, centre (h,k) radius r	$(x-h)^2+(y-k)^2=r^2$

Collinear points make a triangle of zero area. Equivalently, the slopes between pairs are equal.

Lattice points on a segment from (x_1,y_1) to (x_2,y_2), endpoints included: $\gcd(\lvert x_2-x_1\rvert,\lvert y_2-y_1\rvert)+1$.



$A(1,2)$, $B(4,6)$.
Distance $=\sqrt{9+16}=5$.
Midpoint $=(2.5,4)$.
Slope $=\frac{4}{3}$.
Line: $y-2=\frac{4}{3}(x-1)$, or $4x-3y+2=0$.

Example 40.1 (distance to a line)

Find the distance from $(1,1)$ to $3x+4y-10=0$.
Solution. $\frac{\lvert 3+4-10\rvert}{5}=\mathbf{\frac{3}{5}}$.

Example 40.2 (shoelace)

Find the area of the triangle with vertices $(0,0)$, $(4,0)$, $(4,3)$.
Solution. $\frac{1}{2}\lvert 0(0-3)+4(3-0)+4(0-0)\rvert=\mathbf{6}$.

Example 40.3 (completing the square for a circle)

Find the centre and radius of $x^2+y^2-6x+8y=0$.
Solution. $(x-3)^2+(y+4)^2=9+16=25$. Centre $(3,-4)$, radius $\mathbf{5}$.

Example 40.4 (regions from absolute value)

Find the area enclosed by $\lvert x\rvert+\lvert y\rvert=3$.
Solution. It is a square (a diamond) with vertices at $(\pm3,0)$ and $(0,\pm3)$. Its diagonals are 6, so the area is $\frac{1}{2}\cdot 6\cdot 6=\mathbf{18}$.

Exercises 40

40.1 Find the distance between $(-1, 2)$ and $(5, 10)$.

40.2 Find the point dividing the segment from $(1, 2)$ to $(6, 7)$ in the ratio $2 : 3$.

40.3 Find the line through $(2, 3)$ with slope 2, and its x-intercept.

40.4 Find the line through the origin perpendicular to $2x + 3y = 6$.

40.5 For what k are $(1, 2)$, $(3, 6)$ and $(5, k)$ collinear?

40.6 Find the area of the triangle with vertices $(1, 1)$, $(4, 1)$, $(4, 5)$.

40.7 Find the distance between the lines $3x + 4y = 5$ and $3x + 4y = 15$.

40.8 Find the centre and radius of $x^2 + y^2 + 4x - 10y + 13 = 0$.

40.9 Find the area enclosed by $|x| + |y| = 4$.

40.10 How many lattice points lie on the segment from $(1, 1)$ to $(13, 9)$, ends included?

40.11 Where do $x + y = 5$ and $x - y = 1$ meet?

40.12 Find the area of the triangle formed by $3x + 4y = 12$ and the axes.

41. Trigonometry, Heights and Distances

41.1 Ratios

In a right triangle: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$. Reciprocals: csc, sec, cot.

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Memory aid: sin runs $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cos is the same list reversed.

41.2 Identities

$$\sin^2\theta + \cos^2\theta = 1, \quad 1 + \tan^2\theta = \sec^2\theta, \quad 1 + \cot^2\theta = \csc^2\theta,$$

$$\sin(90° - \theta) = \cos\theta, \quad \tan(90° - \theta) = \cot\theta,$$

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B, \quad \cos(A \pm B) = \cos A\cos B \mp \sin A\sin B,$$

$$\sin 2A = 2\sin A\cos A.$$

Max and min: $a\sin\theta + b\cos\theta$ has its maximum at $\sqrt{a^2+b^2}$ and minimum at $-\sqrt{a^2+b^2}$.

Why it works

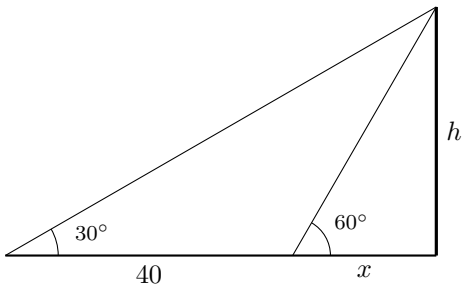
All these follow from the unit circle. A point at angle θ is $(\cos\theta, \sin\theta)$, and its distance from the origin is 1, which is the first identity. Dividing it by $\cos^2\theta$ or $\sin^2\theta$ gives the next two. For the max: $(a, b)\cdot(\sin\theta, \cos\theta)$ is largest when the two vectors align, and then it equals the product of their lengths.

41.3 Any triangle

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$. **Cosine rule:** $c^2 = a^2 + b^2 - 2ab\cos C$.

41.4 Heights and distances

Angle of elevation is measured up from the horizontal. **Angle of depression** is measured down from it. Draw the right triangle, and use tan for most questions.



$h = x\tan 60° = (x+40)\tan 30°$
gives $x = 20$ and $h = 20\sqrt{3} \approx 34.6$.

Example 41.1 (moving closer)

From a point on level ground, the top of a tower has elevation 30°. After walking 40 m towards it, the elevation is 60°. Find the height.
Solution. With x the remaining distance: $h = \sqrt{3}x$, and $h = \frac{x+40}{\sqrt{3}}$. So $3x = x + 40$, giving $x = 20$. $h = 20\sqrt{3} \approx \mathbf{34.64}$ m.
Shortcut: when the angles are θ and $90° - \theta$, $h = \sqrt{d_1 d_2}$ for the two distances from the base: $\sqrt{20 \cdot 60} = \sqrt{1200}$. ✓

Example 41.2 (given one ratio)

If $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ and $\tan\theta$.
Solution. Use the triangle 3-4-5: $\cos\theta = \mathbf{\frac{4}{5}}$, $\tan\theta = \mathbf{\frac{3}{4}}$.

Example 41.3 (cosine rule)

Two roads leave a junction at 60°. Two cyclists ride 8 km and 5 km along them. How far apart are they?
Solution. $c^2 = 64 + 25 - 2 \cdot 8 \cdot 5 \cdot \frac{1}{2} = 49$. So $c = \mathbf{7}$ km.

Exercises 41

41.1 Evaluate $\sin 30° \cos 60° + \cos 30° \sin 60°$.

41.2 If $\tan\theta = \frac{3}{4}$ with θ acute, find $\sin\theta + \cos\theta$.

41.3 Simplify $(1 - \sin^2\theta)\sec^2\theta$.

41.4 A 10 m ladder makes 60° with the ground. How high up the wall does it reach?

41.5 A tower is 30 m tall. How much longer is its shadow when the sun's elevation is 30° than when it is 45°? (Use $\sqrt{3} \approx 1.732$.)

41.6 Find the maximum of $3\sin\theta + 4\cos\theta$.

41.7 In a triangle, two sides are 5 and 8 with a 60° angle between them. Find the third side.

41.8 In triangle ABC, $A = 30°$, $B = 45°$ and $a = 10$. Find b.

41.9 From the top of a 100 m cliff, two boats on the same side of the cliff are seen at depressions of 45° and 30°. How far apart are they?

41.10 Evaluate $\sin^2 1° + \sin^2 2° + \cdots + \sin^2 89°$.

41.11 Evaluate $\tan 1° \tan 2° \cdots \tan 89°$.

Part VII

Data Interpretation and Statistics

42. Data Interpretation: Tables and Charts

Data interpretation (DI) tests whether you can read a chart and do quick percentage and ratio arithmetic on it. The maths is Chapter 22 and Chapter 19. The skill is avoiding unnecessary calculation.

How to attack a DI set

1. **Read the labels first:** units (crores or lakhs?), what the axes are, and what "total" means.
2. **Read the question next**, then return to the data for only what it needs.
3. **Approximate when the options are far apart.** Only compute precisely when they are close.
4. **Compare without computing:** a ratio rises when the top grows by a larger percentage than the bottom.
5. **Choose your sets.** DI sets vary a lot in difficulty. Skim all of them, then start with the cleanest.

Question wording	Calculation
Percentage growth from A to B	$\frac{B-A}{A} \times 100$
A is what percent of B	$\frac{A}{B} \times 100$
A is what percent *more than* B	$\frac{A-B}{B} \times 100$
Share of the total	$\frac{\text{part}}{\text{whole}} \times 100$
Pie sector angle	share $\times\, 360^\circ$
Average annual growth (CAGR)	$\left(\frac{\text{end}}{\text{start}}\right)^{1/n} - 1$
Change in percentage share	Percentage points, *not* percent

42.1 Tables

Data Set A. A company's sales (Rs. crore) by product:

Year	P	Q	R	Total
2021	120	80	100	300
2022	150	90	110	350
2023	180	120	100	400
2024	160	150	140	450
2025	200	180	120	500

Example 42.1 (fast comparison)

In which year was the percentage growth in total sales highest?

Solution. The total rises by exactly 50 every year, but on an ever larger base. So the growth rate is highest in the first year: 2022, at $\frac{50}{300} = \mathbf{16.67}\%$. No division is needed for the comparison.

Example 42.2 (share)

What was product Q's share of sales in 2024? By how many percentage points did it change from 2021?

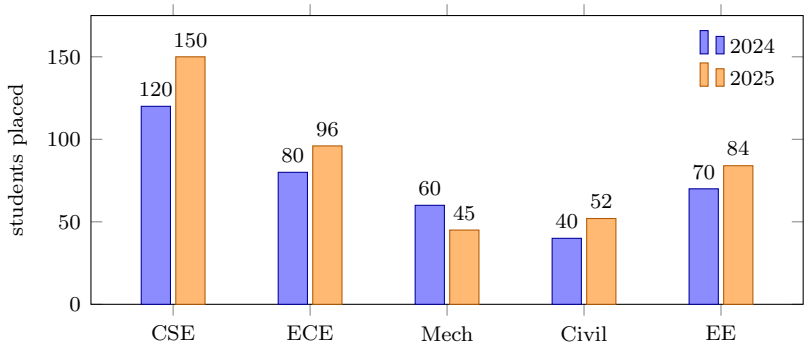
Solution. 2024: $\frac{150}{450} = 33.33\%$. 2021: $\frac{80}{300} = 26.67\%$. Change: $+\mathbf{6.67}$ percentage points.

Example 42.3 (CAGR)

Find the CAGR of product P over 2021–2025.
Solution. $\left(\frac{200}{120}\right)^{1/4} - 1 = 1.667^{0.25} - 1 \approx \mathbf{13.6}\%$. Check: $1.136^4 \approx 1.665$. ✓ (A simple average of $\frac{66.7\%}{4} = 16.7\%$ overstates it.)

42.2 Bar charts

Data Set B. Students placed by branch:

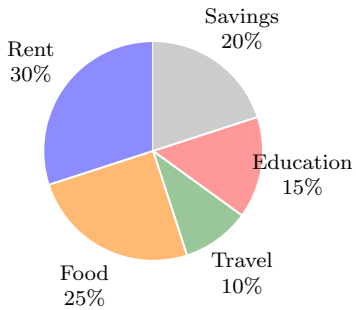


Example 42.4

Which branch had the highest percentage rise in placements?
Solution. CSE +25%; ECE +20%; Mech −25%; Civil +30%; EE +20%. **Civil**, even though it added only 12 students. The largest *number* and the largest *percentage* are different questions, and exam writers exploit this.

42.3 Pie charts

Data Set C. A household's annual income of Rs. 12 lakh is spent as follows:



Example 42.5

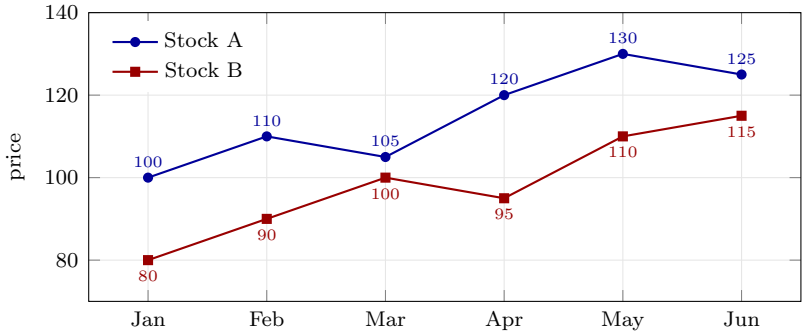
What is the sector angle for Rent, and how much more goes to Food than Travel?
Solution. $0.30 \times 360^\circ = \mathbf{108}^\circ$. Food − Travel = 15% of Rs. 12 lakh = **Rs. 1.8** lakh.

Common trap

You cannot compare pie sectors across two pies with different totals. A smaller share of a larger pie can be a larger amount.

42.4 Line graphs

Data Set D. Month-end prices (Rs.) of two stocks:



Example 42.6

In which month did Stock A have its largest percentage rise?
Solution. Feb: $\frac{10}{100} = 10\%$. Apr: $\frac{15}{105} = 14.29\%$. May: $\frac{10}{120} = 8.33\%$. **April**.

Example 42.7 (real life: portfolio)

Rs. 1,00,000 is split equally between A and B in January. Find its value in June.
Solution. $50000 \times \frac{125}{100} + 50000 \times \frac{115}{80} = 62500 + 71875 = \mathbf{Rs.\,1,34,375}$. That is a 34.375% return.

Exercises 42

42.1 (Set A) Find the average annual sales of product R.

42.2 (Set A) In which year was R's sales lowest as a percentage of the total?

42.3 (Set A) By what percentage did Q's sales grow from 2021 to 2025?

42.4 (Set A) If the profit margin on P is 10%, on Q is 20% and on R is 15%, find the total profit in 2025.

42.5 (Set B) Find the overall percentage rise in placements from 2024 to 2025.

42.6 (Set B) What was Mech's share of total placements in 2024, and in 2025?

42.7 (Set B) What is the ratio of CSE to non-CSE placements in 2025?

42.8 (Set C) Find the annual savings. If rent rises 10% with income fixed, and savings absorb the whole rise, what percentage of income is saved?

42.9 (Set C) What is the combined sector angle of Food and Education?

42.10 (Set D) In which month is the gap between the two prices smallest?

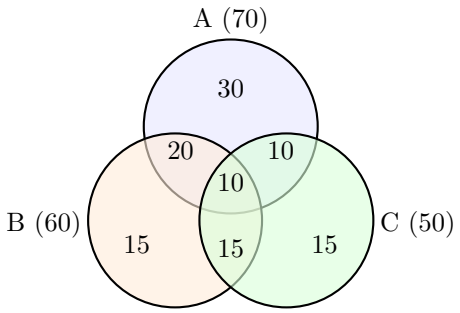
42.11 (Set D) Which stock gained more in percentage terms from January to June, and by how many percentage points?

42.12 (Set D) An investor buys 10 shares of A in March and sells them in May. What is the rupee profit, and the return?

43. Caselets, Venn Logic and Max–Min Overlap

A **caselet** is a data set given as a paragraph. The first step is always to turn the paragraph into a table or a Venn diagram. The second is to fill in the blanks by subtraction.

43.1 Three-set Venn diagrams



Survey of 120 people:
A 70, B 60, C 50;
A&B 30, B&C 25, A&C 20;
all three 10.

Fill from the **centre outwards**:
only A&B $= 30 - 10 = 20$, and so on;
only A $= 70 - 20 - 10 - 10 = 30$.

Example 43.1 (from the centre outwards)

In the survey above, find: the number reading none; exactly one; exactly two.
Solution. At least one: $70 + 60 + 50 - 30 - 25 - 20 + 10 = 115$. None: **5**.
Exactly two: $(30 - 10) + (25 - 10) + (20 - 10) = \mathbf{45}$. Exactly one: $115 - 45 - 10 = \mathbf{60}$.
Only A: $70-20-10-10 = 30$. Only B: $60-20-15-10 = 15$. Only C: $50-15-10-10 = 15$.
Sum $= 60$. ✓

43.2 Max–min overlap

Guaranteed overlap

With a and b out of a total N:

$$\max(0,\ a + b - N) \le |A \cap B| \le \min(a, b).$$

For several sets, count the **misses**. If 90%, 85%, 80%, 75% have properties A, B, C, D, the misses total at most $10 + 15 + 20 + 25 = 70\%$. So at least **30**% have all four.

Why it works

For the minimum overlap, spread out the people missing each property so that no person misses two. This uses up the most people on "missing something". Whoever remains must have everything.

43.3 Two-way tables

Questions about two attributes (such as boys/girls and placed/unplaced) are best done with a table. Each row and column has one sum, so three well-placed facts determine a 2×2 table given the total.

Example 43.2

Of 200 students, 120 are boys. 90 were placed, of whom 50 were boys. What fraction of girls were placed?

Solution.

	Placed	Not placed	Total
Boys	50	70	120
Girls	40	40	80
Total	90	110	200

$\frac{40}{80} = \mathbf{\frac{1}{2}}$.

43.4 Network and flow caselets

Some sets describe flows: people moving between cities, or customers between brands. Use a **conservation check**: for each node, in − out = change.

Example 43.3 (brand switching)

Brand X has 500 customers and Brand Y has 300. Each month, 20% of X's customers switch to Y and 10% of Y's switch to X. Find the numbers after one month and in the long run.

Solution. After one month: $X = 500 - 100 + 30 = \mathbf{430}$ and $Y = 300 + 100 - 30 = \mathbf{370}$. In the long run, flows balance: $0.2X = 0.1Y$, so $X : Y = 1 : 2$, with $X \approx \mathbf{267}$ and $Y \approx \mathbf{533}$. (This is a Markov chain's stationary distribution, Chapter 66.)

Exercises 43

43.1 In a class of 60, 40 like Maths and 35 like Science. Find the minimum and maximum number who like both.

43.2 90%, 85%, 80% and 75% of people own a phone, laptop, tablet and watch respectively. Find the minimum percentage owning all four.

43.3 Out of 100 people: 50 read A, 40 read B, 30 read C; 20 read A&B, 15 read B&C, 10 read A&C; 5 read all three. How many read none? How many read only A?

43.4 Using the two-way table in the text, what percentage of the placed students were girls? What is the placement rate among boys?

43.5 In a group of 70 who speak Hindi or Tamil or both, 50 speak Hindi and 35 speak Tamil. How many speak only Tamil?

43.6 In a group, 40% play cricket, 30% play football and 50% play neither. What percentage play both? What percentage of cricket players also play football?

43.7 A factory has 3 shifts of 40, 50 and 30 workers. 25%, 20% and 10% of them are women, respectively. What percentage of all workers are women?

43.8 Brand X has 1000 customers and Brand Y has 1000. Each year 30% of X's move to Y, and 20% of Y's move to X. Find the numbers after one year, and the long-run split.

43.9 In a 30-person office, 18 drink tea, 15 drink coffee, 5 drink neither. How many drink only one of the two?

44. Descriptive Statistics

44.1 Centre

- **Mean:** the sum over the count. Sensitive to outliers.
- **Median:** the middle value when sorted (the average of the two middle values for an even count). Robust to outliers.
- **Mode:** the most frequent value. Useful for categories.
- **Empirical rule for moderately skewed data:** Mode $\approx 3\,\text{Median} - 2\,\text{Mean}$.

Real life: a few very high salaries pull the mean above the median. That is why "average salary" in a placement brochure can mislead, and why good reports give the median.

44.2 Spread

$$\text{Variance } \sigma^2 = \frac{1}{n}\sum (x_i - \bar{x})^2 = \overline{x^2} - \bar{x}^2, \qquad \text{Standard deviation } \sigma = \sqrt{\sigma^2}.$$

The second form, the mean of squares minus the square of the mean, is usually faster. For a sample used to estimate a population, divide by $n-1$ instead. **Range** is max $-$ min. **Interquartile range** is $Q_3 - Q_1$, with Q_1 and Q_3 the medians of the lower and upper halves.

Why square the deviations?

Plain deviations sum to zero by the definition of the mean, so they say nothing. Squaring makes them all positive, weights large deviations more, and gives the key property that variances of independent things add (Chapter 63).

44.3 Shifting and scaling

Transformation $y = ax + b$	New mean	New standard deviation
Add b to everything	$\bar{x} + b$	unchanged
Multiply everything by a	$a\bar{x}$	$\lvert a\rvert\sigma$
Both	$a\bar{x} + b$	$\lvert a\rvert\sigma$

z-score: $z = \frac{x - \bar{x}}{\sigma}$ says how many standard deviations a value is from the mean, so it can compare marks across different tests.

44.4 Combining groups

For groups of sizes n_1, n_2:

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}, \qquad \sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}, \quad d_i = \bar{x}_i - \bar{x}.$$

So total variance is *within-group* plus *between-group*. This is the same decomposition as the law of total variance (Chapter 65).

44.5 Correlation and regression

With $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$, and likewise S_{xx} and S_{yy}:

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}, \qquad \text{best-fit slope } b = \frac{S_{xy}}{S_{xx}}, \qquad \text{intercept } a = \bar{y} - b\bar{x}.$$

r lies between -1 and 1, and measures only *linear* association. The best-fit line always passes through $(\bar{x}, \bar{y})$.

Example 44.1 (fast variance)

Find the mean, variance and standard deviation of 2, 4, 4, 4, 5, 5, 7, 9.
Solution. Mean $= \frac{40}{8} = 5$. Deviations squared: $9, 1, 1, 1, 0, 0, 4, 16$, totalling 32. Variance $= \frac{32}{8} = \mathbf{4}$; standard deviation $= \mathbf{2}$.

Example 44.2 (comparing across tests)

A student scores 75 in Maths (mean 60, SD 10) and 80 in English (mean 70, SD 8). Which is the better performance?
Solution. z-scores: 1.5 and 1.25. **Maths** is relatively better, despite the lower mark.

Example 44.3 (regression)

Fit a line to $(1,2), (2,3), (3,5), (4,4), (5,6)$.
Solution. $\bar{x} = 3$, $\bar{y} = 4$. $S_{xy} = 4 + 1 + 0 + 0 + 4 = 9$, $S_{xx} = 10$, $S_{yy} = 10$. So $b = 0.9$, $a = 4 - 2.7 = 1.3$, and $r = \frac{9}{10} = 0.9$. Line: $y = \mathbf{1.3 + 0.9x}$.

Common traps: correlation and Simpson's paradox

Correlation does not imply causation: ice-cream sales and drownings both rise in summer. Zero correlation does not imply independence: $y = x^2$ on symmetric data has $r = 0$.
Simpson's paradox: a trend in every group can reverse in the combined data. Treatment A can beat B for both mild and severe cases yet lose overall, if A gets mostly the severe cases. Always ask whether a group mix differs.

Exercises 44

44.1 For 3, 7, 7, 2, 9, 12, 7, 5, find the mean, median and mode.

44.2 Find the standard deviation of 1, 2, 3, 4, 5.

44.3 Data have mean 40 and SD 5. Find the mean and SD of $y = 2x + 10$.

44.4 A data set has mean 50 and SD 6. A new value equal to 50 is added. Does the SD rise, fall or stay the same?

44.5 A mark of 85 on a test with mean 70 and SD 10 has what z-score?

44.6 The mean of 10, 20, x, 40, 50 is 30. Find x. What is the median?

44.7 Two groups of 10 each have means 20 and 30, both with SD 2. Find the combined mean and variance.

44.8 Data have mean 50 and median 48. Estimate the mode with the empirical rule.

44.9 For the regression example, predict y at $x = 6$.

44.10 Find the interquartile range of the integers from 1 to 11.

44.11 Find the mean and variance of a data set in which $\sum x = 100$, $\sum x^2 = 1200$ and $n = 10$.

44.12 A group has 8 people with an average salary of Rs. 50,000 and median Rs. 30,000. One person with a salary of Rs. 1,30,000 leaves. Find the new mean. Can you find the new median?

Part VIII

Logical Reasoning

45. Ordering, Ranking and Linear Arrangements

45.1 Ranking formulas

- Total = (rank from top) + (rank from bottom) -1. (The person is counted in both ranks.)
- Position from the left = Total $-$ (position from the right) $+1$.
- People between A and B (both positions from the left): $|p_A - p_B| - 1$.
- If two people **swap** places, each takes the other's position – use the new position of one to learn the old position of the other.

Example 45.1

In a row of 50 people, A is 15th from the left and B is 20th from the right. How many people sit between them?
Solution. B from the left $= 50 - 20 + 1 = 31$. Between: $31 - 15 - 1 = \mathbf{15}$.

45.2 Linear seating: conventions

Directions in a row

- Facing **north** (up the page, away from you, the usual default): your left is the left of the page.
- Facing **south**: left and right are *reversed* relative to the page.
- "A is second to the right of B" = exactly one person between them, A on B's right.
- "A is to the right of B" (no number) = anywhere to the right, not necessarily adjacent.

45.3 A method that always works

1. Draw blank seats.
2. Place **definite** clues first (extreme ends, fixed positions, "immediately" relations).
3. For clues with two possibilities, draw **one diagram per case** side by side – don't keep cases in your head.
4. Apply negative clues ("not adjacent", "not at an end") last, to kill cases.

Example 45.2 (seven in a row)

Seven friends A–G sit in a row facing north.
(i) D sits at one of the extreme ends.
(ii) Exactly two people sit between D and B.
(iii) E sits immediately to the right of B.
(iv) C sits third to the left of G.
(v) A does not sit at either end.
(vi) Exactly one person sits between D and C.
Find the arrangement.
Solution. Number seats 1 (left) to 7. By (i), D is at 1 or 7. By (vi), C is then at 3 or 5.
- **Case D = 1:** (ii) gives B = 4, (iii) gives E = 5. C = 3 (by (vi)), so G = 6 by (iv).

Remaining A and F take 2 and 7; (v) puts A at 2, F at 7.

- **Case D = 7:** B = 4, E = 5, C = 5 by (vi) – clash with E. Rejected.

Arrangement (left to right): **D A C B E G F**. Every clue checks.

Exercises 45

45.1 In a queue, Ravi is 12th from the front and 18th from the back. How many people are in the queue?

45.2 In a row of 40 students, A is 11th from the left and B is 15th from the right. How many students sit between them?

45.3 In a class, Meera is 7th from the top and 28th from the bottom among students who passed. 5 students failed and 3 were absent. How many students are in the class?

45.4 A is 10th from the left and B is 12th from the right in a row. They swap places, and A is now 17th from the left. How many people are in the row?

45.5 Six people J, K, L, M, N, O sit in a row facing north. K sits second to the right of J. M sits at one of the ends. L is not adjacent to K. N sits next to M. L does not sit at either end.

(a) Who sits at the right end?
(b) How many people sit between M and O?
(c) Who sits immediately to the left of J?
(d) Who sits third to the left of K?

46. Circular Seating Arrangements

46.1 The left/right convention (the most common trap)

Facing centre vs facing outward

Draw the table as seen from above and go round **clockwise**.

- **Facing the centre:** your *left* is the **clockwise** neighbour; your right is anticlockwise.
- **Facing outward:** reversed – your left is **anticlockwise**.
- Mixed facing (some in, some out): apply the rule person by person.

Check it physically: stand at the bottom of a table facing its centre (facing "up" the page). Your left hand points to the left of the page – which is the clockwise direction from the bottom of a circle.

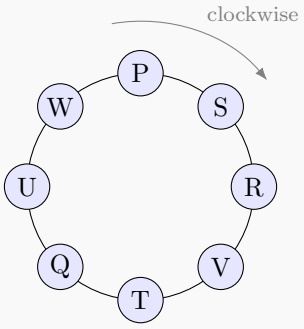
In a circle of $2n$ people, "opposite" means n seats away. "Second to the left" means skip one person. Since only relative positions matter, **anchor one person at the top** and place others relative to them.

Example 46.1 (eight around a table, facing centre)

P, Q, R, S, T, U, V, W sit around a circular table facing the centre.

(i) R sits second to the left of P.
(ii) T sits opposite P.
(iii) V sits immediately to the right of T.
(iv) Q is not an immediate neighbour of P.
(v) U sits immediately to the left of Q.
(vi) W sits immediately to the right of P.
(vii) Exactly one person sits between W and S.

Solution. Number seats 0–7 clockwise with P at 0. Facing centre, left = clockwise (+), right = anticlockwise (−). (i) R = 2. (ii) T = 4. (iii) V = 3. (vi) W = 7. (vii) S is two seats from W: seat 1 or seat 5. Free seats: 1, 5, 6 for Q, S, U. (iv) Q is not at 1. (v) U is at Q+1: Q = 5, U = 6 works (Q = 6 would need U = 7, taken). So S = 1, which satisfies (vii).



Clockwise from P: **P, S, R, V, T, Q, U, W**. Note how the clockwise neighbour of P (S) is on P's left.

Common trap

Many exam takers draw the table, then read "left" as the left of the *page*. For a person at the top of the table facing the centre (facing down the page), their left is the page's *right*. Use the clockwise rule instead of the page.

Exercises 46

46.1 Six people A, B, C, D, E, F sit around a hexagonal table **facing outward**. C sits opposite A. B sits immediately to the left of A. E is not a neighbour of C. D sits second to the right of F.

(a) Who sits immediately to the right of A?
(b) Who sits opposite B?
(c) Who sits second to the left of E?
(d) List everyone in clockwise order starting from A.

46.2 Eight people sit around a table facing the centre. How many people sit between X and Y (counting the shorter way round) if Y is third to the left of X?

46.3 In the worked example, who sits third to the right of U, and what is W's position relative to Q?

47. Multi-Attribute Puzzles (Floors, Schedules, Grids)

These puzzles combine positions (floors 1–5, days Mon–Fri) with attributes (company, city, topic). **Use a grid**: rows are positions, columns are attributes. Fill definite information, then branch into cases.

Habits of fast solvers

- Translate "immediately above" into an equation ($\text{Floor}_X = \text{Floor}_Y + 1$) and "above" into an inequality.
- Look for **pinch points**: a clue that allows only one or two positions ("on an even floor", "on the top floor").
- Combine an attribute clue with a position clue ("the IT person lives immediately below Anil" – so Anil can't be on floor 1).
- When stuck, **try each case of the most constrained person** and run the clues until a contradiction appears.

Example 47.1 (a five-storey building)

Anil, Bina, Chetan, Divya and Esha live on floors 1 (bottom) to 5, one per floor. Each works in a different sector: Banking, Consulting, FMCG, IT, Pharma.

(i) Bina lives immediately above Esha.
(ii) The consultant lives on the top floor.
(iii) The IT professional lives immediately below Anil.
(iv) Divya works neither in Pharma nor in Consulting.
(v) The banker lives on some floor below the FMCG professional.
(vi) Esha lives on one of the two lowest floors.
(vii) Bina works in Pharma.
(viii) Anil lives on an even-numbered floor.

Solution. From (viii), Anil is on 2 or 4. From (i) and (vi), (Esha, Bina) is (1, 2) or (2, 3). If Anil = 2, both options clash with floor 2. So **Anil** = 4, and by (iii) the IT person is on 3. Bina is in Pharma (vii), so Bina is not on 3: hence Esha = 1, Bina = 2. Chetan and Divya take floors 3 and 5. Floor 5 is the consultant (ii) and Divya isn't a consultant (iv), so Divya = 3 (IT) and Chetan = 5 (Consulting). Esha and Anil share Banking and FMCG; (v) puts the banker lower: Esha = Banking, Anil = FMCG.

Floor	Person	Sector
5	Chetan	Consulting
4	Anil	FMCG
3	Divya	IT
2	Bina	Pharma
1	Esha	Banking

(Every one of the eight clues is necessary – removing any one leaves multiple solutions.)

Exercises 47

47.1 Five analysts – Ravi, Sana, Tara, Uday, Veer – each present once from Monday to Friday, on five different topics: Valuation, Credit, Equity, Macro, Risk.

(i) The Macro presentation is on Monday or Friday.

(ii) Sana presents the day after Ravi.
(iii) Tara presents Risk, but not on Wednesday.
(iv) Valuation is presented on Wednesday.
(v) Uday presents on some day before Tara.
(vi) Credit is presented the day immediately after Equity.
(vii) Veer presents on Monday or Friday, but not Macro.

Find the full schedule. Then answer: who presents Equity, and on which day is Credit?

48. Blood Relations

48.1 Method: draw the family tree

- Use a standard notation: + for male, − for female, a horizontal double line for a married couple, a single horizontal line for siblings, a vertical line for parent–child.
- Keep generations on separate levels. Grandparents on top.
- Don't assume a gender unless it's stated or implied ("brother", "mother", "he").

Relation vocabulary: father's/mother's brother = uncle; sibling's son = nephew; spouse's sibling or sibling's spouse = brother/sister-in-law; son's wife = daughter-in-law; "only son of my grandfather" = my father (if paternal).

Example 48.1 (pointing to a photo)

Pointing to a photo, Ramesh says, "She is the daughter of my grandfather's only son." How is she related to Ramesh?
Solution. Grandfather's only son = Ramesh's father. His daughter = Ramesh's **sister**.

Example 48.2 (coded relations)

$A + B$: A is the mother of B. $A - B$: A is the brother of B. $A \times B$: A is the husband of B. $A \div B$: A is the daughter of B. What does $P \times Q + R - S$ mean about P and S?
Solution. P is the husband of Q; Q is the mother of R; R is the brother of S. So R and S are children of P and Q: P is the **father** of S. (S's gender is unknown, so we can't say son or daughter.)

Common trap

"Only daughter of my mother" spoken by a woman is *herself*. And with codes, read strictly left to right in pairs – don't skip intermediate links.

Exercises 48

48.1 Introducing a man, a woman says, "His mother is the only daughter of my mother." How is the woman related to the man?

48.2 A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D?

48.3 Using the codes in the example, what does $M \div N + O \times P$ tell you about M and P?

48.4 Pointing to a boy, Neha says, "He is the son of my paternal grandmother's only son." How is the boy related to Neha?

48.5 P is the father of Q. Q is the sister of R. R is the son of S. How is S related to P? (Assume a single family.)

48.6 "$A \ \$ \ B$" means A is the father of B; "$A \ \# \ B$" means A is the sister of B; "$A \ @ \ B$" means A is the wife of B. In $K \ @ \ L \ \$ \ M \ \# \ N$, how is K related to N?

49. Syllogisms

49.1 The four statement types and their Venn pictures

Statement	Meaning	Converts to
All A are B	circle A inside circle B	Some B are A
No A is B	circles disjoint	No B is A
Some A are B	circles overlap (at least one common member)	Some B are A
Some A are not B	part of A outside B	*no valid conversion*

49.2 Combination rules (premise 1 + premise 2, sharing a middle term B)

Premise 1	Premise 2	Definite conclusion
All A are B	All B are C	All A are C
All A are B	No B is C	No A is C
Some A are B	All B are C	Some A are C
Some A are B	No B is C	Some A are not C
No A is B	All B are C	Some C are not A
All A are B	Some B are C	*no definite conclusion*
Some A are B	Some B are C	*no definite conclusion*

Three question types

1. **Definite conclusions:** a conclusion follows only if it is true in *every* possible Venn diagram consistent with the statements.
2. **Either–or:** if two conclusions are about the same pair of terms, neither follows on its own, and together they cover all cases ("Some A are B" / "No A is B"; or "All A are B" / "Some A are not B"), then answer "either I or II follows".
3. **Possibility:** "X is a possibility" follows if at least *one* valid diagram makes X true.

Why it works

A syllogism question is really asking about the set of all diagrams compatible with the premises. "Follows" = true in all of them; "is a possibility" = true in at least one. The minimal diagram (draw only what's forced) is your tool for finding counterexamples.

Example 49.1

Statements: All analysts are graduates. All graduates are readers. Conclusions: I. All analysts are readers. II. Some readers are analysts.
Solution. I follows (All+All). II is the conversion of I. **Both follow.**

Example 49.2

Statements: Some bankers are traders. No trader is an auditor. Conclusions: I. Some bankers are not auditors. II. No banker is an auditor.
Solution. Some+No ⇒ Some bankers are not auditors: I follows. II is not forced (other bankers might be auditors). **Only I.**

Example 49.3

Statements: Some pens are books. Some books are bags. Conclusions: I. Some pens are bags. II. No pen is a bag.
Solution. No definite conclusion links pens and bags. I and II form a complementary pair: **either I or II follows.**

Example 49.4 (possibility)

Statements: All cats are dogs. Some dogs are rats. Conclusions: I. Some cats being rats is a possibility. II. All rats being cats is a possibility.
Solution. I: draw rats overlapping the cat circle – valid. II: put the rat circle inside the cat circle (inside dogs) – still "some dogs are rats". **Both follow.**

Exercises 49

49.1 All managers are leaders. No leader is lazy. I. No manager is lazy. II. Some lazy people are leaders.

49.2 Some apples are mangoes. All mangoes are fruits. I. Some apples are fruits. II. All fruits are mangoes.

49.3 No stock is a bond. Some bonds are funds. I. Some funds are not stocks. II. Some stocks are funds.

49.4 All rings are circles. All squares are rings. I. All squares are circles. II. Some circles are squares.

49.5 Some desks are chairs. Some chairs are tables. I. Some desks are tables. II. No desk is a table.

49.6 All cars are bikes. No bike is a truck. I. No car is a truck. II. Some trucks being cars is a possibility.

49.7 Some X are Y. No Y is Z. I. Some X are not Z. II. All Z being X is a possibility.

For each, state: only I / only II / both / neither / either I or II.

50. Data Sufficiency

Format. A question is followed by two statements. Decide whether the data is enough to answer it – you usually don't need the answer itself. Standard options:

A	Statement I alone is sufficient; II alone is not
B	Statement II alone is sufficient; I alone is not
C	Each statement alone is sufficient
D	Both together are sufficient, neither alone is
E	Not sufficient even together

(Some papers use a different letter order. Read the options every time.)

Discipline

1. Test I alone, *forgetting* II. Then II alone, forgetting I. Only then combine.
2. "Sufficient" means **exactly one** answer. For a yes/no question, a definite "no" is sufficient too.
3. Try to break sufficiency with two different numbers that satisfy the statement but give different answers.
4. Don't solve to the end – just check that a unique answer exists.

Example 50.1

Is the integer x even? I. x^2 is even. II. $3x$ is even.
Solution. I: x^2 even $\Rightarrow$ x even. Sufficient. II: $3x$ even $\Rightarrow$ x even (3 is odd). Sufficient. Answer **C**.

Example 50.2

What is the value of n? I. n is a prime between 20 and 30. II. $n+2$ is also prime.
Solution. I: 23 or 29 – not sufficient. II: many twin primes – not sufficient. Together: $23+2=25$ is not prime, $29+2=31$ is. Unique. Answer **D**.

Example 50.3

How many people sit around a circular table? I. A sits exactly opposite B. II. There are exactly 3 people between A and B going clockwise from A.
Solution. I alone gives no count. II alone tells one side only. Together: "opposite" means equal numbers on both sides, so $3+3+2=8$. Answer **D**.

Common trap

Don't let what you learn from one statement leak into your test of the other alone. Also beware "it seems sufficient because the obvious case works" – look for a second case.

Exercises 50

50.1 What is the cost price of an item? I. Its selling price is Rs. 1,200. II. The profit is 20% of the cost price.

50.2 Is the integer n divisible by 6? I. n is divisible by 3. II. n is divisible by 4.

50.3 What is x? I. $x^2 = 9$. II. $x > 0$.

50.4 In a row of 30 people, what is P's position from the left? I. P is 12th from the right. II. Q is 5th from the left and P is 14 places to Q's right.

50.5 Is $x + y$ odd, for integers x and y? I. x is even. II. y is odd.

50.6 What is the area of a rectangle? I. Its perimeter is 28 cm. II. Its diagonal is 10 cm.

51. Number and Letter Series, Analogies, Odd One Out

51.1 A checklist for number series

Work down this list, and stop when a rule fits every term:

1. **Differences:** constant (an AP), or forming a pattern themselves (second differences).
2. **Ratios:** constant (a GP), or changing steadily ($\times 2, \times 3, \times 4$).
3. **Squares, cubes, primes, factorials, Fibonacci**, or these $\pm$ a small number.
4. $ax + b$ **rules:** each term is built from the last, such as $2x + 1$.
5. **Alternating:** two series interleaved. Suspect this if the terms jump up and down, or if the series is long.
6. n**-based:** $n(n+1)$, $n^2 + 1$, $n^3 - n$.

Series	Rule	Next
3, 8, 15, 24, 35	$n^2 - 1$ from $n = 2$; differences $5, 7, 9, 11$	48
2, 5, 10, 17, 26	$n^2 + 1$	37
1, 3, 7, 15, 31	$2x + 1$, or $2^n - 1$	63
1, 2, 6, 24, 120	Factorials; ratios $\times 2, \times 3, \times 4$	720
2, 3, 5, 7, 11, 13	Primes	17
1, 1, 2, 3, 5, 8	Fibonacci	13
4, 9, 7, 12, 10, 15	$+5, -2$ alternating	13
1, 2, 4, 3, 9, 4, 16	Alternating: $1, 4, 9, 16$ and $2, 3, 4$	5

Speed shortcut

Find the **wrong term** the same way: the rule works everywhere except one spot. If differences look almost regular, a single wrong term spoils two consecutive differences. So look for two neighbouring differences that break the pattern.

51.2 Letter series

Memorise letter positions in fives: **E**=5, **J**=10, **O**=15, **T**=20, **Y**=25 ("EJOTY"). A letter's opposite has the position $27 - p$: A↔Z, B↔Y, M↔N.

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

Convert letters to numbers, then use the number-series checklist. Remember to wrap around after Z if the pattern requires.

51.3 Analogies and odd one out

Analogy: state the relationship between the first pair as a sentence or formula, then apply it. $3 : 27 :: 4 : 64$ is $x : x^3$. Doctor : Hospital :: Teacher : School is "works in".

Odd one out: look for what all but one share: prime, square, even, divisible by something, digit sum, or category. If several properties fit, choose the one that isolates *exactly one* item.

Exercises 51

51.1 Find the next term: 2, 6, 12, 20, 30, ?

51.2 Find the next term: 1, 4, 27, 16, 125, 36, ?

51.3 Find the next term: 3, 7, 15, 31, 63, ?

51.4 Find the next term: 2, 3, 5, 9, 17, 33, ?

51.5 Find the next term: 5, 11, 23, 47, ?

51.6 Find the wrong term: 1, 8, 27, 65, 125, 216.

51.7 Find the next term: 2, 5, 11, 20, 32, ?

51.8 Find the next term: 1, 2, 6, 15, 31, ?

51.9 Find the next term: B, E, H, K, ?

51.10 Find the next term: AZ, BY, CX, ?

51.11 Find the next term: ACE, GIK, MOQ, ?

51.12 Find the odd one out: 121, 144, 169, 196, 225, 250.

51.13 Complete: 8 : 27 :: 64 : ?

51.14 Find the odd one out: 2, 3, 5, 9, 11, 13.

52. Coding–Decoding and Direction Sense

52.1 Coding patterns

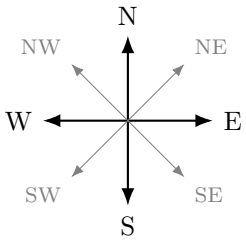
Type	Example
Shift each letter	CAT $\to$ DBU (+1 each)
Position-dependent shift	$+1, +2, +3$ by position: ABC $\to$ BDF
Opposite letter	$p \to 27 - p$: ROSE $\to$ ILHV
Reversal	COMPUTER $\to$ RETUPMOC
Letters to numbers	Sum of positions: CAB $\to 3 + 1 + 2 = 6$
Common-word elimination	Compare sentences: the word in both has the code in both
Substitution	"Sky is called sea": answer the *real* question, then translate

Example 52.1 (common-word elimination)

"sky is blue" = *la pa ra*; "is blue sea" = *pa ra mo*; "sea sky" = *mo la*. Find the code for "sea".

Solution. Sentences 2 and 3 share only "sea", and share only the code *mo*. So sea = *mo*. Likewise sky = *la*. But "is" and "blue" can't be told apart from each other: *pa* and *ra* might go either way. In an exam, "cannot be determined" can be the right option.

52.2 Direction sense



Turning right (clockwise) goes N $\to$ E $\to$ S $\to$ W.
Turning left goes N $\to$ W $\to$ S $\to$ E.
Facing south, your right hand points west.

Method: track coordinates. Start at $(0, 0)$ facing a known way. North adds to y, and east adds to x. The final distance is by Pythagoras.

Shadows: the sun rises in the east, so in the **morning** shadows fall to the **west**, and in the **evening** to the **east**. (At noon in India the shadow is short and near north, and is rarely used.)

Example 52.2 (walking)

A person walks 5 km north, turns right and walks 3 km, turns right and walks 5 km, then turns left and walks 2 km. How far and in what direction are they from the start?

Solution. North: $(0, 5)$. Right, so east: $(3, 5)$. Right, so south: $(3, 0)$. Left from south is east: $(5, 0)$. **5 km east.**

Example 52.3 (shadow)

One morning, Ravi faces a pole, and the pole's shadow falls exactly to Ravi's right. Which way is Ravi facing?

Solution. Morning shadows point west, so west is on Ravi's right. Facing south, your right

is west. **South.**

Exercises 52

52.1 If CAT is coded DBU, how is DOG coded?

52.2 Using the opposite-letter code, write LOVE.

52.3 If each letter's value is its position, find the value of PLACE.

52.4 In the common-word example, can you work out the code for "blue"? What additional sentence would settle it?

52.5 If COMPUTER is written RETUPMOC, how is LOGIC written?

52.6 If MANGO is coded NBOHP, how is TIGER coded?

52.7 If sky is called sea, sea is called water, and water is called air, where do fish live?

52.8 A person walks 3 km north and then 4 km east. How far are they from the start?

52.9 A person walks 10 km east, turns left and walks 6 km, turns left and walks 10 km, and turns right and walks 3 km. Find the final distance and direction from the start.

52.10 One evening, two friends stand face to face, and A's shadow falls exactly to A's left. Which way is A facing?

52.11 A person facing north-east turns 135° clockwise. Which way do they face now?

52.12 Starting at a point, you walk 2 km south, 3 km east, 6 km north and 3 km east. How far are you from the start?

53. Cubes, Dice and Counting in Figures

53.1 Painted cube

A cube of side n is painted on the outside and cut into n^3 unit cubes.

Painted faces	Where	Count
3	Corners	8
2	Edges, excluding corners	$12(n-2)$
1	Faces, excluding edges	$6(n-2)^2$
0	Inside	$(n-2)^3$

Check: for $n=4$, the total is $8+24+24+8=64$. ✓

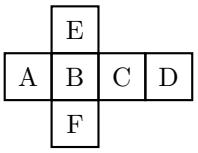
For an $a \times b \times c$ cuboid with all sides at least 2: corners 8; two faces $4[(a-2)+(b-2)+(c-2)]$; one face $2[(a-2)(b-2)+(b-2)(c-2)+(c-2)(a-2)]$; inside $(a-2)(b-2)(c-2)$.

Maximum pieces from k straight cuts of a cube: split k as evenly as possible among the three directions, $k=p+q+r$. Pieces $=(p+1)(q+1)(r+1)$. This is AM–GM in action (Chapter 34).

53.2 Dice

- **Standard die:** opposite faces sum to 7.
- **Two views sharing one face:** the faces adjacent to a common face in both views give four neighbours, and so the remaining number is opposite it.
- Faces shown together on one view are always adjacent, so they cannot be opposite.

53.3 Nets



In a straight run of squares, alternate squares are opposite: A opp. C, and B opp. D. The two side flaps are opposite: E opp. F. A row of 5 or more squares cannot fold into a cube.

53.4 Counting in figures

Be systematic: count by size. Common results:

Figure	Count
Squares on an $n \times n$ board	$\frac{n(n+1)(2n+1)}{6}$
Rectangles on an $m \times n$ board	$\binom{m+1}{2}\binom{n+1}{2}$
Triangles in a triangle fan with n lines from the apex to the base	$\binom{n}{2}$
Triangles in a triangular grid of side n (both orientations)	$\left\lfloor \frac{n(n+2)(2n+1)}{8} \right\rfloor$

Example 53.1 (painted)

A cube of side 4 is painted and cut into unit cubes. How many have exactly 2 painted faces? None?
Solution. $12 \times 2 = \mathbf{24}$; $2^3 = \mathbf{8}$.

Example 53.2 (dice)

Three views of one die show $(1, 2, 3)$, $(1, 3, 5)$ and $(1, 4, 2)$. Which number is opposite 1?
Solution. The faces seen next to 1 are 2, 3, 5, 4. Only 6 remains, so it is opposite **1**.

Exercises 53

53.1 A cube of side 5 is painted and cut. Find the numbers of unit cubes with 3, 2, 1 and 0 painted faces.

53.2 A painted cube is cut into 1000 unit cubes. How many have exactly 2 faces painted? How many have none?

53.3 What is the maximum number of pieces obtainable from a cube using 6 straight cuts? Using 7?

53.4 A $4 \times 3 \times 2$ cuboid is painted and cut into unit cubes. Find how many have 3, 2, 1 and 0 painted faces.

53.5 How many unit cubes of a $5 \times 5 \times 5$ cube can be seen from one corner view (with 3 faces visible)?

53.6 A cube is cut into pieces so that exactly 8 unit cubes have no paint. What is its side?

53.7 In the net shown, which face is opposite E? And which is opposite B?

53.8 How many triangles are in a triangular grid of side 3? Of side 4?

53.9 How many squares are on a 5×5 board? How many rectangles?

53.10 A standard die shows 3 on top and 2 facing you. A third visible face shows 1. What is the total on the three hidden faces?

54. Formal Logic, Truth-Tellers and Weighing

54.1 Implications

"If P then Q" is false in only one case: P true and Q false.

Given: If P then Q	Conclusion	Valid?
P is true	Q is true	Yes (affirming)
Q is false	P is false	Yes (the **contrapositive**)
Q is true	P is true	**No** (the converse error)
P is false	Q is false	**No** (the inverse error)

"Only if": "P only if Q" means "if P then Q". **"Unless":** "P unless Q" means "if not Q then P".

Negations: the negation of "All A are B" is "Some A are not B". Of "Some A are B": "No A is B". Of "P and Q": "not P or not Q". Of "P or Q": "not P and not Q".

54.2 Truth-tellers and liars

Method

Assume a statement is true and follow the consequences. If you reach a contradiction, it was false. When a puzzle says "exactly one is telling the truth", look for two statements that contradict each other. One of them must be true, and so all the rest are false.

Example 54.1 (knights and knaves)

Knights always tell the truth; knaves always lie. A says, "We are both knaves." What are A and B?

Solution. If A were a knight, the statement would be true, making A a knave: a contradiction. So A is a knave, and the statement is false, so they are not both knaves. So **B is a knight.**

Example 54.2 (one truth-teller)

Exactly one of four suspects tells the truth, and exactly one is guilty. A: "B did it." B: "D did it." C: "I didn't do it." D: "B is lying."

Solution. B and D contradict each other, so one is the truth-teller. Thus A and C are lying. C's lie means **C did it**. Check: A false, B false, D true, C false. ✓

54.3 Critical reasoning formats

- **Statement and assumption:** an assumption is what must be taken for granted for the statement to make sense. "Use Brand X for whiter teeth" assumes people want whiter teeth, but it does not assume that other brands fail.
- **Statement and conclusion:** a conclusion must follow from the statement alone, without outside facts. Beware absolute words like "all", "only" and "best".

- **Course of action:** a course of action must be practical and must address the stated problem, without being extreme.
- **Cause and effect:** decide which event logically comes first, or if both have a common cause.

54.4 Weighing puzzles

Information counting

A balance has **three outcomes** (left heavy, right heavy, balanced), so k weighings distinguish at most 3^k cases.

- One heavy coin among N: split into three groups. 3^k coins need k weighings.
- One odd coin of unknown weight direction: at most $\frac{3^k-3}{2}$ coins, so 12 coins in 3 weighings.

Example 54.3 (8 coins, one heavy)

What is the minimum number of weighings on a balance?
Solution. $3^1 < 8 \leq 3^2$, so at least 2. It is achieved by weighing 3 vs 3. If they balance, compare the remaining 2. Otherwise take the heavier 3 and weigh 1 vs 1. **2 weighings.**

Measuring with weights: with weights on one pan only, use powers of 2 (1, 2, 4, 8, ...), which is binary. With weights allowed on both pans, use powers of 3 (1, 3, 9, 27, ...), since each weight can be on the left, on the right, or unused. This is called balanced ternary.

Exercises 54

54.1 "If it rains, the match is cancelled. The match was not cancelled." What can you conclude?

54.2 "If it rains, the match is cancelled. The match was cancelled." Can you conclude it rained?

54.3 Write the negation of "All bankers are rich".

54.4 A says, "B is a knave." B says, "A and I are the same type." What are they?

54.5 Exactly one of these is true. I: "Exactly one of these statements is false." II: "Exactly two of these statements are false." III: "All three are false." Which is true?

54.6 Minimum weighings to find a heavy coin among 27? Among 28?

54.7 What is the minimum number of weights to weigh every integer from 1 to 40 kg on one pan? With both pans usable?

54.8 Statement: "The government has cut fuel tax to help consumers." Assumptions: I. Fuel prices affect consumers. II. Fuel tax is the only way to help consumers. Which is implicit?

54.9 "No one gets in unless they have a ticket." Priya got in. What follows? Someone has a ticket; does it follow they got in?

54.10 Three boxes are labelled Apples, Oranges and Mixed, and every label is wrong. You may take one fruit from one box. Which box do you pick from, and how does it help?

Part IX

Probability and Quant Interviews

55. Foundations

55.1 The language

An **experiment** has a **sample space** Ω of outcomes; an **event** is a subset. When outcomes are equally likely,

$$P(A) = \frac{|A|}{|\Omega|}$$

– which is why Parts I and II matter: most aptitude probability is counting in disguise.

Rules you use constantly

- Complement: $P(\text{not } A) = 1 - P(A)$. "At least one" $\Rightarrow 1 - P(\text{none})$.
- Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$; multiplication: $P(A \cap B) = P(B)\,P(A \mid B)$.
- Independence: $P(A \cap B) = P(A)P(B)$, i.e. knowing B doesn't change the chance of A.
- Law of total probability: $P(A) = \sum_i P(A \mid B_i)P(B_i)$ over a partition $B_1, B_2, \ldots$

Common trap: mutually exclusive $\neq$ independent

Mutually exclusive events (can't both happen) with positive probabilities are *never* independent: if one happens, the other's probability drops to zero.

Common trap: make outcomes equally likely

"Two dice: the sum is 2, 3, …, 12, eleven outcomes, so $P(7) = \frac{1}{11}$" is wrong. The 36 ordered pairs are equally likely; the sums are not. Always label the dice (red/blue) to count ordered outcomes.

Example 55.1

Two dice are rolled. Probability that the sum is 7? That at least one shows a 6?
Solution. Sum 7: $(1,6),(2,5),\ldots,(6,1) \to \frac{6}{36} = \frac{1}{6}$. At least one 6: $1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$.

Example 55.2 (real life: two independent approvals)

A loan file needs approval from Credit (probability 0.8) and Compliance (0.9), decided independently. Probability it gets through? Probability that exactly one approves?
Solution. Both: $0.8 \times 0.9 = 0.72$. Exactly one: $0.8 \times 0.1 + 0.2 \times 0.9 = 0.26$.

Exercises 55

55.1 Two dice are rolled. What is the probability that the sum is a prime number?

55.2 From a bag of 5 red and 3 blue balls, two are drawn without replacement. Probability they are of different colours?

55.3 A 3-digit number is chosen at random. Probability that its digits are all distinct?

55.4 $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(\text{neither})$. Are A and B independent?

55.5 A card is drawn from a standard deck. Probability it is a king or a heart?

55.6 Five people are chosen at random. Probability that at least two share a birth month (months equally likely)?

56. Conditional Probability and Bayes' Theorem

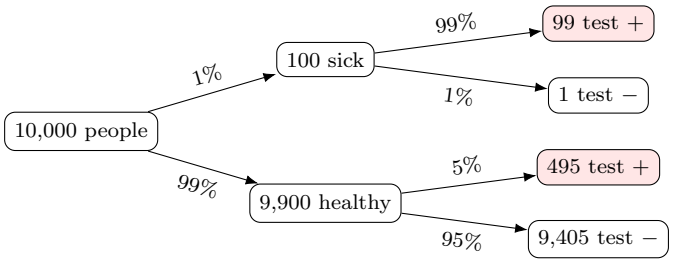
56.1 Bayes' theorem

$$P(H \mid E) = \frac{P(E \mid H)\,P(H)}{P(E \mid H)P(H) + P(E \mid \bar{H})P(\bar{H})}.$$

In words: **posterior** $\propto$ **likelihood** $\times$ **prior**. The denominator just rescales so the posteriors add to 1.

Speed shortcut: natural frequencies

Imagine 10,000 people (or 1,000 items) and push them through the tree. Count, don't multiply fractions. This avoids almost every Bayes error and is how strong interview candidates talk through these problems.



$$P(\text{sick} \mid +) = \frac{99}{99+495} = \frac{1}{6}$$

Example 56.1 (the false-positive paradox)

A disease affects 1% of people. A test catches 99% of true cases but also gives a positive result for 5% of healthy people. You test positive. Probability you have the disease?
Solution. Of 10,000 people: 100 are sick, 99 test positive. 9,900 are healthy, 495 test positive. Total positives 594, of which 99 are sick:

$$P = \frac{99}{594} = \frac{1}{6} \approx \mathbf{16.7}\%.$$

The low base rate dominates. This is the same logic as fraud-detection alerts in banking: most flagged transactions are false alarms when fraud is rare.

56.2 Classic puzzles

Example 56.2 (Monty Hall)

Three doors: a car behind one, goats behind two. You pick door 1. The host, who *knows* where the car is, opens another door showing a goat and offers a switch. Should you switch?
Solution. Your first pick is right with probability $\frac{1}{3}$; then switching loses. With probability $\frac{2}{3}$ your pick is wrong; the host is *forced* to reveal the only other goat, and switching wins. Switch: $P(\text{win}) = \mathbf{\frac{2}{3}}$.

Why it works: the host's knowledge carries the information

The host's action depends on where the car is – that's what makes it informative. If the host opened a door at random and it merely *happened* to show a goat, switching would win only $\frac{1}{2}$ of the time (see Exercise 56.5). With 100 doors and the host opening 98 goat doors, the intuition becomes obvious: switch.

Example 56.3 (Bertrand's box)

Three boxes: gold–gold, silver–silver, gold–silver. You pick a box at random and draw a coin: it's gold. Probability the other coin in that box is gold?

Solution. Count gold *coins*, not boxes: there are 3 gold coins, equally likely to be drawn, and 2 of them sit in the gold–gold box. Answer $\mathbf{\frac{2}{3}}$ (not $\frac{1}{2}$).

Example 56.4 (the two-child problems)

A family has two children. (a) At least one is a boy. Probability both are boys? (b) At least one is a boy born on a Tuesday. Probability both are boys?

Solution. (a) Equally likely: BB, BG, GB, GG. "At least one boy" leaves 3, one of which is BB: $\mathbf{\frac{1}{3}}$. (b) Each child is one of 14 equally likely (sex, day) types, so $14^2 = 196$ families. Families with at least one Tuesday-boy: $196-13^2 = 27$. Among them, both boys: $7^2-6^2 = 13$. Answer $\mathbf{\frac{13}{27}}$ – close to $\frac{1}{2}$, because the extra detail almost identifies *which* child we're talking about.

Common trap

The answer depends on *how the information was obtained*. "The older child is a boy" gives $\frac{1}{2}$; "at least one is a boy" gives $\frac{1}{3}$. In interviews, state your assumption about how you learned the fact.

Exercises 56

56.1 A family has two children and the older one is a girl. Probability both are girls? What if you only know at least one is a girl?

56.2 Of three coins, one is double-headed and two are fair. A coin is picked at random and tossed; it shows heads. Probability it is the double-headed coin?

56.3 Machines A, B, C produce 50%, 30% and 20% of a factory's output, with defect rates 1%, 2% and 3%. A randomly chosen item is defective. Probability it came from C?

56.4 Monty Hall with 4 doors: you pick one, the host (who knows) opens one goat door, and you switch to one of the two other closed doors at random. Probability of winning? Compare with staying.

56.5 Monty Hall where the host does *not* know where the car is, opens one of the other two doors at random, and it happens to show a goat. Probability of winning if you switch?

56.6 A test is 95% sensitive (catches 95% of true cases) and 90% specific (10% false positives). Prevalence is 2%. Probability of disease given a positive test?

56.7 A coin has an unknown bias p, uniformly distributed on $[0,1]$. It lands heads in all of 3 tosses. Probability that the next toss is heads? (Laplace's rule of succession: $\frac{h+1}{n+2}$.)

57. Counting-Based Probability

57.1 Cards, dice, birthdays

Deck facts: 52 cards, 4 suits of 13, 12 face cards. $\binom{52}{5} = 2{,}598{,}960$ five-card hands.

Example 57.1 (the birthday problem)

How many people do you need for a better than even chance that two share a birthday (365 equally likely days)?
Solution. $P(\text{all distinct among } n) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-n+1}{365}$. At $n = 23$ this drops to 0.493, so the chance of a match is **50.7**%. **Intuition:** with 23 people there are $\binom{23}{2} = 253$ *pairs*, each matching with probability $\frac{1}{365}$. Rough estimate: $P(\text{no match}) \approx e^{-253/365} \approx 0.50$. With 70 people it's 99.9%.

Example 57.2 (poker hands)

Probability of a full house (three of one rank, two of another)?
Solution. Choose the rank of the triple (13) and its suits ($\binom{4}{3}$), then the pair's rank (12) and suits ($\binom{4}{2}$): $13 \cdot 4 \cdot 12 \cdot 6 = 3744$. $P = \frac{3744}{2598960} \approx \mathbf{0.144}\%$.

Example 57.3 (de Méré's paradox)

Which is more likely: at least one six in 4 rolls of a die, or at least one double-six in 24 rolls of two dice?
Solution. $1 - \left(\frac{5}{6}\right)^4 = 0.518$ vs $1 - \left(\frac{35}{36}\right)^{24} = 0.491$. The first. (Gamblers in the 1650s noticed this from experience; the puzzle helped launch probability theory.)

57.2 The hypergeometric pattern

Drawing n items without replacement from N, of which K are "special":

$$P(\text{exactly } k \text{ special}) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Exercises 57

57.1 What is the probability that a 5-card hand contains exactly 2 aces?

57.2 What is the probability that a 5-card hand is a flush (all one suit) but not a straight flush? (There are 40 straight flushes, including royal flushes.)

57.3 (Newton–Pepys) Which is most likely: at least one six with 6 dice, at least two sixes with 12 dice, or at least three sixes with 18 dice?

57.4 A committee of 4 is chosen at random from 6 men and 4 women. Probability it has at least 2 women?

57.5 8 people sit randomly around a round table. Probability that A and B sit together?

57.6 The letters of BANANA are arranged at random. Probability that the two N's are not adjacent?

58. Distributions Toolkit

Distribution	Story	$P(X=k)$	Mean, variance
Bernoulli(p)	one trial	p or $1-p$	p, $p(1-p)$
Binomial(n,p)	successes in n independent trials	$\binom{n}{k}p^k(1-p)^{n-k}$	np, $np(1-p)$
Geometric(p)	trials until first success	$(1-p)^{k-1}p$	$\frac{1}{p}$, $\frac{1-p}{p^2}$
Hypergeometric	successes without replacement	see Ch. 57	$n\frac{K}{N}$
Poisson(λ)	count of rare events in an interval	$e^{-\lambda}\frac{\lambda^k}{k!}$	λ, λ
Uniform(a,b)	continuous, flat	density $\frac{1}{b-a}$	$\frac{a+b}{2}$, $\frac{(b-a)^2}{12}$
Exponential(λ)	waiting time between Poisson events	density $\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$, $\frac{1}{\lambda^2}$

Why the geometric mean is $1/p$

Let E be the expected number of trials. The first trial succeeds with probability p (done in 1); otherwise you've used 1 trial and are back at the start: $E = 1 + (1-p)E \Rightarrow E = \frac{1}{p}$. This "first-step" argument is the engine of Chapter 60.

Memorylessness. Geometric and exponential waiting times "forget" the past: having waited 10 minutes for a Poisson-process bus, your expected remaining wait is unchanged. Real buses aren't Poisson – that's the inspection paradox (Chapter 61).

Example 58.1 (real life: call centre)

A support desk gets 3 calls per hour on average (Poisson). Probability of no calls in an hour? At least 2?

Solution. $P(0) = e^{-3} \approx \mathbf{0.050}$. $P(\geq 2) = 1 - e^{-3}(1+3) \approx \mathbf{0.801}$.

Exercises 58

58.1 A family has 4 children (boys and girls equally likely). Probability of exactly 2 boys?

58.2 A student guesses all 10 questions of an MCQ test, each with 4 options. Probability of at least 8 correct? Expected number correct?

58.3 Probability that the first six appears on the third roll of a die?

58.4 Using the call-centre numbers, what is the probability of exactly 3 calls in an hour?

58.5 Expected number of rolls to get a six, and the variance of that number?

59. Expectation and the Linearity Trick

$$\mathbb{E}[X] = \sum_x x\,P(X = x), \qquad \mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] \ \textbf{always, even if } X, Y \textbf{ are dependent.}$$

Indicator variables

To find the expected *number* of things that happen, write $X = I_1 + I_2 + \cdots + I_n$, where $I_j = 1$ if thing j happens. Then

$$\mathbb{E}[X] = P(\text{thing } 1) + P(\text{thing } 2) + \cdots + P(\text{thing } n).$$

You never need the distribution of X. This turns many "impossible" puzzles into one-liners.

Example 59.1 (hat-check)

n people's hats are returned at random. Expected number of people who get their own hat?
Solution. Each person gets their own hat with probability $\frac{1}{n}$. Sum over n people: **1**, for every n. (Compare with derangements: the *probability* of zero matches is about $\frac{1}{e}$.)

Example 59.2 (distinct faces)

A die is rolled 6 times. Expected number of distinct faces seen?
Solution. Face j is seen with probability $1 - \left(\frac{5}{6}\right)^6$. Six faces: $6\left(1 - \left(\frac{5}{6}\right)^6\right) \approx \mathbf{3.99}$.

Example 59.3 (coupon collector)

Expected number of rolls to see all six faces?
Solution. After seeing k faces, each roll shows a new face with probability $\frac{6-k}{6}$, so the wait is geometric with mean $\frac{6}{6-k}$. Add the stages:

$$6\left(\frac{1}{6} + \frac{1}{5} + \frac{1}{4} + \frac{1}{3} + \frac{1}{2} + 1\right) = \mathbf{14.7}.$$

In general $nH_n \approx n \ln n$. (Real life: how many packs of cricket cards until you complete the set.)

Example 59.4 (maximum of two dice)

Expected value of the larger of two dice?
Solution. $P(\max \le k) = \left(\frac{k}{6}\right)^2$, so $P(\max = k) = \frac{k^2-(k-1)^2}{36} = \frac{2k-1}{36}$. $\mathbb{E} = \frac{1\cdot1+2\cdot3+3\cdot5+4\cdot7+5\cdot9+6\cdot11}{36} = \frac{161}{36} \approx \mathbf{4.47}$.

Example 59.5 (couples at a round table)

Four couples sit at random around a round table. Expected number of couples sitting together?
Solution. A given person's partner is one of 7 others, 2 of whom are neighbours: probability $\frac{2}{7}$. Four couples: $\frac{8}{7} \approx \mathbf{1.14}$.

Speed shortcut: the tail-sum formula

For a non-negative integer variable, $\mathbb{E}[X] = \sum_{k\geq 1} P(X \geq k)$. For a non-negative continuous variable, $\mathbb{E}[X] = \int_0^\infty P(X > x)\,dx$.

Exercises 59

59.1 Expected value of the smaller of two dice?

59.2 10 balls are thrown at random into 5 boxes. Expected number of empty boxes?

59.3 Cards are drawn one by one without replacement from a shuffled deck. Expected number of cards drawn up to and including the first ace? (Hint: the 4 aces split the 48 other cards into 5 gaps, each of expected size $\frac{48}{5}$.)

59.4 Expected number of runs (maximal blocks of identical outcomes) in 10 fair coin tosses?

59.5 A die is rolled until a six appears. Expected sum of all the rolls, including the six?

59.6 Among 30 people, expected number of pairs who share a birthday?

60. First-Step Analysis and Markov Chains

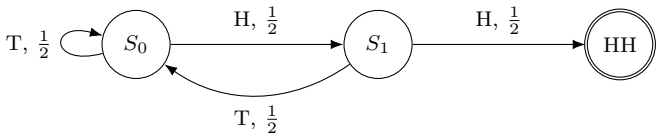
60.1 The method

First-step analysis

Define a **state** that captures everything relevant about the past. Let E_s be the expected time (or p_s the probability of winning) from state s. Condition on the *first step*:

$$E_s = 1 + \sum_t P(s \to t)\, E_t, \qquad p_s = \sum_t P(s \to t)\, p_t.$$

Solve the linear equations.



A **state diagram** makes the equations automatic: there is one equation per state and one term per outgoing arrow.

Example 60.1 (waiting for HH)

Expected number of fair-coin tosses until two heads in a row?
Solution. States: S_0 (no progress), S_1 (last toss H).

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}E_0.$$

From the first, $E_0 = 2 + E_1$; substituting, $E_1 = 1 + \frac{1}{2}(2 + E_1) \Rightarrow E_1 = 4$, $E_0 = \mathbf{6}$.

Example 60.2 (HT is faster than HH)

Expected tosses until HT?
Solution. Wait for the first H (expected 2), then wait for a T (expected 2): **4**. Unlike HH, a failure (another H) doesn't destroy progress – you're still one step away.

Speed shortcut: Conway's formula for pattern waiting times

For a fair coin, the expected wait for a pattern is $\sum 2^k$ over all k such that the first k symbols of the pattern equal its last k symbols (always including $k =$ full length). HH: $2 + 4 = 6$. HT: 4. HHH: $2 + 4 + 8 = 14$. HTH: $2 + 8 = 10$. With a die, replace 2 by 6: two consecutive sixes take $6 + 36 = 42$ rolls.

60.2 Races between patterns (Penney's game)

Example 60.3

A fair coin is tossed until either HH or TH appears. Probability HH comes first?
Solution. If the first two tosses are HH (probability $\frac{1}{4}$), HH wins. Otherwise a T appears before any HH, and from then on the first HH must be preceded by a T, so TH appears first. $P = \mathbf{\frac{1}{4}}$.

Penney's game: for any pattern of length 3 your opponent picks, there is a pattern that beats it. If they pick HHT, you pick THH and win with probability $\frac{3}{4}$. Patterns are *non-transitive*, like rock–paper–scissors.

60.3 Gambler's ruin

You have Rs. i; each round you win Rs. 1 with probability p or lose Rs. 1 with probability $q = 1 - p$. You stop at Rs. 0 or Rs. N.

$$P(\text{reach } N) = \begin{cases} \dfrac{i}{N} & p = \frac{1}{2} \\ \dfrac{1-(q/p)^i}{1-(q/p)^N} & p \neq \frac{1}{2} \end{cases} \qquad \mathbb{E}[\text{duration}] = i(N-i) \ \text{ when } p = \tfrac{1}{2}.$$

Why it works

For $p = \frac{1}{2}$, first-step analysis gives $p_i = \frac{1}{2}p_{i+1} + \frac{1}{2}p_{i-1}$, so the p_i lie on a straight line from $p_0 = 0$ to $p_N = 1$. A fair game keeps your expected wealth constant: $i = N \cdot P(\text{reach } N)$.

Real life: This is why a small edge for the house is fatal over many rounds: at $p = 0.4$, starting with Rs. 5 and aiming for Rs. 10, you reach your target only about 12% of the time.

Example 60.4 (ant on a cube)

An ant walks along the edges of a cube, choosing one of the 3 edges at random at each vertex. Expected number of moves from one vertex to the opposite vertex?
Solution. Use distance from the target as the state: $E_3 = 1 + E_2$; $E_2 = 1 + \frac{2}{3}E_1 + \frac{1}{3}E_3$; $E_1 = 1 + \frac{1}{3}\cdot 0 + \frac{2}{3}E_2$. Solving: $E_1 = 7$, $E_2 = 9$, $E_3 = \mathbf{10}$.

Exercises 60

60.1 Expected number of fair-coin tosses until HHH?

60.2 Expected number of die rolls until two consecutive sixes?

60.3 Expected tosses until HTH? Until HHT?

60.4 You start with Rs. 5 and bet Rs. 1 per round on a fair coin, stopping at Rs. 0 or Rs. 10. Probability of reaching Rs. 10? Expected number of rounds?

60.5 Same as above, but you win each round with probability 0.4. Probability of reaching Rs. 10?

60.6 A random walk starts at 0 and moves ± 1 with equal probability. Probability it hits $+3$ before -2?

60.7 Your opponent chooses HHT in Penney's game. Which pattern should you choose, and what is your probability of winning?

61. Continuous and Geometric Probability

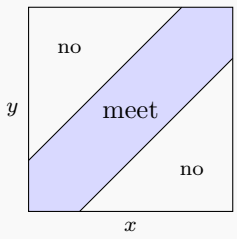
When outcomes are points chosen uniformly in a region, **probability = favourable area (or length) ÷ total area**.

Example 61.1 (the meeting problem)

Two friends agree to meet between 5 and 6 pm. Each arrives at a uniformly random time and waits 15 minutes, then leaves. Probability they meet?

Solution. Plot arrival times (x, y) in a 60×60 square. They meet if $|x - y| \le 15$. The "no meeting" region is two corner triangles, each with legs 45: total area 45^2.

$$P = 1 - \frac{45^2}{60^2} = 1 - \frac{9}{16} = \boxed{\frac{7}{16}}.$$



Example 61.2 (broken stick)

A stick is broken at two uniformly random points. Probability the three pieces form a triangle?

Solution. A triangle needs every piece shorter than $\frac{1}{2}$. With break points (x, y) in the unit square, the valid region consists of two triangles of total area $\frac{1}{4}$. $P = \boxed{\frac{1}{4}}$.

Example 61.3 (points on a circle)

Three points are chosen uniformly on a circle. Probability they all lie in some semicircle?

Solution. For each point, the event "the other two lie within the semicircle starting at this point and going clockwise" has probability $\left(\frac{1}{2}\right)^2$. These three events are disjoint (at most one point can be the "first" of a semicircle containing all). Total $3 \times \frac{1}{4} = \boxed{\frac{3}{4}}$. In general, n points: $\frac{n}{2^{n-1}}$. (The triangle they form is acute exactly when they are *not* in a semicircle: probability $1/4$.)

Order statistics of uniforms: for n independent Uniform$(0, 1)$ variables, $\mathbb{E}[\max] = \frac{n}{n+1}$, $\mathbb{E}[\min] = \frac{1}{n+1}$. The n points split $[0, 1]$ into $n + 1$ pieces of equal expected length.

Bertrand's paradox

"A random chord of a circle" has no single meaning: choosing random endpoints, a random midpoint, or a random distance from the centre give different answers ($\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{2}$ for "longer than the side of the inscribed triangle"). In an interview, define your randomness before computing.

Exercises 61

61.1 Two friends arrive uniformly at random between 5 and 6 pm and each waits 20 minutes. Probability they meet?

61.2 A stick of length 1 is broken at a uniformly random point. Expected length of the longer piece?

61.3 Expected value of the maximum of three independent Uniform$(0, 1)$ variables?

61.4 X, Y are independent Uniform$(0, 1)$. Find $P(X + Y < 1)$ and $P(X^2 + Y^2 < 1)$.

61.5 Four points are chosen uniformly on a circle. Probability they all lie in some semicircle?

61.6 A point is chosen uniformly in a unit square. Probability that it lies within 0.1 of the nearest side?

61.7 (Inspection paradox) Buses come exactly every 10 minutes and you arrive at a random time: expected wait? If instead buses arrive as a Poisson process averaging one every 10 minutes: expected wait?

62. Optimal Stopping, Games and Betting

62.1 Backward induction

Example 62.1 (the reroll game)

Roll a die and receive its face value in rupees – but you may reroll once (and must accept the second roll). What's the game worth, and when should you reroll?

Solution. Work backwards. The final roll is worth 3.5. On the first roll, keep it if it beats 3.5, i.e. keep 4, 5, 6; reroll 1, 2, 3.

$$V_2 = \frac{4+5+6}{6} + \frac{3}{6} \times 3.5 = 2.5 + 1.75 = \mathbf{4.25}.$$

With up to three rolls, keep the first roll only if it beats 4.25 (5 or 6): $V_3 = \frac{11}{6} + \frac{4}{6} \times 4.25 = \frac{14}{3} \approx 4.67$.

Why it works

At each decision, compare what you have with the *expected value of continuing optimally*. That continuation value is computed from the end first, hence "backward" induction. It's the same logic as valuing an American option on a binomial tree: at each node, take the maximum of exercising now and the discounted value of continuing.

62.2 The secretary problem

n candidates arrive in random order; you see each one's relative rank and must accept or reject on the spot. To maximise the chance of picking the very best: **reject the first $n/e \approx 37\%$, then take the first candidate better than all of them**. Success probability $\to \frac{1}{e} \approx \mathbf{37}\%$.

62.3 Betting: expectation is not everything

- **St Petersburg paradox:** a coin is tossed until the first head; if that takes k tosses you win Rs. 2^k. The expected payout is $\sum_k \frac{1}{2^k} \cdot 2^k = \infty$, yet nobody would pay much to play. Resolution: utility (diminishing value of money) and finite bankrolls.
- **Martingale (doubling) strategy:** doubling after every loss "guarantees" a Rs. 1 profit, but a finite bankroll means a rare, huge loss exactly cancels the frequent small wins. Expected profit stays 0 in a fair game and is negative in an unfair one.
- **Kelly criterion:** for a bet paying b-to-1 that you win with probability p, the fraction of wealth that maximises long-run growth is

  $$f^* = \frac{bp - q}{b} = p - \frac{q}{b}.$$

  Bet nothing if $f^* \le 0$. Many traders use a fraction (like half-Kelly) because the true p is uncertain.

Example 62.2 (Russian roulette decision)

A 6-chamber revolver has 2 bullets in *adjacent* chambers. The cylinder is spun, the trigger pulled on you, and it clicks (empty). It's your turn again: is it better to pull again immediately or to spin first?

Solution. Spinning: survival $\frac{4}{6} = \frac{2}{3}$. Not spinning: the 4 empty chambers form a block; the one just fired is one of these 4, and only the last of the block is followed by a bullet. Survival $\frac{3}{4}$. **Don't spin.**

The two-envelope paradox

One envelope has twice the money of the other. You hold one with amount x. "The other has $2x$ or $x/2$ with equal probability, so its expected value is $1.25x$; always switch" – and then switch back, forever. The flaw: x is not fixed while the "other" varies; the equal-probability assumption can't hold for every x under any proper prior. Switching gains nothing on average.

Exercises 62

62.1 In the reroll game with up to 3 rolls, state the full strategy and check the value $\frac{14}{3}$.

62.2 You may roll a die repeatedly, paying Rs. 1 per roll (including the first), and you may stop at any time and receive the face value of your last roll. What is the game worth, and when should you stop?

62.3 Kelly fractions: (a) an even-money bet you win with probability 0.55; (b) a bet paying 2-to-1 that you win with probability 0.4.

62.4 With 100 candidates in the secretary problem, roughly how many should you reject outright, and what is your chance of picking the best?

62.5 A martingale gambler has Rs. 1023, starts with a Rs. 1 bet on a fair coin, and doubles after each loss. How many consecutive losses can he cover, and what is the probability of going broke in one "cycle"? What is his expected profit per cycle?

63. Variance, Covariance and Correlation

Expectation tells you the centre, and variance tells you how far you stray from it. Quant and risk interviews care at least as much about the second.

63.1 Definitions

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$
$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \qquad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

The rules

- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$. Constants do not add spread, and scaling squares.
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$.
- Independent $\Rightarrow$ Cov $= 0$, so variances add. **Standard deviations never simply add**, except when $\rho = 1$.
- For the mean of n independent copies: $\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$, so $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- $\mathrm{Cov}(aX+bY,\ cU+dV)$ expands like a product of brackets.
- $-1 \le \rho \le 1$, with ± 1 exactly when Y is a linear function of X.

Why the $\sqrt{n}$ matters

Add n independent bets. Their total's mean grows like n, but its standard deviation only grows like $\sqrt{n}$, since variances add and not standard deviations. Relative risk shrinks like $\frac{1}{\sqrt{n}}$. This single fact underlies casinos, insurance and diversification. It is also why reducing a poll's error by half takes 4 times the sample.

Common trap

Zero covariance does not mean independent. Take X uniform on $\{-1,0,1\}$ and $Y = X^2$. $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$ and $\mathbb{E}[X] = 0$, so Cov $= 0$. But Y is completely determined by X.

63.2 Variance by indicators

For $X = \sum I_j$:

$$\mathrm{Var}(X) = \sum_j \mathrm{Var}(I_j) + \sum_{i \ne j} \mathrm{Cov}(I_i, I_j), \quad \mathrm{Var}(I_j) = p_j(1-p_j).$$

Example 63.1 (hat-check variance)

n hats returned at random: find the variance of the number of matches.
Solution. $\mathrm{Var}(I_j) = \frac{1}{n}\left(1 - \frac{1}{n}\right)$. For $i \ne j$: $\mathbb{E}[I_i I_j] = \frac{1}{n(n-1)}$, so Cov $= \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)}$. There are $n(n-1)$ ordered pairs. Total $= \left(1 - \frac{1}{n}\right) + \frac{1}{n} = \mathbf{1}$, for every $n \ge 2$. Both mean and variance are 1, just like a Poisson(1), which is indeed the limit.

63.3 Two-asset portfolios

Example 63.2 (real life: diversification)

Assets A and B have volatilities 20% and 30%, with correlation 0.5. Find the volatility of a 50/50 portfolio.
Solution. $\sigma_p^2 = 0.25(0.04) + 0.25(0.09) + 2(0.25)(0.5)(0.2)(0.3) = 0.01 + 0.0225 + 0.015 = 0.0475$. So $\sigma_p \approx \mathbf{21.8}\%$. That beats the 25% that plain averaging would suggest, and the gap is the benefit of diversification. It disappears only when $\rho = 1$.

Minimum-variance weight in A: $w^* = \dfrac{\sigma_B^2 - \rho\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}$. (Differentiate the variance in w, and set it to 0.)

Exercises 63

63.1 Find the variance of one die roll, and of the sum of two independent dice.

63.2 $X \sim \text{Binomial}(100, 0.3)$. Find its mean, variance and standard deviation.

63.3 $\text{Var}(X) = 4$, $\text{Var}(Y) = 1$ and $\text{Cov}(X, Y) = 1$. Find $\text{Var}(2X - 3Y + 5)$.

63.4 X and Y are independent with the same variance. Find the correlation of X and $X + Y$.

63.5 Using the data of the portfolio example, find the minimum-variance weight in A.

63.6 100 independent measurements each have SD 15. Find the SD of their mean.

63.7 Two dice are rolled. Find Cov(max, min). (Hint: max $\times$ min = the product of the dice.)

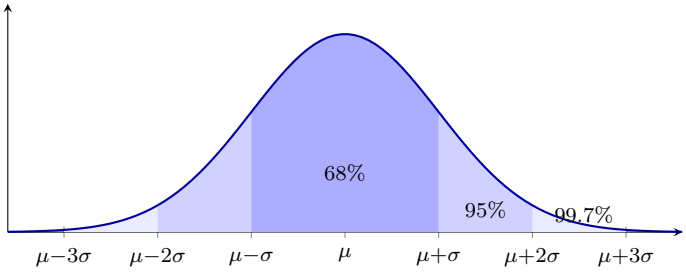
63.8 X has mean 2 and variance 5. Find $\mathbb{E}[X^2]$.

63.9 A portfolio holds n independent assets, each with volatility 30%, in equal weights. How large must n be for the portfolio volatility to fall to 10%?

63.10 Toss a fair coin 10 times; let X be heads and Y tails. Find the correlation of X and Y.

64. The Normal Distribution, LLN and CLT

64.1 The bell curve



Standardise: if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$.

z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.33	3.0
$P(Z > z)$	30.9%	15.9%	10%	6.7%	5%	2.5%	2.3%	1%	0.13%

Sums: independent normals add to a normal, with means and variances added.

64.2 Law of large numbers

The average of n independent copies of X converges to $\mathbb{E}[X]$. That is what makes expected value practical: in the long run, your average per-play result equals the expectation. It does *not* say that a losing streak is "due" to reverse. Later plays just swamp the early ones.

Common trap: gambler's fallacy

After 10 reds, the next spin of a fair wheel is still 50–50. The law of large numbers works by dilution, not compensation. The excess of reds stays about the same size while the total count grows.

64.3 Central limit theorem

CLT

Add up many independent, identically distributed variables with mean μ and variance σ^2. Then, whatever their original shape,

$$S_n \approx N(n\mu,\ n\sigma^2), \qquad \bar{X} \approx N\left(\mu,\ \frac{\sigma^2}{n}\right).$$

Normal approximation to binomial: $\mathrm{Bin}(n,p) \approx N(np,\ np(1-p))$, good when np and $n(1-p)$ are both at least about 10. For discrete counts, use the **continuity correction**: $P(X \geq 60) \approx P(Y \geq 59.5)$.

Example 64.1 (coin)

In 100 fair tosses, estimate the chance of at least 60 heads.
Solution. Mean 50, SD 5. $z = \frac{59.5-50}{5} = 1.9$. $P \approx \mathbf{2.9}\%$. (The exact value is 2.84%.)

Example 64.2 (real life: scaling risk)

A stock has a daily volatility of 1%. Estimate its annual volatility, with 252 trading days and independent days.
Solution. $1\% \times \sqrt{252} \approx \mathbf{15.9}\%$. Variances add over time, so volatility scales with $\sqrt{\text{time}}$.

64.4 Confidence intervals and testing

95% confidence interval for a mean: $\bar{x} \pm 1.96\dfrac{s}{\sqrt{n}}$; for a proportion: $\hat{p} \pm 1.96\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$.

Sample size for margin m on a proportion (worst case $p = \frac{1}{2}$): $n \approx \frac{1.96^2 \cdot 0.25}{m^2} \approx \frac{1}{m^2}$.

Example 64.3 (poll)

How many people must be polled for a $\pm 3\%$ margin at 95% confidence?
Solution. $\frac{0.9604}{0.0009} \approx \mathbf{1067}$. This is why national polls so often use about 1,000 people, however large the country.

Hypothesis test: assume the boring explanation (the null), and calculate how surprising the data would be. A **p-value** is the probability of data at least this extreme *if the null is true*. It is *not* the probability that the null is true. Getting those two confused is a Bayes error (Chapter 56).

Exercises 64

64.1 IQ scores are $N(100, 15^2)$. What fraction exceed 130? What fraction lie between 85 and 115?

64.2 Marks have mean 60 and SD 8. Roughly what score marks the top 16%?

64.3 In 400 fair tosses, estimate the chance of between 190 and 210 heads inclusive, with the continuity correction.

64.4 100 dice are rolled. Estimate the chance that the total exceeds 380.

64.5 A survey wants its error cut in half. By what factor must the sample grow?

64.6 In a poll of 400, 55% favour a proposal. Give the 95% confidence interval.

64.7 $X \sim N(10, 9)$ and $Y \sim N(20, 16)$ are independent. Find $P(X + Y > 35)$ and $P(X > Y)$.

64.8 Daily volatility is 1.5%. Estimate annual volatility (252 days).

64.9 A coin gives 60 heads in 100 tosses. Is this significant at the 5% level (two-sided, no continuity correction)?

64.10 Height is $N(170, 10^2)$. Find the probability that a person exceeds 185. Among 100 people, how many would you expect?

65. Conditional Expectation and Recursion

65.1 The tower law

Conditional expectation

Break a hard average into easier conditional averages:

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \sum_y \mathbb{E}[X \mid Y = y]\, P(Y = y).$$

Total variance: $\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y])$, or within-group plus between-group.

Example 65.1 (a random number of coin flips)

Roll a die, then flip that many fair coins. Find the mean and variance of the number of heads.

Solution. Given N, heads is $\operatorname{Bin}(N, \frac{1}{2})$: conditional mean $\frac{N}{2}$, conditional variance $\frac{N}{4}$. $\mathbb{E} = \frac{3.5}{2} = \mathbf{1.75}$. $\operatorname{Var} = \mathbb{E}\left[\frac{N}{4}\right] + \operatorname{Var}\left(\frac{N}{2}\right) = \frac{3.5}{4} + \frac{1}{4} \cdot \frac{35}{12} \approx 0.875 + 0.729 = \mathbf{1.604}$.

65.2 Random sums (Wald)

If N is independent of the i.i.d. X_i, then for $S = X_1 + \cdots + X_N$:

$$\mathbb{E}[S] = \mathbb{E}[N]\,\mathbb{E}[X], \qquad \operatorname{Var}(S) = \mathbb{E}[N]\operatorname{Var}(X) + \operatorname{Var}(N)\,(\mathbb{E}X)^2.$$

Real life: insurance claims. The number of claims and the size of each claim both contribute to the risk.

65.3 Who shoots first

Example 65.2 (alternating turns)

A and B alternate tossing a coin (A first). The first to get heads wins. Find A's chance.

Solution. Let w be the probability that whoever goes first wins. $w = \frac{1}{2} + \frac{1}{2}(1 - w)$, since if the first player misses, they are now effectively second. So $w = \mathbf{\frac{2}{3}}$.

With general hit rates a and b: $P(\text{A wins}) = \dfrac{a}{1 - (1-a)(1-b)}$. The denominator is the chance that the round ends.

65.4 Famous recursive results

Example 65.3 (uniform sums reach 1)

Add Uniform$(0,1)$ numbers until the sum exceeds 1. Find the expected count.
Solution. $P(N > n) = P(U_1 + \cdots + U_n \le 1) = \frac{1}{n!}$, the volume of a simplex. By the tail-sum formula, $\mathbb{E}[N] = \sum_{n\ge 0} \frac{1}{n!} = \mathbf{e}$.

Example 65.4 (Pólya's urn)

An urn holds 1 red and 1 blue ball. Draw one, and return it with another of the same colour. Repeat. Find the chance that the second draw is red, and describe the long-run share of red.
Solution. $\frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{1}{3} = \mathbf{\frac{1}{2}}$. By symmetry, every draw has the same chance of red, though draws are far from independent. Remarkably, the long-run proportion of red is itself *uniform* on $(0,1)$. Early luck gets locked in, which is a toy model of "rich get richer".

Example 65.5 (first success without replacement)

An urn has r red and b blue balls. Find the expected draws to the first red.
Solution. Each blue ball is drawn before all the reds with probability $\frac{1}{r+1}$, since among it and the r reds, each is equally likely to be first. So expected blues before the first red $= \frac{b}{r+1}$. Answer: $1 + \frac{b}{r+1} = \mathbf{\frac{r+b+1}{r+1}}$.

Common trap: conditioning changes the distribution

"Roll a die until a 6 appears. Given that all rolls were even, find the expected number of rolls." There are two tempting answers: 6 (the usual wait for a six), or 3 (one six among the three even faces). The truth is $\frac{3}{2}$. Conditioning on "no odd rolls" discards all the long sequences, as those are the most likely to include an odd roll. Then $P(N = n \mid \text{all even}) \propto \left(\frac{1}{3}\right)^{n-1}$, which is geometric with success rate $\frac{2}{3}$.

Exercises 65

65.1 Roll a die, then roll that many dice. Find the expected total on those dice.

65.2 Toss a coin until the first head. Then roll a die that many times. Find the expected sum of the rolls.

65.3 A and B take alternate shots, A first, with hit rates $\frac{1}{3}$ and $\frac{1}{2}$. Find A's chance of hitting first.

65.4 An urn has 3 red and 7 blue balls. Find the expected draws without replacement to the first red.

65.5 A Pólya urn starts with 2 red and 1 blue. Find the chance that the second draw is red.

65.6 A stick of length 1 is broken at a random point and the left part is kept. This is done twice. Find the expected final length.

65.7 Customers per day are Poisson(10), spending a mean of Rs. 200 with SD Rs. 50, independently. Find the mean and SD of daily revenue.

65.8 Pick U_1 uniform on $(0,1)$, then U_2 uniform on $(0, U_1)$. Find $\mathbb{E}[U_2]$.

65.9 Roll a die until a 6 appears. How many 2s do you expect before that 6? How many non-six rolls?

65.10 A coin is chosen at random from 2 fair coins and 1 double-headed coin, and tossed twice. Find the expected number of heads.

66. Random Walks and Markov Chains in the Long Run

66.1 Symmetric random walk

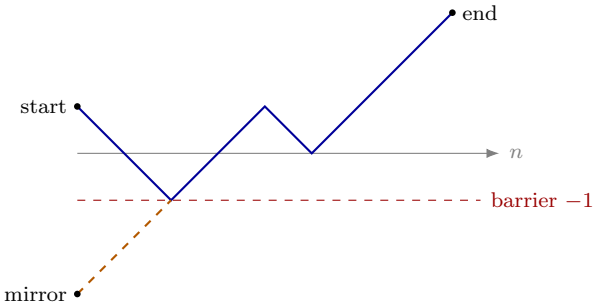
S_n moves ± 1 with equal probability.

- $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$. So the typical distance from the start grows like $\sqrt{n}$.
- $P(S_{2n} = 0) = \frac{\binom{2n}{n}}{4^n} \approx \frac{1}{\sqrt{\pi n}}$.
- Hits $+a$ before $-b$ with probability $\frac{b}{a+b}$, after ab expected steps.
- In one or two dimensions the walk returns to the origin with certainty; in three dimensions it may never come back.

66.2 The reflection principle

Reflection

The number of paths from x to y that touch a barrier equals the number of *all* paths from x's mirror image to y. Reflect the part of the path before the first touch.



This is the same tool that gave us Catalan numbers (Chapter 10), and it gives the **ballot theorem**: if A gets a votes and B gets $b < a$, A leads throughout with probability $\frac{a-b}{a+b}$.

66.3 Martingales and optional stopping

A **martingale** is a fair-game process: the expected next value equals the current value. **Optional stopping:** under reasonable conditions, $\mathbb{E}$[value at a stopping time] = starting value. No stopping rule can turn a fair game into a winning one.

Example 66.1 (two lines of proof)

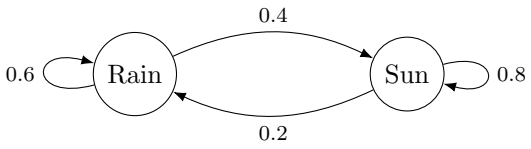
A walk starts at 0 and stops at $+a$ or $-b$. Find the probability of $+a$ and the expected duration.

Solution. S_n is a martingale: $0 = a\,p - b(1-p)$, so $p = \frac{b}{a+b}$. And $S_n^2 - n$ is a martingale: $\mathbb{E}[S_T^2] = \mathbb{E}[T]$, so $\mathbb{E}[T] = a^2 p + b^2(1-p) = ab$.

Biased walk: with $p > \frac{1}{2}$, the chance of ever dropping to $-k$ is $\left(\frac{q}{p}\right)^k$. That is the risk of ruin for someone with a positive edge.

66.4 Stationary distributions

A Markov chain's next state depends only on the current state. Under mild conditions, it settles into a **stationary distribution** π with $\pi = \pi P$, whatever the start.



Example 66.2 (weather)

Using the chain above, find the long-run fraction of rainy days.
Solution. Flows balance: $\pi_R \times 0.4 = \pi_S \times 0.2$. So $\pi_R : \pi_S = 1 : 2$ and $\pi_R = \mathbf{\frac{1}{3}}$. A rainy day recurs on average every $\frac{1}{\pi_R} = 3$ days: the expected return time to a state is $\frac{1}{\pi}$.

Random walk on a graph: π at each vertex is proportional to its degree. On a regular graph, such as a cycle or a cube, the walk is uniform, so the expected return time equals the number of vertices. PageRank is a stationary distribution on the graph of web links.

Hitting times on a cycle of n nodes: from distance k, the expected time to reach the target is $k(n-k)$. This is gambler's ruin in disguise.

Exercises 66

66.1 Find the stationary distribution of the chain with $P(A \to A) = 0.9$, $P(A \to B) = 0.1$, $P(B \to A) = 0.5$, $P(B \to B) = 0.5$.

66.2 A symmetric walk takes 6 steps. Find the probability that it ends where it began.

66.3 Among 6-step walks from 0 ending at $+2$, how many touch -1 along the way?

66.4 A symmetric walk starts at 0 and stops at $+4$ or -6. Find the probability of $+4$, and the expected duration.

66.5 A walk has $p = 0.6$ up and $q = 0.4$ down. Find the probability of ever reaching -1, and of ever reaching -3.

66.6 You start with Rs. 10 in a fair game, stopping at Rs. 0 or Rs. 30. Find your chance of reaching Rs. 30.

66.7 A fly walks randomly on the 8 vertices of a cube. Find the expected time to return to its starting vertex.

66.8 A walker on a cycle of 8 nodes starts opposite the target. Find the expected hitting time.

66.9 In an election A gets 5 votes and B gets 3. Find the chance that A leads throughout the count.

66.10 A symmetric walk stops on first reaching $+5$ or -5. Find the expected number of steps.

67. Trading, Options and Pricing Questions

Trading firms ask quick probability questions in a market setting. The underlying idea is usually a single one: **price equals expected value under the right probabilities**.

67.1 Expected value games and market making

How to quote a market

Asked to "make a market" on an unknown quantity (the sum of 3 dice, the number of cars in a city):

1. Estimate the fair value (the mean).
2. Quote a bid below it and an ask above it. The width reflects your uncertainty.
3. If someone trades with you, **update**: they may know something, so move your quote their way. This is adverse selection.

Example 67.1 (sum of three dice)

Make a market on the sum of 3 dice.

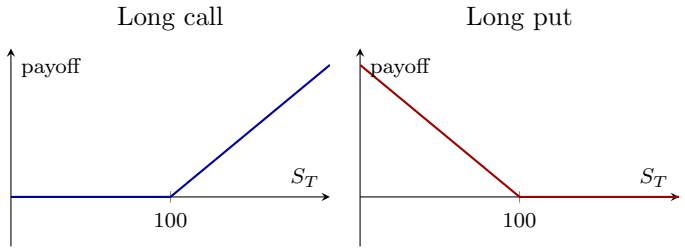
Solution. Mean = 10.5, and SD = $\sqrt{3 \cdot \frac{35}{12}} \approx 2.96$. A sensible tight quote is 10–11. If you sell at 11 and the counterparty has seen one die, you can guess it was high. If they buy again at 12, assume it was a 5 or 6.

Example 67.2 (convex payoffs)

A die pays you the square of its face. What is it worth? Is that more or less than the square of the average face?

Solution. $\frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx \mathbf{15.17}$, which is more than $3.5^2 = 12.25$. For a convex payoff f, Jensen's inequality gives $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. The gap is why options have value.

67.2 Options



Call: right to buy at strike K, with payoff $\max(S_T - K, 0)$. **Put:** right to sell at K, with payoff $\max(K - S_T, 0)$.

Put–call parity (European, no dividends): holding a call and shorting a put has payoff $S_T - K$ in every case, the same as a forward. So:

$$C - P = S_0 - Ke^{-rT}.$$

Bounds and intuition: options gain value with higher volatility, since the payoff is one-sided. A call's price lies between $\max(S_0 - Ke^{-rT}, 0)$ and S_0.

67.3 One-step binomial pricing

Replication, not prediction

The stock is at S_0, and next period is uS_0 or dS_0. Gross riskless return R. Then the option's price is its discounted expectation under the **risk-neutral probability**:

$$q = \frac{R-d}{u-d}, \qquad \text{Price} = \frac{q\,V_{\text{up}} + (1-q)\,V_{\text{down}}}{R}.$$

The real-world chance of going up does not appear, because you can build the option from stock and cash.

Example 67.3 (pricing a call)

$S_0 = 100$, moving to 120 or 90, with zero interest. Price a call with $K = 100$.
Solution. Payoffs 20 and 0. $q = \frac{100-90}{120-90} = \frac{1}{3}$. Price $= \frac{20}{3} \approx \mathbf{6.67}$.
Replication check: hold $\Delta = \frac{20-0}{120-90} = \frac{2}{3}$ shares, and borrow 60. The value next period is $\frac{2}{3}\cdot 120 - 60 = 20$ or $\frac{2}{3}\cdot 90 - 60 = 0$. Cost today $= \frac{2}{3}\cdot 100 - 60 = 6.67$. ✓

Why the real probability doesn't matter

If anyone could quote a different price, you could buy the cheaper of the option and its replicating portfolio and sell the other, for a riskless profit in both scenarios. The real probabilities are already built into today's stock price.

67.4 Arbitrage and odds

Decimal odds o imply a probability $\frac{1}{o}$. If the implied probabilities over all outcomes sum to less than 1, you can bet on all of them for a guaranteed profit, staking in proportion to $\frac{1}{o}$. Normally they sum to more than 1, and the excess is the bookmaker's margin.

67.5 Risk-adjusted performance

Sharpe ratio	$\frac{\text{return}-\text{risk-free}}{\text{volatility}}$
Beta	$\beta = \frac{\text{Cov}(r, r_m)}{\text{Var}(r_m)} = \rho\frac{\sigma}{\sigma_m}$
CAPM	$\mathbb{E}[r] = r_f + \beta(\mathbb{E}[r_m] - r_f)$
Kelly fraction	$f^* = p - \frac{q}{b}$, for a bet paying b-to-1
Kelly growth rate	$g = p\ln(1+bf) + q\ln(1-f)$

Exercises 67

67.1 Two dice are rolled and you receive $\max(S-7, 0)$, where S is their sum. What is it worth?

67.2 $S_0 = 50$, moving to 60 or 40, with zero interest. Price the $K = 50$ call and the $K = 50$ put.

67.3 $S_0 = 100$, moving to 110 or 90, with $R = 1.02$. Price a $K = 100$ call.

67.4 A stock is at 100, a call with strike 105 costs 10, and the discount factor is 0.95. What should the same-strike put cost?

67.5 A bookmaker offers 1.9 on team A and 2.2 on team B (only two outcomes). Is there an arbitrage? If so, what is the guaranteed return?

67.6 A fund returns 15% with 20% volatility. The risk-free rate is 5%. Find its Sharpe ratio.

67.7 A stock has $\mathrm{Cov}(r, r_m) = 0.02$ and $\mathrm{Var}(r_m) = 0.04$. With $r_f = 4\%$ and $\mathbb{E}[r_m] = 10\%$, find its beta and CAPM return.

67.8 For an even-money bet with $p = 0.6$, find the Kelly fraction and the growth rate per bet.

67.9 What position has the payoff $S_T - K$? What does a long call plus a long put at the same strike (a straddle) pay? If the two cost 10 in total at $K = 100$, where does it break even?

67.10 You may roll a die and either take the face value in rupees or pay Rs. 1 to reroll, as often as you like. Find the value of the game.

67.11 A game: toss a coin until it lands tails. You receive Rs. n, where n is the number of tosses. What is its fair price? What would you pay if the payoff were 2^n?

67.12 A market is quoted at 49–51 on the number of heads in 100 tosses. Someone offers to buy from you at 51 after peeking at the first 10 tosses. What should you worry about?

68. Maths Refresher for Quant Interviews

Interviewers expect quick, reliable fluency with basic calculus and linear algebra.

68.1 Approximations

Rule	Example	True value
$(1+x)^n \approx 1+nx$ for small nx	$1.01^5 \approx 1.05$	1.0510
$\sqrt{a^2+b} \approx a + \frac{b}{2a}$	$\sqrt{17} \approx 4 + \frac{1}{8} = 4.125$	4.1231
$e^x \approx 1 + x + \frac{x^2}{2}$	$e^{0.1} \approx 1.105$	1.1052
$\ln(1+x) \approx x - \frac{x^2}{2}$	$\ln 1.1 \approx 0.095$	0.0953
$\left(1-\frac{1}{n}\right)^n \approx \frac{1}{e}$	$0.99^{100} \approx 0.368$	0.366
Stirling: $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$	$10! \approx 3.60 \times 10^6$	3.63×10^6
$\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$	$\frac{\binom{100}{50}}{2^{100}} \approx 0.0798$	0.0796

68.2 Calculus essentials

$f(x)$	$f'(x)$	$\int f\,dx$
x^n	nx^{n-1}	$\frac{x^{n+1}}{n+1}$ $(n \neq -1)$
e^{ax}	ae^{ax}	$\frac{e^{ax}}{a}$
$\ln x$	$\frac{1}{x}$	$x\ln x - x$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\ln\lvert x\rvert$

- **Chain rule:** $(f(g(x)))' = f'(g(x))g'(x)$. **Product rule:** $(fg)' = f'g + fg'$. **Parts:** $\int u\,dv = uv - \int v\,du$.
- **Log-differentiation** for x^x: $y = e^{x\ln x}$, so $y' = x^x(\ln x + 1)$.
- **Optimise:** set $f' = 0$; a negative second derivative means a maximum.
- **Gamma:** $\int_0^\infty x^n e^{-x}dx = n!$; $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$.

68.3 Series

$$\sum_{k\geq 0} x^k = \frac{1}{1-x}, \qquad \sum_{k\geq 1} kx^k = \frac{x}{(1-x)^2}, \qquad e^x = \sum_{k\geq 0}\frac{x^k}{k!},$$

$$\ln(1+x) = \sum_{k\geq 1}\frac{(-1)^{k+1}x^k}{k}, \qquad \sum_{k\geq 1}\frac{1}{k^2} = \frac{\pi^2}{6}.$$

Harmonic sum: $H_n \approx \ln n + 0.577$. So the sum diverges, but extremely slowly.

68.4 Linear algebra essentials

- For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: determinant $ad - bc$; inverse $\frac{1}{\det}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- Eigenvalues solve $\det(A - \lambda I) = 0$. Their **sum is the trace** and their **product is the determinant**.

- A covariance matrix is symmetric and positive semi-definite, so all its eigenvalues are real and at least 0. This gives $|\rho| \le 1$.
- $A^n = PD^nP^{-1}$ for diagonalisable A. So recurrences grow like the largest eigenvalue. Fibonacci comes from $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, whose largest eigenvalue is the golden ratio.

Example 68.1 (which is bigger)

Which is larger, e^π or π^e?
Solution. Compare $\frac{\ln x}{x}$ at e and π: taking logs, $e^\pi > \pi^e \iff \frac{\ln e}{e} > \frac{\ln \pi}{\pi}$. The function $\frac{\ln x}{x}$ has derivative $\frac{1-\ln x}{x^2}$, so its maximum is at $x = e$. So $\mathbf{e^\pi}$ is larger. (They are 23.14 and 22.46.)

Example 68.2 (integration by parts)

Evaluate $\int_0^1 x \ln x \, dx$.
Solution. $\left[\frac{x^2}{2} \ln x\right]_0^1 - \int_0^1 \frac{x}{2} dx = 0 - \frac{1}{4} = -\mathbf{\frac{1}{4}}$.

Exercises 68

68.1 Differentiate x^x and e^{x^2}.

68.2 Evaluate $\int_0^\infty x^2 e^{-x} dx$.

68.3 Estimate $\sqrt{101}$ without a calculator, to 3 decimal places.

68.4 Estimate 1.02^{10} by the binomial approximation, then by $e^{0.2}$.

68.5 Find the eigenvalues of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

68.6 Find the maximum of xe^{-x} for $x > 0$.

68.7 Find $\lim_{n\to\infty} \left(1 + \frac{2}{n}\right)^n$.

68.8 Evaluate $\int_0^1 x^2 \, dx$, and use it to find $\mathbb{E}[U^2]$ and $\mathrm{Var}(U)$ for $U \sim \mathrm{Uniform}(0, 1)$.

68.9 Roughly what is the chance of exactly 500 heads in 1000 fair tosses?

68.10 Solve $x = 1 + \frac{1}{x}$ for $x > 0$. What does this number have to do with Fibonacci?

68.11 Roughly what is $\ln(1000)$? Use $\ln 2 \approx 0.693$ and $2^{10} \approx 1000$.

68.12 Roughly how many people does a coupon-collector problem with 100 types need to complete a set?

69. Brainteasers and Fermi Estimates

69.1 Thinking habits

What interviewers want to see

1. **Try small cases** to find a pattern. Then try to prove it.
2. **Work backwards** from the end state, as with the pirates and in strategy games.
3. **Look for invariants:** something that never changes, such as parity or a colouring.
4. **Change the point of view:** ants pass through each other; the fly's distance is just time $\times$ speed.
5. **Count the information:** each test or weighing has a limited number of outcomes.
6. **Talk through your reasoning:** a clean approach with a small slip beats a silent guess.

Example 69.1 (invariant: the mutilated chessboard)

Remove two opposite corners from a chessboard. Can you cover the remaining 62 squares with 1×2 dominoes?

Solution. Every domino covers one black and one white square. Opposite corners have the same colour, so 32 of one colour and 30 of the other remain. **Impossible.**

Example 69.2 (work backwards: a stone game)

Two players take turns removing 1, 2 or 3 stones from a pile of 21. Whoever takes the last stone wins. Who wins?

Solution. A pile of 4 is lost for the player to move: whatever they take, the other finishes. Likewise every multiple of 4. From 21, the first player takes 1 and leaves 20, and then always restores a multiple of 4. **The first player wins.**

Example 69.3 (change of view: ants on a stick)

100 ants walk on a 1 m stick at 1 cm/s, in random directions. When two meet, both reverse. What is the longest time before all have fallen off?

Solution. Two ants bouncing look exactly like two ants passing through each other. So each "ant identity" walks straight on, and the longest is one walking the full length: **100** seconds.

69.2 Fermi estimates

Method

Break the unknown into factors you can estimate. State assumptions clearly, round freely, and sanity-check the answer from another direction. Errors in factors tend to partly cancel. Being within a factor of 2 or 3 is usually enough.

Example 69.4 (smartphones sold each year in India)

One approach. Population about 1.4 billion. Perhaps 750 million own a smartphone. If a phone lasts about 3–4 years, about 200–250 million phones are replaced a year; adding first-time buyers gives somewhat more. Actual market figures are of order $1.5\text{–}2 \times 10^8$ phones a year, so the estimate has the right order of magnitude. A well-explained estimate matters more than the final number.

Exercises 69

69.1 Bridge. Four people with crossing times of 1, 2, 5 and 10 minutes must cross a bridge at night with one torch. At most two can cross at once, at the slower person's pace. What is the minimum total time?

69.2 Ropes. You have 2 ropes that each take 60 minutes to burn, at uneven rates. How do you measure 45 minutes?

69.3 Horses. 25 horses, a track for 5 at a time, no stopwatch. What is the minimum number of races to find the top 3?

69.4 Pirates. 5 ordered, perfectly logical pirates split 100 coins. The most senior proposes a split; if at least half (including the proposer) agree, it passes; otherwise the proposer is thrown overboard and the next proposes. What should the first propose?

69.5 Lockers. 100 lockers are closed. Person k toggles every kth locker, for $k = 1, \dots, 100$. Which lockers end up open?

69.6 Hats. 100 prisoners stand in a line, each wearing a black or white hat, seeing all those ahead. From the back, each calls a colour. How many can be guaranteed to survive with a pre-agreed strategy?

69.7 Two eggs. With two eggs and 100 floors, what is the minimum number of drops, in the worst case, to find the highest safe floor?

69.8 Poisoned wine. One of 1000 bottles is poisoned, and the poison acts in about a day. How many test prisoners do you need to identify it in a single round?

69.9 Jugs. With a 3 L and a 5 L jug, how do you measure exactly 4 L?

69.10 Bulbs. Three switches outside a closed room control 3 bulbs inside. You may enter once. How do you match them up?

69.11 Gold bar. You must pay a worker one-seventh of a gold bar per day for 7 days. What is the minimum number of cuts?

69.12 Wine and water. Move a spoon of wine into a glass of water, stir, and move a spoon of the mix back. Is there more water in the wine or wine in the water?

69.13 Missing number. All numbers from 1 to 100 but one are given in random order. How do you find the missing one with very little memory?

69.14 Handshakes. At a party, prove that the number of people who shook an odd number of hands is even.

69.15 Fermi. Estimate the number of piano tuners in a city of 1 crore people, in the style of the example above.

70. Famous Puzzles Collection

Quick-fire puzzles that appear in quant and consulting interviews. Try each before reading the solution at the end. Most have a short, clever argument – find it.

Exercises 70

70.1 Airplane seating. 100 passengers board a plane with 100 seats, each with an assigned seat. The first passenger has lost their boarding pass and sits in a random seat. Each later passenger takes their own seat if free, otherwise a random free seat. Probability that the last passenger gets their own seat?

70.2 Ants on a triangle. Three ants sit at the corners of a triangle. Each picks a direction at random and walks along an edge. Probability that no two ants collide? What about 4 ants on a square?

70.3 Ballot problem. In an election, A gets 6 votes and B gets 4. The votes are counted in random order. Probability that A stays strictly ahead throughout the count? (Use the ballot theorem: $\frac{a-b}{a+b}$.)

70.4 Socks. A drawer contains red and blue socks. Two socks drawn at random are both red with probability exactly $\frac{1}{2}$. What is the smallest possible number of socks? The smallest if the number of blue socks is even?

70.5 100 prisoners. Each of 100 prisoners may open 50 of 100 boxes looking for their own number. All must succeed. With the "follow the cycle" strategy (open the box with your own number, then the box numbered by what you found, and so on), roughly what is the success probability?

70.6 Fair coin from a biased coin. You have a coin with unknown bias p. How do you simulate a fair coin toss?

70.7 Coprime integers. Two positive integers are chosen "at random". Roughly what is the probability that they are coprime?

70.8 Boy born on Tuesday. Re-derive the $\frac{13}{27}$ answer from Chapter 56 by listing what the 27 families look like.

70.9 Die from a coin. You have a fair coin. How do you simulate a fair 6-sided die as efficiently as possible? What is the expected number of tosses?

70.10 Card colours. A deck is shuffled. You turn cards over one at a time and may say "stop" at any moment; you win if the next card is red. Is there any strategy that beats $\frac{1}{2}$?

70.11 Three-way duel. A, B and C hit with probabilities $\frac{1}{3}$, 1 and $\frac{1}{2}$. They shoot in turn starting with A, and always at a target of their choice, until one survives. Should A aim at B, at C, or deliberately miss? Explain.

70.12 Longest run. Roughly how long is the longest run of heads in 1000 fair tosses?

70.13 Birthday pairs. How many people are needed for a better-than-even chance that someone shares *your* birthday? Compare with the 23 in the classic problem.

70.14 Six before odd. What is the chance that a 6 turns up before an odd number, in repeated rolls of a die?

Part X

Mock Tests and Formula Sheets

71. Mock Test 1: Screening Paper

25 questions in 30 minutes. No calculator. Score $+1$ for a correct answer and $-\frac{1}{4}$ for a wrong one. A score of 15 or more is solid for a typical screen. Use the error-log method from Chapter 18 afterwards.

Exercises 71

71.1 35% of 240 plus 15% of 120 is: (a) 96 (b) 102 (c) 108 (d) 112

71.2 A price rises 25%. To keep spending fixed, consumption must fall by: (a) 20% (b) 25% (c) 30% (d) 33%

71.3 Cost Rs. 400, marked Rs. 600, discount 10%. Profit is: (a) 25% (b) 30% (c) 35% (d) 40%

71.4 Compound interest on Rs. 5,000 at 10% for 2 years is: (a) Rs. 900 (b) Rs. 950 (c) Rs. 1,000 (d) Rs. 1,050

71.5 $A : B = 2 : 3$ and $B : C = 4 : 5$. Then $A : C$ is: (a) $2 : 5$ (b) $8 : 15$ (c) $3 : 5$ (d) $4 : 9$

71.6 Five numbers average 24. With one removed, the average is 22. The number removed is: (a) 28 (b) 30 (c) 32 (d) 34

71.7 A does a job in 10 days and B in 15. Together they take: (a) 4 (b) 5 (c) 5.5 (d) 6 days

71.8 A 200 m train at 72 km/h crosses a 300 m platform in: (a) 15 (b) 20 (c) 25 (d) 30 seconds

71.9 A boat at 8 km/h in still water goes 30 km downstream in a 2 km/h stream. It takes: (a) 3 (b) 3.75 (c) 5 (d) 7.5 hours

71.10 The LCM of 84 and 126 is: (a) 168 (b) 210 (c) 252 (d) 504

71.11 The unit digit of 7^{35} is: (a) 1 (b) 3 (c) 7 (d) 9

71.12 The remainder when 2^{31} is divided by 5 is: (a) 1 (b) 2 (c) 3 (d) 4

71.13 If α, β are the roots of $x^2 - 7x + 12 = 0$, then $\alpha^2 + \beta^2$ is: (a) 13 (b) 19 (c) 23 (d) 25

71.14 The sum of the first 20 terms of $5, 8, 11, \ldots$ is: (a) 620 (b) 650 (c) 670 (d) 700

71.15 A rectangle has area 96 and length 12. Its perimeter is: (a) 28 (b) 32 (c) 36 (d) 40

71.16 A cylinder of radius 7 and height 10 has volume ($\pi = \frac{22}{7}$): (a) 440 (b) 770 (c) 1540 (d) 3080

71.17 The angle between the clock hands at 4:20 is: (a) 10° (b) 15° (c) 20° (d) 25°

71.18 1 January of an ordinary year is a Monday. Then 31 December is: (a) Sunday (b) Monday (c) Tuesday (d) Wednesday

71.19 The number of arrangements of the letters of LEADER is: (a) 360 (b) 480 (c) 600 (d) 720

71.20 A committee of 3 men and 2 women from 5 men and 4 women can be chosen in: (a) 40 (b) 60 (c) 84 (d) 126 ways

71.21 With two dice, the probability of a total of 9 is: (a) $\frac{1}{12}$ (b) $\frac{1}{9}$ (c) $\frac{1}{6}$ (d) $\frac{5}{36}$

71.22 The next term of 3, 5, 9, 17, 33 is: (a) 49 (b) 57 (c) 65 (d) 66

71.23 A person walks 6 km north and 8 km east. The straight-line distance from the start is: (a) 10 (b) 12 (c) 14 (d) 7 km

71.24 The pass mark is 40%. A student scores 180 and fails by 20. The maximum marks are: (a) 500 (b) 550 (c) 600 (d) 650

71.25 40 L of a mixture has milk and water in the ratio $3:2$. To make them equal, the water to add is: (a) 2 (b) 4 (c) 6 (d) 8 L

72. Mock Test 2: Harder Paper

25 questions in 45 minutes. No calculator. This is closer to a tough screen or a first-round quant test. Answers are numerical.

Exercises 72

72.1 How many positive divisors does 2520 have?

72.2 How many trailing zeros does 125! have?

72.3 Find the least number that leaves remainder 5 when divided by 6, 8 or 12, and is divisible by 7.

72.4 Find the remainder when 3^{100} is divided by 7.

72.5 Find the amount on Rs. 10,000 for 1 year at 12% compounded half-yearly.

72.6 A loan of Rs. 11,000 is repaid in 2 equal annual instalments at 20% compound interest. Find the instalment.

72.7 Goods are marked 30% above cost, then sold at a 15% discount. Find the profit percentage.

72.8 A, B and C do a job alone in 20, 30 and 60 days. A works every day and is helped by both B and C on every third day. How long does it take?

72.9 Three equal stretches are covered at 20, 30 and 60 km/h. Find the average speed.

72.10 Trains of 110 m and 90 m run in opposite directions at 40 and 32 km/h. How long do they take to pass?

72.11 In a 1 km race A beats B by 100 m, and B beats C by 100 m. By how many metres does A beat C?

72.12 If α, β are the roots of $2x^2 - 5x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

72.13 How many integers satisfy $x^2 - 7x + 10 < 0$?

72.14 Evaluate $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

72.15 An infinite GP has first term 8 and sum 12. Find its ratio.

72.16 Find the inradius of a triangle with sides 9, 40, 41.

72.17 A cone has radius 6 and height 8. Find its curved surface area, in terms of π.

72.18 Find the area enclosed by $|x - 2| + |y - 3| = 3$.

72.19 How many arrangements does BALLOON have?

72.20 How many positive integer solutions does $x + y + z = 10$ have?

72.21 Factory A makes 40% of output with 2% defects, and factory B makes 60% with 1% defects. Given a defective item, find the probability it came from A.

72.22 What is the expected number of tosses of a fair coin until HT first appears?

72.23 Find the population standard deviation of 2, 4, 6, 8, 10.

72.24 $X \sim N(100, 10^2)$. Roughly what is $P(X > 110)$?

72.25 Revenue grows from Rs. 400 crore at a 20% margin to Rs. 500 crore at an 18% margin. Find the percentage change in profit.

73. Formula Sheets

Numbers

Divisors of $\prod p_i^{a_i}$	Count $\prod(a_i+1)$; sum $\prod(1+p_i+\cdots+p_i^{a_i})$; product $N^{d/2}$
Totient	$\varphi(N) = N\prod(1-\frac{1}{p})$
HCF, LCM	$\text{HCF}\cdot\text{LCM} = ab$ for two numbers
Fermat, Euler	$a^{p-1}\equiv 1 \pmod p$; $a^{\varphi(n)}\equiv 1 \pmod n$
Unit digit	Cycle of length 1, 2 or 4
Legendre	Power of p in $n!$ is $\sum_k \lfloor n/p^k \rfloor$
Digits of N	$\lfloor \log_{10} N \rfloor + 1$

Arithmetic

Successive changes	$a+b+\frac{ab}{100}$ %
Undoing a change	$+\frac{1}{n}$ is undone by $-\frac{1}{n+1}$
Mark-up and discount	$\frac{MP}{CP} = \frac{1+p}{1-d}$; margin $= \frac{\text{markup}}{1+\text{markup}}$
False weight	Profit $= \frac{1000-w}{w}$
Interest	SI $= \frac{PRT}{100}$; $A = P(1+\frac{R}{100})^T$; CI $-$ SI $= Pr^2$ (2 years), $Pr^2(3+r)$ (3 years)
Annuity, EMI	PV $= \frac{C}{r}(1-(1+r)^{-n})$; EMI $= \frac{Pr(1+r)^n}{(1+r)^n-1}$
Perpetuity; CAGR	$\frac{C}{r}$, $\frac{C}{r-g}$; $(\frac{\text{End}}{\text{Start}})^{1/n}-1$
Alligation	$\frac{\text{cheap}}{\text{dear}} = \frac{d-m}{m-c}$
Replacement	$V(1-\frac{x}{V})^n$
Together time	$\frac{xy}{x+y}$; man-days: $\frac{M_1D_1H_1}{W_1} = \frac{M_2D_2H_2}{W_2}$
Average speed	Equal distances $\frac{2ab}{a+b}$; equal times $\frac{a+b}{2}$
After meeting	$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$
Boats	$u = \frac{d+e}{2}$; $v = \frac{d-e}{2}$
Clock	Angle $= \lvert 30H - 5.5M\rvert$; hands meet every $65\frac{5}{11}$ min
Calendar	Odd days: year 1 (leap 2); 100 yrs 5; 200 yrs 3; 300 yrs 1; 400 yrs 0

Algebra

Quadratic	$\alpha+\beta = -\frac{b}{a}$; $\alpha\beta = \frac{c}{a}$; $D = b^2-4ac$; vertex at $-\frac{b}{2a}$
Power sums	$\alpha^2+\beta^2 = s^2-2p$; $\alpha^3+\beta^3 = s^3-3ps$
$a+b+c=0$	$a^3+b^3+c^3 = 3abc$
AM–GM	$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$; $\text{GM}^2 = \text{AM}\cdot\text{HM}$
Logs	$\log_a b = \frac{\log b}{\log a}$; $\log 2 \approx .301$, $\log 3 \approx .477$, $\log 7 \approx .845$
AP	$a_n = a+(n-1)d$; $S_n = \frac{n}{2}(\text{first}+\text{last})$
GP	$S_n = a\frac{r^n-1}{r-1}$; $S_\infty = \frac{a}{1-r}$
Power sums of naturals	$\frac{n(n+1)}{2}$; $\frac{n(n+1)(2n+1)}{6}$; $\left(\frac{n(n+1)}{2}\right)^2$

Geometry

Triangle area	$\frac{1}{2}bh$; $\frac{1}{2}ab\sin C$; $\sqrt{s(s-a)(s-b)(s-c)}$; rs; $\frac{abc}{4R}$
Equilateral	$\frac{\sqrt{3}}{4}a^2$
Polygon	Angle sum $(n-2)180°$; diagonals $\frac{n(n-3)}{2}$
Circles	Chord^2: $r^2 = d^2 + (c/2)^2$; $PA \cdot PB = PC \cdot PD$; $PT^2 = PA \cdot PB$
Common tangents	$\sqrt{d^2-(r_1-r_2)^2}$; $\sqrt{d^2-(r_1+r_2)^2}$
Volumes	Cylinder $\pi r^2 h$; cone $\frac{1}{3}\pi r^2 h$; sphere $\frac{4}{3}\pi r^3$; frustum $\frac{1}{3}\pi h(R^2+r^2+Rr)$
Surfaces	Cylinder $2\pi rh$; cone πrl; sphere $4\pi r^2$
Scaling	Lengths k; areas k^2; volumes k^3
Point to line	$\frac{\lvert ax_0+by_0+c\rvert}{\sqrt{a^2+b^2}}$
Trig	$\sin^2+\cos^2=1$; $c^2 = a^2+b^2-2ab\cos C$; $\frac{a}{\sin A} = 2R$

Counting

Permutations	${}^nP_r = \frac{n!}{(n-r)!}$; repeats $\frac{n!}{p!q!\cdots}$; circle $(n-1)!$; necklace $\frac{(n-1)!}{2}$
Stars and bars	Non-negative: $\binom{n+r-1}{r-1}$; positive: $\binom{n-1}{r-1}$
Derangements	D_n: 0, 1, 2, 9, 44, 265, 1854; $D_n \approx \frac{n!}{e}$
Onto functions	$\sum_k (-1)^k \binom{r}{k}(r-k)^n$
Catalan	$\frac{1}{n+1}\binom{2n}{n}$: 1, 1, 2, 5, 14, 42, 132
No two consecutive	$\binom{n-k+1}{k}$
Grid	Rectangles $\binom{m+1}{2}\binom{n+1}{2}$; paths $\binom{m+n}{m}$

Probability and statistics

Bayes	$P(H \mid E) = \frac{P(E\lvert H)P(H)}{P(E)}$
Linearity	$\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$, always
Variance	$\mathbb{E}[X^2] - (\mathbb{E}X)^2$; $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}$
Distributions	Bin: np, npq. Geo: $\frac{1}{p}$, $\frac{q}{p^2}$. Poisson: λ, λ. Exp: $\frac{1}{\lambda}$, $\frac{1}{\lambda^2}$. Unif: $\frac{a+b}{2}$, $\frac{(b-a)^2}{12}$
Coupon collector	nH_n
Conway	Pattern wait: $\sum 2^k$ over self-overlaps k
Gambler's ruin	Fair: $\frac{i}{N}$, duration $i(N-i)$; biased: $\frac{1-(q/p)^i}{1-(q/p)^N}$
Tower, Wald	$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$; $\mathbb{E}[S] = \mathbb{E}[N]\mathbb{E}[X]$
Normal	68–95–99.7; $z = 1.645$ (5% one tail), 1.96 (2.5% one tail)
CLT	$\bar{X} \approx N(\mu, \frac{\sigma^2}{n})$; 95% CI: $\bar{x} \pm 1.96\frac{s}{\sqrt{n}}$
Kelly	$f^* = p - \frac{q}{b}$
Binomial pricing	$q = \frac{R-d}{u-d}$; price $= \frac{qV_u+(1-q)V_d}{R}$
Parity	$C - P = S_0 - Ke^{-rT}$

Solutions

Solutions to Exercises

Final answers are in **bold** at the start of each solution, so you can use this section as an answer key. Solutions are grouped by chapter number. If your method differs but the answer matches, your method is probably fine: many of these have several good routes.

Part I – Counting Fundamentals

Chapter 2: Fundamental Principles

2.1 120. Main: 6. Starter: 4 options or none = 5. Dessert: 3 or none = 4. $6 \cdot 5 \cdot 4 = 120$.

2.2 252. Total 3-digit numbers: 900. With no 7: first digit 8 choices (1–9 except 7), others 9 each: $8 \cdot 9 \cdot 9 = 648$. $900 - 648 = 252$.

2.3 1232. First digit $\in \{3, 4, 5, 6, 7\}$, last digit odd, all distinct. First digit odd (3, 5, 7): last digit is one of the other 4 odd digits; middle two: $8 \cdot 7$. $3 \cdot 4 \cdot 56 = 672$. First digit even (4, 6): last digit any of 5 odd digits: $2 \cdot 5 \cdot 56 = 560$. Total 1232. (8000 is even, so the upper bound doesn't matter.)

2.4 243. Each letter independently chooses a box: 3^5. (The letters choose boxes, not the other way round.)

2.5 819. 1-digit: 9; 2-digit: $9^2 = 81$; 3-digit: $9^3 = 729$.

Chapter 3: Permutations

3.1 60; 40. $\frac{6!}{3!\,2!} = 60$. N's together: block [NN] + B, A, A, A $\to \frac{5!}{3!} = 20$. Not adjacent: $60 - 20 = 40$.

3.2 309. Sorted letters: E, H, M, O, R, T. Before M: E, H $\to 2 \cdot 5! = 240$. Fix M; next O – smaller available: E, H $\to 2 \cdot 4! = 48$ (288). Fix MO; next T – smaller: E, H, R $\to 3 \cdot 3! = 18$ (306). Fix MOT; next H – smaller: E $\to 2! = 2$ (308). Then E, R: nothing smaller. MOTHER is next: 309.

3.3 30. Need the last two digits divisible by 4: possible endings 04, 12, 20, 24, 32, 40. Endings containing 0 (04, 20, 40): the other 3 non-zero digits fill the first three places: $3! = 6$ each $\to 18$. Endings without 0 (12, 24, 32): the remaining 3 digits include 0, which can't lead: $3! - 2! = 4$ each $\to 12$.

3.4 277200. E×3, N×3, G×2, I×2, R: $\frac{11!}{3!\,3!\,2!\,2!}$.

3.5 7560. A×4, L×2, H, B, D: $\frac{9!}{4!\,2!}$.

3.6 2520. By symmetry, half of 7!.

Chapter 4: Combinations

4.1 66. $\binom{12}{2}$.

4.2 110. $\binom{5}{3}\binom{5}{4} + \binom{5}{4}\binom{5}{3} + \binom{5}{5}\binom{5}{2} = 50 + 50 + 10$.

4.3 54. $\frac{12 \cdot 9}{2}$.

4.4 127. Mangoes 0–3 (4 choices) $\times\ 2^5$ for the fruits, minus the empty selection: $128 - 1$.

4.5 570. 10 odd and 10 even numbers. The sum is even iff the number of odd numbers is 0 or 2: $\binom{10}{3} + \binom{10}{2}\binom{10}{1} = 120 + 450$.

4.6 136. EXAMINATION: E, X, M, T, O once; A, I, N twice – 8 distinct letters, 3 of them doubled. All different: $\binom{8}{4} = 70$. One pair + 2 others: $3 \cdot \binom{7}{2} = 63$. Two pairs: $\binom{3}{2} = 3$.

Chapter 5: Circular Arrangements

5.1 **30240.** Block of 3 + 7 others = 8 units: $7! \times 3!$.

5.2 **2520.** $7!/2$.

5.3 **72.** Total $5! = 120$; together $4! \cdot 2 = 48$.

5.4 **2880.** $4! \cdot 5!$.

5.5 **(a) 96; (b) 144.** (a) 4 couple-blocks: 3!, internal 2^4: $6 \cdot 16$. (b) Men 3!, women in gaps 4!.

Chapter 6: Distributions

6.1 **84.** $\binom{9}{3}$.

6.2 **286.** Pre-give: $a' + b' + c + d = 10$: $\binom{13}{3}$.

6.3 **280.** $\frac{9!}{(3!)^3\,3!}$.

6.4 **8.** All in one box: 1; split 3+1: 4; split 2+2: $\binom{4}{2}/2 = 3$. (Or $2^4/2$.)

6.5 **165.** $i + j + k + l = 8$: $\binom{11}{3}$.

6.6 **54.** $a + b + c = 10$, $a \geq 1$, all ≤ 9. With $a' = a - 1$: $\binom{11}{2} = 55$, minus the one case $a' = 9$.

6.7 **10.** $\binom{8}{2} = 28$ minus $3 \cdot \binom{4}{2} = 18$ (one variable ≥ 4).

Chapter 7: Inclusion–Exclusion

7.1 **20.** $100 - (60 + 45 - 25)$.

7.2 **286.** Multiples of 3: 166, of 7: 71, of 21: 23. $500 - (166 + 71 - 23)$.

7.3 **540.** $3^6 - 3 \cdot 2^6 + 3$.

7.4 **78.** $120 - (24 + 24 - 6)$.

7.5 **10.** $\binom{17}{2} - 3\binom{10}{2} + 3\binom{3}{2} = 136 - 135 + 9$.

Chapter 8: Derangements

8.1 **44.** D_5.

8.2 **40.** $\binom{6}{3} D_3 = 20 \cdot 2$.

8.3 **455.** $6! - D_6 = 720 - 265$.

8.4 **1854.** $D_7 = 7D_6 - 1$ (using $D_n = nD_{n-1} + (-1)^n$).

Chapter 9: Geometry and Grids

9.1 **210.** $\binom{12}{3} - \binom{5}{3} = 220 - 10$.

9.2 **150.** $\binom{6}{2}\binom{5}{2}$.

9.3 **50.** Total 120; sharing exactly one side with the decagon: $10 \times 6 = 60$; sharing two sides: 10. $120 - 70 = 50$. (Shortcut: $\frac{n(n-4)(n-5)}{6}$.)

9.4 **60.** $\binom{4}{2}\binom{5}{2}$.

9.5 **524.** $\binom{12}{6} - \binom{6}{3}^2 = 924 - 400$.

Chapter 10: Catalan Numbers

10.1 14. $n+2=6 \Rightarrow C_4$.

10.2 42. C_5.

10.3 42. C_5.

10.4 5. C_3.

Chapter 11: Pigeonhole

11.1 25. Worst case: 2 per month = 24.

11.2 Boxes $\{1,10\},\{2,9\},\{3,8\},\{4,7\},\{5,6\}$; 6 numbers in 5 boxes force a full box, whose two numbers sum to 11. (With 5 numbers it can fail: $\{1,2,3,4,5\}$.)

11.3 (a) 10; (b) 17. (a) 3 of each colour = 9, then one more. (b) All 9 green and 7 blue = 16, then one more.

Chapter 12: Number-Based Counting

12.1 3108. $2! \cdot 14 \cdot 111$.

12.2 62. $50+10+2$.

12.3 26. $15+7+3+1$.

12.4 22. $16+5+1$.

12.5 20. Units place: 10; tens place (30–39): 10.

12.6 642. $9+90 \cdot 2+151 \cdot 3$.

Part II – Advanced Counting

Chapter 13: Recurrences

13.1 81. $a_n=a_{n-1}+a_{n-2}+a_{n-3}$, $a_0=1$: $1,1,2,4,7,13,24,44,81$.

13.2 504. A valid string starts with 0, 10 or 110, followed by a valid shorter string: $b_n=b_{n-1}+b_{n-2}+b_{n-3}$ with $b_0=1, b_1=2, b_2=4$: $7,13,24,44,81,149,274,504$.

13.3 1224. $a_n=2a_{n-1}+2a_{n-2}$, $a_1=3$, $a_2=8$: $22,60,164,448,1224$.

13.4 37. $1+\frac{8\cdot 9}{2}$.

13.5 127. 2^7-1.

13.6 22. 5^2-5+2.

13.7 41. $f(6)=4\cdot 11-3$.

Chapter 14: Generating Functions

14.1 140. Coefficient of x^{11} in $(1-x^6)^4(1-x)^{-4}$: $\binom{14}{3}-4\binom{8}{3}=364-224$.

14.2 40. Case on Rs. 10 coins: 0 tens → ways to make 20 from {1,2,5} = 29; 1 ten → make 10 = 10; 2 tens → 1. (Making 20 from {1,2,5}: fives = 0..4 leave $20,15,10,5,0$, with $11,8,6,3,1$ ways using 1s and 2s.)

14.3 375. $\binom{23}{3} - 4\binom{14}{3} + 6\binom{5}{3} = 1771 - 1456 + 60$.

14.4 25 (vs 27). Coefficient of x^6 in $(1-x^6)^3(1-x)^{-3}$: $\binom{8}{2} - 3 = 25$.

14.5 6 each. Distinct: 8, 7+1, 6+2, 5+3, 5+2+1, 4+3+1. Odd: 7+1, 5+3, 5+1+1+1, 3+3+1+1, 3+1×5, 1×8.

14.6 16. $(1 + x + x^2)^4$ has coefficients $1, 4, 10, 16, 19, 16, 10, 4, 1$.

Chapter 15: Burnside's Lemma

15.1 51. $\frac{1}{5}(3^5 + 4 \cdot 3) = \frac{255}{5}$ (the 4 non-trivial rotations of a 5-cycle each have 1 cycle).

15.2 39. Add 5 reflections, each with 3 cycles ($3^3 = 27$): $\frac{1}{10}(255 + 135)$.

15.3 57. $\frac{1}{24}(3^6 + 6 \cdot 3^3 + 3 \cdot 3^4 + 8 \cdot 3^2 + 6 \cdot 3^3) = \frac{729+162+243+72+162}{24}$.

15.4 140. Identity $2^9 = 512$; 90°/270°: centre + two 4-cycles → $2^3 = 8$ each; 180°: centre + four 2-cycles → $2^5 = 32$. $\frac{512+8+8+32}{4}$.

15.5 4. Identity: $\binom{6}{3} = 20$. Rotations by 1, 5: none (would need one colour). Rotations by 2, 4 (two 3-cycles): one cycle red, one blue: 2 each. Rotation by 3 (three 2-cycles): red count would be even: 0. $\frac{20+2+2}{6}$.

15.6 21. Rotations: $81 + 3 + 9 + 3 = 96$. Reflections: 2 through opposite beads ($3^3 = 27$ each), 2 through edges ($3^2 = 9$ each): 72. $\frac{168}{8}$.

Chapter 16: Bijections, Identities, Stirling Numbers

16.1 5120. $10 \cdot 2^9$.

16.2 12870. $\binom{16}{8}$.

16.3 495. $\binom{15-4+1}{4} = \binom{12}{4}$.

16.4 301. $S(7, 3)$.

16.5 52. Bell number $B_5 = 1 + 15 + 25 + 10 + 1$.

16.6 50. $c(5, 2)$: split as 4+1: $5 \times 3! = 30$; 3+2: $\binom{5}{3} \times 2! \times 1! = 20$.

16.7 42. $p(10)$.

16.8 1287. Hockey stick: $\binom{13}{5}$.

16.9 50. $\frac{10}{7}\binom{7}{3}$. (Same as Exercise 9.3 – triangles with no side on the decagon. Same problem in disguise.)

Chapter 17: Mixed Counting Practice

17.1 360. [AAAA] + L, L, H, B, D: 6!/2!.

17.2 84. 0 can't appear (it would have to lead), so choose 6 of 1–9: $\binom{9}{6}$.

17.3 3003. Multisets of 6 digits from 1–9: $\binom{14}{6}$.

17.4 10. 4 reds make 5 gaps; choose 3: $\binom{5}{3}$.

17.5 262495. $26^4 - 21^4$.

17.6 35. $\binom{8}{4}/2$.

17.7 1568. $64 \cdot 49/2$.

17.8 2520. ${}^7P_5$.

17.9 630. Choose the two digits $\binom{10}{2} = 45$; fill 4 places using both: $2^4 - 2 = 14$.

17.10 377. Fibonacci-type: $2, 3, 5, \ldots$, the 12th term.

Part III – Number Sense and Speed

Chapter 19: Speed Maths and Estimation

19.1 9506. Shortfalls 2 and 3: $98 - 3 = 95$, $2 \cdot 3 = 06$.

19.2 4225; 13225. $6 \cdot 7 = 42$; $11 \cdot 12 = 132$.

19.3 1591. $40^2 - 3^2$.

19.4 123. 16.67% is $\frac{1}{6}$; 738/6.

19.5 150000. $25 \times 4 = 100$ and $48 = 4 \times 12$, then $12 \times 125 = 1500$.

19.6 $\frac{15}{19}$. Gaps from 1: $\frac{2}{9} = 0.222$, $\frac{3}{14} = 0.214$, $\frac{4}{19} = 0.211$. The smallest gap wins.

19.7 10. $2^{30} \approx 10^9$ and a bit more: $1024^3 \approx 1.07 \times 10^9$, which has 10 digits.

19.8 About 800. $\frac{1}{2} \cdot 1200 + \frac{1}{4} \cdot 800 = 800$. (Exact: 798.8.)

19.9 998001. $(1000 - 1)^2$.

19.10 It passes the check, and in fact it is correct. $738 \to 0$, so the product must be $\equiv 0$. $188928 \to 36 \to 0$. ✓

Chapter 20: Number System

20.1 30; 2418. $720 = 2^4 \cdot 3^2 \cdot 5$. $d = 5 \cdot 3 \cdot 2$. $\sigma = 31 \cdot 13 \cdot 6$.

20.2 26. Multiples of 2, 3 or 5: $50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$. $100 - 74$.

20.3 127. HCF$(1651, 2032)$; see Euclid above.

20.4 548. LCM$(12, 15, 20, 54) = 540$, plus 8.

20.5 166. A gap of 2 each time: LCM$(6, 7, 8) - 2 = 168 - 2$.

20.6 (15, 180) and (45, 60). $xy = \frac{180}{15} = 12$ with x, y coprime: $(1, 12)$ and $(3, 4)$.

20.7 1. $\frac{36+63}{99}$.

20.8 16. LCM = 120 s; 1800/120 = 15 intervals, plus the start.

20.9 No; no. $437 = 19 \times 23$ and $221 = 13 \times 17$.

20.10 6^9. $36^{9/2} = 6^9$, since $d(36) = 9$.

20.11 (a) 2; (b) 15. $1800 = 2^3 \cdot 3^2 \cdot 5^2$. For a square, make the power of 2 even. For a cube, you need one more 3 and one more 5.

20.12 4. Exactly 3 divisors means the square of a prime: 4, 9, 25, 49.

20.13 72 cm; 63 tiles. In centimetres, HCF$(648, 504) = 72$. The floor takes 9×7 tiles.

Chapter 21: Remainders and Cyclicity

21.1 8. $1234 \to 1$, $5678 \to 8$.

21.2 2. $5^6 \equiv 1$; $100 \equiv 4 \pmod 6$; $5^4 = 625 \equiv 2$.

21.3 **7.** 3^{2023}: exponent $\equiv 3$, unit digit 7. 7^{2024}: exponent $\equiv 0$, unit digit 1.

21.4 **41.** $(20+1)^{37} \equiv 1 + 37 \cdot 20 = 741 \pmod{100}$; higher terms contain $20^2 = 400 \equiv 0$.

21.5 **Yes to all three.** $2^2 \equiv 1 \pmod 3$, $2^4 \equiv 1 \pmod 5$, $2^3 \equiv 1 \pmod 7$, and 60 is a multiple of 2, 4 and 3.

21.6 **1.** $4! = 24$ and all later factorials are divisible by 8. $1 + 2 + 6 = 9 \equiv 1$.

21.7 **59.** $n + 1$ is a multiple of $\mathrm{LCM}(2, \ldots, 6) = 60$.

21.8 $(10011100)_2$**;** $(234)_8$**.** $156 = 128 + 16 + 8 + 4$.

21.9 **3.** $7 \equiv -1 \pmod 4$, and the exponent is odd, so $-1 \equiv 3$.

21.10 **4.** $2^4 \equiv 1 \pmod 5$; $2026 \equiv 2$; $2^2 = 4$.

21.11 **219.** $10^{25} - 7$ is 24 nines followed by a 3. $24 \cdot 9 + 3$.

21.12 **7.** $3^3 = 27 \equiv 3 \pmod 4$, so the unit digit is that of 3^3.

21.13 **3.** $1 + 4 + 27 + 256 + 3125 = 3413$.

21.14 **3; no.** Since 4 divides 12, $n = 12k + 7 \equiv 3 \pmod 4$. Since 8 does not divide 12, the remainder mod 8 depends on k: 19 gives 3, and 7 gives 7.

Part IV – Arithmetic Aptitude

Chapter 22: Percentages

22.1 **20%.** $\frac{25}{125}$.

22.2 **+8%.** $1.2 \times 0.9 = 1.08$.

22.3 **28%.** $1 - 0.8 \times 0.9$.

22.4 **27.5%.** $\frac{0.3 \times 1.1}{1.2}$.

22.5 **1050.** Net multiplier $1.4 \times 0.6 = 0.84$, a 16% fall = 168.

22.6 **500.** Pass mark $= 178 + 22 = 200 = 40\%$.

22.7 **0.5 percentage points; 7.69%.** $\frac{0.5}{6.5}$.

22.8 **90%.** $1.2 \times 0.75 = 0.9$.

22.9 **55%.** Fail in at least one: $35 + 25 - 15 = 45$.

22.10 **26,620.** 20000×1.1^3.

22.11 **Rs. 5,78,000.** 800000×0.85^2.

22.12 **Rs. 16 and Rs. 20.** A 20% fall buys $\frac{5}{4}$ as much, so 5 kg is a quarter of the old quantity: old 20 kg, new 25 kg. $\frac{400}{25}$ and $\frac{400}{20}$.

22.13 **Down 1%; a fall of 9.09%.** $1.1 \times 0.9 = 0.99$. To undo $+\frac{1}{10}$, apply $-\frac{1}{11}$.

Chapter 23: Profit, Loss and Discount

23.1 **Rs. 900.** The swing from -12% to $+8\%$ is 20% of CP = 180.

23.2 **40%.** $\frac{MP}{CP} = \frac{1.26}{0.9} = 1.4$.

23.3 **25%.** He gives 800 g for the price of 1000 g: $\frac{200}{800}$.

23.4 50%. Per "kg" sold: revenue $= 1.5 \times 0.8 = 1.2$ (units of cost per true kg); cost of the 800 g given $= 0.8$. $\frac{1.2}{0.8} = 1.5$.

23.5 21%. CP per banana $\frac{10}{11}$, SP $\frac{11}{10}$: ratio $\frac{121}{100}$.

23.6 38.8%. $1 - 0.8 \times 0.85 \times 0.9 = 1 - 0.612$.

23.7 Rs. 945. CP $= 840/1.2 = 700$; 700×1.35.

23.8 20%. You pay for 4 and take 5, so $\frac{1}{5}$ is free.

23.9 Rs. 56. Target revenue $800 \times 1.26 = 1008$ from 18 kg.

23.10 20%. Pre-tax price $= \frac{944}{1.18} = 800$.

23.11 Rs. 500. $\frac{660}{1.2 \times 1.1}$.

23.12 37.5%. Per Rs. 100 marked: buys at 80, sells at 110. $\frac{30}{80}$.

23.13 20%. $\frac{0.25}{1.25}$.

Chapter 24: Interest

24.1 Rs. 15,000. $P(0.1)^2 = 150$.

24.2 Rs. 248. $8000 \times 0.01 \times 3.1$.

24.3 10%; Rs. 1,000. $\frac{1331}{1210} = 1.1$; $P = \frac{1210}{1.21}$.

24.4 16%. Tripling needs 200% in 12.5 years.

24.5 12.68%. $1.01^{12} - 1$.

24.6 Rs. 3,310. $\frac{1331}{1.1} + \frac{1331}{1.21} + \frac{1331}{1.331} = 1210 + 1100 + 1000$.

24.7 About 8 years. $72/9 = 8$; exact $\frac{\ln 2}{\ln 1.09} \approx 8.04$ years.

24.8 Rs. 11,236 vs Rs. 11,200. 10000×1.06^2; half-yearly earns Rs. 36 more.

24.9 Rs. 27,720. $20000 \times 1.05 \times 1.1 \times 1.2$.

24.10 10%. $1.61051 = 1.1^5$.

24.11 +Rs. 41.32; yes. $\frac{600}{1.1} + \frac{600}{1.21} = 545.45 + 495.87$, less 1000.

24.12 Rs. 62,500. $\frac{5000}{0.08}$.

24.13 About Rs. 8,885. $\frac{100000 \times 0.01 \times 1.12683}{0.12683}$.

24.14 Rs. 1,000; about Rs. 966.20. At 10%, coupon rate equals yield. At 12%: $\frac{100}{1.12} + \frac{1100}{1.2544} = 89.29 + 876.91$.

Chapter 25: Ratio, Proportion, Partnership, Ages

25.1 3:5. $A : B : C : D = 3 : 4 : \frac{14}{3} : 5$ (scale $B : C = 6 : 7$ to $4 : \frac{14}{3}$, then $C : D = 14 : 15 = \frac{14}{3} : 5$).

25.2 Rs. 3,600, Rs. 2,400, Rs. 1,800. Ratio $6 : 4 : 3$ (sum 13), $7800/13 = 600$.

25.3 15 and 25. $\frac{3k+10}{5k+10} = \frac{5}{7} \Rightarrow k = 5$.

25.4 Rs. 17,000 each. $40000 \times 12 = 480000$; $60000 \times 8 = 480000$.

25.5 Son 15, father 45. $3s + 15 = 2(s + 15)$.

25.6 Rs. 40,000 and Rs. 50,000. $4x - 7y = 5000$, $5x - 9y = 5000 \Rightarrow x = 2y$, $y = 5000$, $x = 10000$.

25.7 A Rs. 38,400; B Rs. 21,600. Salary Rs. 6,000 leaves Rs. 54,000, shared 3 : 2: Rs. 32,400 and Rs. 21,600.

25.8 40 km/h. $v = \frac{k}{\sqrt{w}}$: $k = 120$; $\frac{120}{3}$.

25.9 A 35, B 15. $A - 5 = 3(B - 5)$, $A + 5 = 2(B + 5)$.

25.10 9. $4 : 6 = 6 : x$.

25.11 30. Value $= (5 \cdot 2 + 2 \cdot 3 + 1 \cdot 5)k = 21k = 210$, so $k = 10$ and Rs. 2 coins $= 3k$.

Chapter 26: Averages

26.1 73.5. $7 \times \frac{21}{2}$.

26.2 31. The middle term is 27; the terms rise by 2 each.

26.3 36. $13 \cdot 32 + 13 \cdot 40 - 25 \cdot 36 = 416 + 520 - 900$.

26.4 39. New value = old average $+17 \times 3$: $87 = (x - 3) + 51$, so $x = 39$.

26.5 15.5. $\frac{600+144}{48}$.

26.6 31. $A + B = 80$, $B + C = 86$, $A + B + C = 135$, so $B = 80 + 86 - 135$.

26.7 Rs. 36,000. $15000 + 21 \times 1000$.

26.8 3:2. Alligation: $(85 - 76) : (76 - 70) = 9 : 6$.

26.9 20. $\frac{160-40}{6}$. (Removing two numbers that average 20 leaves the average unchanged.)

26.10 90. $75 \times 6 - 72 \times 5$.

26.11 2 years. The five were 85 in total, and are 100 today. Six people averaging 17 total 102.

Chapter 27: Mixtures and Alligation

27.1 7:3. $(420 - 350) : (350 - 320) = 70 : 30$.

27.2 4 L. Milk 48 L must be 75%: new total 64 L.

27.3 81:19. Wine left $80 \times 0.9^2 = 64.8$; water 15.2.

27.4 7:5. Gold fraction $\frac{7}{9}$ and $\frac{7}{18}$; equal weights: gold $\frac{1}{2}(\frac{7}{9} + \frac{7}{18}) = \frac{7}{12}$.

27.5 85 kg. Total rises by $8 \times 2.5 = 20$.

27.6 12 L. Total cost $48 \times 60 = 2880$ must equal 48×(new volume): 60 L.

27.7 Rs. 4,000 at 6%, Rs. 6,000 at 9%. Mean rate 7.8%: $(9 - 7.8) : (7.8 - 6) = 2 : 3$.

27.8 60 L. Milk stays 40 L; water must become 80 L, from 20 L.

27.9 72.9%. 0.9^3.

27.10 30 L of 30%; 10 L of 50%. $(50 - 35) : (35 - 30) = 3 : 1$.

27.11 1:6. A profit of $\frac{1}{6}$ on cost: for each 6 units of liquid, 1 unit of free water.

27.12 $6\frac{2}{3}$ L. Milk 15 L must become 10 L: $15\left(1 - \frac{x}{20}\right) = 10$.

Chapter 28: Time and Work

28.1 21 days. Rates 2 and 1 (units/day); total $3 \times 14 = 42$ units; A alone: 42/2.

28.2 15 days. $\frac{T-5}{20} + \frac{T}{30} = 1$.

28.3 6 days. 60 units; rates 5, 4, 3. Two days $\times 12 = 24$; remaining 36 at rate 9: 4 days.

28.4 24 hours. $\frac{1}{6} - \frac{1}{8} = \frac{1}{24}$.

28.5 8 days. $15 \cdot 12 \cdot 8 = 20 \cdot D \cdot 9$.

28.6 10 days; A alone 30 days. $2(A + B + C) = \frac{1}{12} + \frac{1}{15} + \frac{1}{20} = \frac{1}{5}$; $A = \frac{1}{10} - \frac{1}{15}$.

28.7 13 min 15 s. 60 units; rates 5, 4; each 2-minute cycle does 9. After 12 min: 54. A's minute: 59. B needs $\frac{1}{4}$ min for the last unit.

28.8 15 days. A did $\frac{4}{10}$; B did $\frac{6}{10}$ in 9 days.

28.9 4 days. 1 man $= \frac{5}{3}$ women; or 5 women $= 3$ men, so the crew is $6 + 3 = 9$ men. 3 men take 12 days: $\frac{3 \cdot 12}{9}$.

28.10 15 days. $\frac{21}{1.4}$.

28.11 15 days. With 60 units, the rates are 3, 2, 1. Each 3-day cycle does $3 + 3 + 6 = 12$ units; 5 cycles.

28.12 5760 L. Leak rate $\frac{C}{6}$ per hour. $\frac{C}{6} - 240 = \frac{C}{8}$, so $\frac{C}{24} = 240$.

28.13 10 more. 200 person-days did $\frac{1}{4}$, so the job is 800 person-days. Remaining 600 over 20 days needs 30 people.

28.14 120 L. $\frac{1}{20} + \frac{1}{24} - \frac{1}{15} = \frac{1}{40}$ of the tank per minute must be what the drain takes away, so $\frac{3}{C} = \frac{1}{40}$.

Chapter 29: Time, Speed and Distance

29.1 20 m/s; 90 km/h.

29.2 14.4 km/h. $\frac{2 \cdot 12 \cdot 18}{30}$.

29.3 48 minutes. $\frac{1}{3}$ of usual $= 16$.

29.4 60 km. They meet in $\frac{72}{18} = 4$ hours; the bird flies for 4 hours at 15 km/h. No infinite series is needed.

29.5 6 km. $\frac{D}{5} - \frac{D}{6} = \frac{12}{60}$.

29.6 400 m; 3 minutes. $\frac{0.1}{2}$ h is 3 min; 8×0.05 km.

29.7 15 minutes. $\frac{80-60}{80} = \frac{1}{4}$ of each hour.

29.8 3:4. $\sqrt{\frac{9}{16}}$.

29.9 Every 300 s; every 75 s; after 600 s. $\frac{600}{2}$; $\frac{600}{8}$; $\mathrm{LCM}(120, 200)$.

29.10 $\frac{180}{11} \approx 16.36$ **km/h.** $\frac{3}{\frac{1}{10} + \frac{1}{20} + \frac{1}{30}}$.

29.11 80 km. By the second meeting A has covered $3 \times 60 = 180$ km: 100 km to B's end, and 80 km back.

29.12 50 km/h; 48 km/h. Equal times give the arithmetic mean, equal distances the harmonic mean. Equal time is faster, because more of the distance is done at the higher speed.

Chapter 30: Trains, Boats, Races and Escalators

30.1 9 s. $\frac{180}{20}$.

30.2 72 km/h. $\frac{600}{30} = 20$ m/s.

30.3 10 s. $\frac{250}{25}$.

30.4 84 km/h; 225 m. $(v - 3) \cdot 10 = (v + 6) \cdot 9$, so $v = 84$. $L = 81 \cdot \frac{5}{18} \cdot 10$.

30.5 5 hours. $\frac{36}{18} + \frac{36}{12}$.

30.6 3:1. $u + v = 2(u - v)$.

30.7 8 km. $\frac{D}{8} + \frac{D}{4} = 3$.

30.8 19 m. $0.9 \times 0.9 = 0.81$.

30.9 32 m. B runs $\frac{3}{5} \times 80 = 48$ m in A's time.

30.10 36 s. Rates add: $\frac{1}{60} + \frac{1}{90} = \frac{1}{36}$.

30.11 1.5 km/h. Rowing back also takes 20 minutes, by the frame argument. So the hat drifts for 40 minutes: $\frac{1}{2/3}$ km/h.

30.12 2 hours. Boats' speeds relative to the water add, and the current cancels: $\frac{60}{12+18}$.

Chapter 31: Clocks and Calendars

31.1 $100°$**.** $|210 - 110|$.

31.2 $15°$**.** $|150 - 165|$.

31.3 $21\frac{9}{11}$ **and** $54\frac{6}{11}$ **minutes past 7.** $210 - 5.5M = \pm 90$.

31.4 Fast, gaining about $10\frac{10}{143}$ **minutes a day.** A correct clock has hands meeting every $\frac{720}{11}$ of its own minutes. Here that takes 65 true minutes, a rate of $\frac{144}{143}$. Over 1440 min, the gain is $\frac{1440}{143}$.

31.5 3:35. 11:60 − 8:25.

31.6 10:48 am. It shows 50 min per true 60. 240 shown is 288 true.

31.7 Saturday. 1600 years: 0. 1601–1900: 1 odd day. 1901–1999 is 99 years with 24 leap years: $123 \equiv 4$. Plus 1 for the date: $0 + 1 + 4 + 1 = 6$.

31.8 Wednesday.

31.9 Friday. 2026 is an ordinary year.

31.10 2037. Weekdays of 1 Jan: 2026 Thu, 2027 Fri, 2028 Sat, 2029 Mon, 2030 Tue, 2031 Wed, 2032 Thu (but leap), 2033 Sat, 2034 Sun, 2035 Mon, 2036 Tue, 2037 Thu.

31.11 2/7. 366 days is 52 weeks and 2 extra consecutive days. 2 of the 7 possible pairs contain a Sunday.

31.12 3. For example 2026 has three (Feb, Mar, Nov). More than 3 never happens: there are only 14 possible calendars, and a computer check of all of them gives a maximum of 3.

Part V – Algebra

Chapter 32: Linear Equations and Word Problems

32.1 36. $a + b = 9$, $9(b - a) = 27$, so $b - a = 3$.

32.2 $x = 3$**,** $y = 2$**.** Substitute $y = 3x - 7$.

32.3 $k = 4$**.** $\frac{3}{6} = \frac{k}{8} \neq \frac{7}{5}$.

32.4 30 of Rs. 2, 10 of Rs. 5. All Rs. 2 would be Rs. 80. Each Rs. 5 coin adds 3: $\frac{30}{3} = 10$.

32.5 3. $7x$ must be divisible by 4, so $x = 4, 8, 12$ with $y = 18, 11, 4$.

32.6 100. All child tickets: Rs. 6,000. Each adult adds 20: $\frac{2000}{20}$.

32.7 $\frac{1}{4}$**.** $2(n + 1) = d$ and $3n = d - 1$, so $n = 1$, $d = 4$.

32.8 $x = 1, y = 2, z = 3$**.** Subtract the third from the first: $2z = 6$. Then $x + y = 3$ and $x + 2y = 5$.

32.9 **35.** All correct $= 150$. Each wrong costs 4: $\frac{150-90}{4} = 15$ wrong.

32.10 $x = 7$**,** $y = 3$**.** $x - y = \frac{40}{10} = 4$.

32.11 **8.** Adding: $6a + 6b + 6c = 48$.

32.12 **2.** $8x + 5y = 100$ needs x divisible by 5: $x = 5, y = 12$ and $x = 10, y = 4$. $x = 0$ and $x = 12.5$ are out.

Chapter 33: Quadratics and Polynomials

33.1 $x = 3$ **or** $\frac{1}{2}$**.** $(2x - 1)(x - 3)$.

33.2 **28; 144.** $36 - 8$; $216 - 3 \cdot 4 \cdot 6$.

33.3 **4.** $16 - 4k = 0$.

33.4 $6x^2 - 5x + 1 = 0$**.** Reverse the coefficients.

33.5 **13 and 14.**

33.6 **12 and 5.** $l + w = 17$, $lw = 60$.

33.7 **14; 52.** $16 - 2$; $64 - 12$.

33.8 -11**.** $f(-1) = -2 - 3 - 1 - 5$.

33.9 **4, at** $x = 4$**.** $(x - 4)^2 + 4$.

33.10 **1560.** $13 + (-8) + (-5) = 0$, so the sum is $3 \cdot 13 \cdot (-8)(-5)$.

33.11 **45 km/h.** $\frac{180}{v} - \frac{180}{v+15} = 1 \Rightarrow v^2 + 15v - 2700 = 0$. (Check: 4 h vs 3 h.)

33.12 **Sum 6, product 6; roots 1, 2, 3.**

Chapter 34: Inequalities, Functions and Optimisation

34.1 $x < 2$ **or** $x > 3$**.**

34.2 **3.** $2 < x < 6$: 3, 4, 5.

34.3 **9.** $-1 \le x \le 7$.

34.4 **6.** $2x = 3y = 6$, so $x = 3$, $y = 2$.

34.5 **12.** $2\sqrt{36}$, at $x = \frac{3}{2}$.

34.6 **14.** Any $x \in [6, 7]$: $4 + 0 + 1 + 9$.

34.7 **90.** Total 100, with the others at $1, 2, 3, 4$.

34.8 **100; 99.** 10×10; 9×11.

34.9 **19; 49.**

34.10 **50.** At $x = y = 5$.

34.11 **38.** $1 \times 3 + 2 \times 5 + 3 \times 7 + 4 \times 1$.

34.12 $x^2 < x < \sqrt{x} < \frac{1}{x}$**. No:** for $x > 1$ it is $\frac{1}{x} < \sqrt{x} < x < x^2$.

34.13 $x \le -2$ **or** $1 \le x < 3$**.** Zeros at $-2, 1, 3$; signs from the right: $+, -, +, -$.

34.14 **64.** $4 \times 4 \times 4$.

Chapter 35: Indices, Surds and Logarithms

35.1 **8.**

35.2 **2.** $3x = 2x + 2$.

35.3 3^{20}**.** As tenth powers: 8^{10}, 9^{10}, 5^{10}.

35.4 $\sqrt{5} + 2$**.**

35.5 $\sqrt{6} + \sqrt{5}$**.** $x + y = 11$, $xy = 30$.

35.6 **4.** $x^2 = 12 + x$.

35.7 **9.** $6 + 3$.

35.8 **3.** $\log_{10} 1000$.

35.9 **2.** $x^2 - 1 = 3$; $x = -2$ is outside the domain.

35.10 **48.** $100 \times 0.477 = 47.7$.

35.11 $\frac{3}{2}$**.** $\log_{3^2} 3^3$.

35.12 **About 14.3 years.** $\frac{0.477}{0.0334}$.

35.13 **2.** $xy = \log_2 3 \cdot \log_3 4 = \log_2 4$.

35.14 **3.** The chain gives $\log_2 8$.

Chapter 36: Sequences and Series

36.1 **79; 820.** $3 + 19 \cdot 4$; $10(3 + 79)$.

36.2 **1665.** $12, 15, \ldots, 99$: 30 terms, 15×111.

36.3 **4374; 6560.** $2 \cdot 3^7$; $3^8 - 1$.

36.4 $\frac{3}{2}$**.**

36.5 **4, 6, 9.** The middle term is $\sqrt[3]{216} = 6$. $\frac{6}{r} + 6r = 13$ gives $r = \frac{3}{2}$.

36.6 **2870; 3025.** $\frac{20 \cdot 21 \cdot 41}{6}$; 55^2.

36.7 $\frac{2}{5}$**.** $\frac{1}{2} - \frac{1}{10}$.

36.8 **Rs. 18,600.** $12\,(400 + 23 \cdot 50)$.

36.9 **6.4; 4 and 16.** $\frac{64}{10}$; sum 20 and product 64.

36.10 $\frac{9}{4}$**.** $\frac{1}{(1-1/3)^2}$.

36.11 **6172839500.** $\frac{5}{9}(11111111110 - 10) = \frac{5}{9} \times 11111111100$.

36.12 **165.** 11 rows of average 15.

36.13 **9.** Telescopes to $\sqrt{100} - \sqrt{1}$.

36.14 $a = 4$**,** $d = 4$**,** $S_{15} = 480$**.** $7d = 28$. $\frac{15}{2}(4 + 60)$.

Part VI – Geometry and Mensuration

Chapter 37: Lines, Angles and Triangles

37.1 $40^\circ, 60^\circ, 80^\circ$**.** $\frac{180}{9} = 20$.

37.2 **15; 54; 7.2.** $3 \times (3,4,5)$. Altitude $= \frac{9 \cdot 12}{15}$.

37.3 $12\sqrt{5} \approx 26.83$**.** $s = 12$; $\sqrt{12 \cdot 5 \cdot 4 \cdot 3}$.

37.4 **7.** $7 < c < 15$: 8 to 14.

37.5 $9\sqrt{3}$**;** $3\sqrt{3}$**;** $\sqrt{3}$**;** $2\sqrt{3}$**.** The inradius is $\frac{1}{3}$ of the height and the circumradius $\frac{2}{3}$, by the centroid split.

37.6 **42.** Length ratio $5 : 7$.

37.7 **4:21.** $AD : AB = 2 : 5$, so areas $4 : 25$.

37.8 **6 and 9.** $BD : DC = 2 : 3$.

37.9 **20.** One third of the total.

37.10 **7 m.** Height is 12 at first, and becomes 5, so the foot moves from 5 to 12 out.

37.11 **3.** $(2,5,5)$, $(3,4,5)$, $(4,4,4)$. Each side must be less than 6.

37.12 $r = 4$**;** $R = 20.5$**.** It is a right triangle. $r = \frac{9+40-41}{2}$, $R = \frac{41}{2}$.

Chapter 38: Circles, Quadrilaterals and Polygons

38.1 $55°$**.**

38.2 $105°$**.**

38.3 **8.** $\sqrt{100 - 36}$.

38.4 **6.** $\sqrt{4 \cdot 9}$.

38.5 **13.** The radius is perpendicular to the tangent.

38.6 $24\sqrt{3}$**.** $\frac{3\sqrt{3}}{2} \cdot 16$.

38.7 **9 sides; 27 diagonals.** Exterior $40°$; $\frac{9 \cdot 6}{2}$.

38.8 **13; 120.** Half-diagonals 5 and 12.

38.9 $20\sqrt{3}$**.** $5 \cdot 8 \cdot \sin 60°$.

38.10 **91.** $\frac{1}{2} \cdot 26 \cdot 7$.

38.11 **12.** $2\sqrt{36}$.

38.12 **10.** $AB + CD = BC + DA$: $16 = 6 + DA$.

Chapter 39: Mensuration

39.1 $r = 7$**; area 154.**

39.2 **72.** $\frac{1}{2}d^2$.

39.3 **462.** $\frac{22}{7}(196 - 49)$.

39.4 **440; 748; 1540.** $2\pi rh$; $2\pi r(r + h)$; $\pi r^2 h$.

39.5 5**;** 15π**;** 12π**.**

39.6 **125.** $6a^2 = 150$.

39.7 $\frac{\pi}{6} \approx 52.4\%$**.** $\frac{\frac{4}{3}\pi(a/2)^3}{a^3}$.

39.8 **2:1.** One big sphere of radius $2r$ has area $16\pi r^2$. Eight small ones have $32\pi r^2$.

39.9 **10,00,000 L.** $20000 \times 0.05 = 1000$ m^3.

39.10 **77 m^3.** $\frac{22}{7} \times 0.07^2 \times 5000$.

39.11 **72.8%; 44%.** 1.2^3; 1.2^2.

39.12 **500.** Circumference 2.2 m; $\frac{1100}{2.2}$.

39.13 **Arc 22; area 231.** $\frac{1}{6} \cdot 2\pi \cdot 21$; $\frac{1}{2} \cdot 21 \cdot 22$.

39.14 $\frac{1}{8}$**.** The top is a similar cone with $k = \frac{1}{2}$, and volume scales by k^3.

Chapter 40: Coordinate Geometry

40.1 **10.** A 6-8-10 triangle.

40.2 $(3, 4)$**.** $\left(\frac{2\cdot 6+3\cdot 1}{5}, \frac{2\cdot 7+3\cdot 2}{5}\right)$.

40.3 $y = 2x - 1$**;** $x = \frac{1}{2}$**.**

40.4 $3x - 2y = 0$**.** Slope $-\frac{2}{3}$ becomes $+\frac{3}{2}$.

40.5 **10.** Slope 2 throughout.

40.6 **6.** A right triangle with legs 3 and 4.

40.7 **2.** $\frac{|15-5|}{5}$.

40.8 **Centre** $(-2, 5)$**; radius 4.** $(x+2)^2 + (y-5)^2 = 4 + 25 - 13$.

40.9 **32.** Diagonals of 8: $\frac{1}{2} \cdot 8 \cdot 8$.

40.10 **5.** $\gcd(12, 8) + 1$.

40.11 $(3, 2)$**.**

40.12 **6.** Intercepts 4 and 3.

Chapter 41: Trigonometry, Heights and Distances

41.1 **1.** It is $\sin 90°$.

41.2 $\frac{7}{5}$**.** $\frac{3}{5} + \frac{4}{5}$.

41.3 **1.** $\cos^2\theta \sec^2\theta$.

41.4 $5\sqrt{3} \approx 8.66$ **m.**

41.5 **About 21.96 m.** $30\sqrt{3} - 30$.

41.6 **5.** $\sqrt{9 + 16}$.

41.7 **7.** $\sqrt{25 + 64 - 40}$.

41.8 $10\sqrt{2}$**.** $b = \frac{10 \sin 45°}{\sin 30°}$.

41.9 **About 73.2 m.** $100\sqrt{3} - 100$.

41.10 **44.5.** Pair $\sin^2 k° + \sin^2(90-k)° = 1$: 44 pairs plus $\sin^2 45° = \frac{1}{2}$.

41.11 **1.** $\tan k° \tan(90-k)° = 1$, and $\tan 45° = 1$.

Part VII – Data Interpretation and Statistics

Chapter 42: Data Interpretation

42.1 **Rs. 114 crore.** $\frac{100+110+100+140+120}{5}$.

42.2 2025. $\frac{120}{500} = 24\%$. Others: 33.3%, 31.4%, 25%, 31.1%.

42.3 125%. $\frac{180-80}{80}$.

42.4 Rs. 74 crore. $20 + 36 + 18$.

42.5 About 15.4%. $\frac{427-370}{370}$.

42.6 16.2%; 10.5%. $\frac{60}{370}$; $\frac{45}{427}$.

42.7 150:277. Non-CSE: $427 - 150$.

42.8 Rs. 2.4 lakh; 17%. The rent rise is 10% of 30%, or 3% of income: $20\% - 3\%$.

42.9 $144°$**.** $40\% \times 360°$.

42.10 March. Gaps: 20, 20, 5, 25, 20, 10.

42.11 B, by 18.75 points. A: +25%. B: $+\frac{35}{80} = +43.75\%$.

42.12 Rs. 250; about 23.8%. $10 \times (130 - 105)$; $\frac{25}{105}$.

Chapter 43: Caselets, Venn Logic and Overlap

43.1 15; 35. $40 + 35 - 60$; $\min(40, 35)$.

43.2 30%. Misses total 70%.

43.3 20; 25. Union $= 120 - 45 + 5 = 80$. Only A $= 50 - 20 - 10 + 5$.

43.4 44.4%; 41.7%. $\frac{40}{90}$; $\frac{50}{120}$.

43.5 20. Both $= 50 + 35 - 70 = 15$. Only Tamil $= 35 - 15$.

43.6 20%; 50%. At least one: 50%. Both $= 40 + 30 - 50$. $\frac{20}{40}$.

43.7 About 19.2%. $\frac{10+10+3}{120}$.

43.8 X 900, Y 1100; long run X 800, Y 1200. $1000-300+200$. Balance: $0.3X = 0.2Y$, so $X : Y = 2 : 3$.

43.9 17. At least one: 25. Both: $18 + 15 - 25 = 8$. Only one: $25 - 8$.

Chapter 44: Descriptive Statistics

44.1 6.5; 7; 7. Sorted: 2, 3, 5, 7, 7, 7, 9, 12.

44.2 $\sqrt{2} \approx 1.41$**.**

44.3 90; 10.

44.4 It falls. The squared deviations' total is unchanged, but the count is larger.

44.5 1.5.

44.6 $x = 30$**; median 30.**

44.7 25; 29. Within-group 4, between-group 25.

44.8 44. $3 \cdot 48 - 2 \cdot 50$.

44.9 6.7. $1.3 + 0.9 \cdot 6$.

44.10 6. $Q_1 = 3$, $Q_3 = 9$.

44.11 Mean 10; variance 20. $\frac{1200}{10} - 10^2$.

44.12 Rs. 38,571; not from the given data. $\frac{400000-130000}{7}$. The median depends on the two middle values, which we don't know.

Part VIII – Logical Reasoning

Chapter 45: Ordering, Ranking and Linear Arrangements

45.1 29. $12 + 18 - 1$.

45.2 14. B from the left $= 40 - 15 + 1 = 26$; $26 - 11 - 1$.

45.3 42. Passed $= 7 + 28 - 1 = 34$; plus 5 and 3.

45.4 28. After the swap A is at B's old position, which is 17th from the left and 12th from the right: $17 + 12 - 1$.

45.5 Arrangement M N L J O K. (a) K; (b) 3 (N, L, J); (c) L; (d) L. Since K = J+2, J is at 1–4. M is at an end, with N next to M. Try cases: if M = 6, then N = 5, and J, K must fit with K = J+2 among seats 1–4: (J, K) = (1, 3) or (2, 4). L can't be at an end and can't touch K: with (1, 3), L must be at 2 or 4 – both touch K; with (2, 4), the free seats are 1 and 3, L = 3 touches K. So M = 1, N = 2; the free seats 3–6 take J, K, L, O with K = J+2: (J, K) = (3, 5) puts L at 4 or 6 – 4 touches K, 6 is an end; (J, K) = (4, 6) leaves 3 and 5: L = 5 touches K, so L = 3, O = 5.

Chapter 46: Circular Seating

46.1 Clockwise from A: A, E, F, C, D, B. (a) E; (b) F; (c) B. Facing outward, left = anticlockwise (-1) and right = clockwise ($+1$). Put A at 0 (seats 0–5 clockwise). C = 3. B is A's immediate left: seat 5. D = F+2 with F, D, E filling seats 1, 2, 4: the only option is F = 2, D = 4, leaving E = 1. E's neighbours are A and F, so E is not next to C, as required. (a) A's right = seat 1 = E. (b) Opposite seat 5 is seat 2 = F. (c) Second to the left of E (seat 1) is seat $1 - 2 = 5$, which is B.

46.2 2. Going left there are 2 people between them (the other way, 4).

46.3 V; W is second to the left of Q. U is at seat 6, and right $= -1$: $6 - 3 = 3$, which is V. Q is at 5 and W at 7: W is two places clockwise from Q, i.e. second to Q's left.

Chapter 47: Multi-Attribute Puzzles

47.1 Mon: Uday (Macro); Tue: Tara (Risk); Wed: Ravi (Valuation); Thu: Sana (Equity); Fri: Veer (Credit). Veer and the Macro presenter (someone other than Veer) occupy Monday and Friday. Tara is neither, and not Wednesday, so Tara is on Tuesday or Thursday. *Tara on Thursday:* Ravi and Sana need consecutive days from the remaining slots. If they take Tue–Wed, Uday is the Macro presenter and must come before Tara, so Monday; Veer is Friday; Equity and Credit go to Ravi (Tue) and Veer (Fri) – not consecutive, contradiction. If Ravi takes Monday (as Macro) and Sana Tuesday, Uday takes Wednesday and Equity/Credit fall on Tuesday and Friday – contradiction again. *Tara on Tuesday:* Uday is before her, so Monday – he is the Macro presenter; Veer is Friday. Ravi and Sana take Wed and Thu, in that order. Wednesday is Valuation (Ravi). Equity then Credit on consecutive days: Sana (Thu) Equity, Veer (Fri) Credit. **Equity: Sana; Credit: Friday.**

Chapter 48: Blood Relations

48.1 Mother. The only daughter of her mother is herself.

48.2 Granddaughter. A is C's daughter, and C is D's child.

48.3 M is P's sister-in-law. M is N's daughter; N is O's mother; so M is O's sister; O is P's husband.

48.4 Brother. Her paternal grandmother's only son is her father; his son is her brother.

48.5 Wife. R and Q are P's children; R is S's son; in a single family, S is the mother, i.e. P's wife.

48.6 Mother. K is L's wife; L is M's father; M is N's sister, so N is also L's child and K's child.

Chapter 49: Syllogisms

49.1 Only I. All+No ⇒ No manager is lazy. II contradicts "No leader is lazy".

49.2 Only I. Some+All ⇒ Some apples are fruits. II is not forced.

49.3 Only I. Some funds are bonds, and no bond is a stock ⇒ some funds are not stocks. II is not forced.

49.4 Both. Squares ⊂ rings ⊂ circles; II is the conversion of I.

49.5 Either I or II. A complementary pair with no definite link.

49.6 Only I. Cars ⊂ bikes, and bikes and trucks are disjoint, so cars and trucks are disjoint – II is impossible.

49.7 Both. Some+No gives I. Z must avoid Y, but X can extend outside Y far enough to contain all of Z.

Chapter 50: Data Sufficiency

50.1 D. Together: $1.2\,CP = 1200$.

50.2 D. Divisible by 3 and by 4 ⇒ by 12 ⇒ by 6. Neither alone settles it (e.g. 3 vs 6; 4 vs 12).

50.3 D. $x = \pm 3$; II picks 3.

50.4 C. I: $30 - 12 + 1 = 19$. II: $5 + 14 = 19$.

50.5 D. Even + odd = odd; either statement alone leaves the other variable's parity open.

50.6 D. $l + w = 14$ and $l^2 + w^2 = 100 \Rightarrow lw = \frac{196-100}{2} = 48$ (no need to find l and w).

Chapter 51: Series, Analogies, Odd One Out

51.1 42. $n(n+1)$.

51.2 343. Odd positions are cubes of odd numbers; even positions are squares of even numbers.

51.3 127. $2x + 1$.

51.4 65. $2x - 1$.

51.5 95. $2x + 1$.

51.6 65 should be 64. Cubes.

51.7 47. Differences 3, 6, 9, 12, 15.

51.8 56. Differences are squares: 1, 4, 9, 16, 25.

51.9 N. +3.

51.10 DW.

51.11 SUW. Each group starts 6 after the last; inside each, +2.

51.12 250. The others are squares.

51.13 125. $2^3 : 3^3 :: 4^3 : 5^3$.

51.14 9. The rest are prime.

Chapter 52: Coding–Decoding and Direction Sense

52.1 EPH.

52.2 OLEV. L→O, O→L, V→E, E→V.

52.3 37. $16 + 12 + 1 + 3 + 5$.

52.4 No. *pa* and *ra* go with "is" and "blue" in some order. A sentence containing "blue" but not "is" would settle it.

52.5 CIGOL.

52.6 UJHFS. +1 each.

52.7 In "air". Fish live in water, and water is called air.

52.8 5 km.

52.9 9 km north. East to $(10, 0)$; left is north to $(10, 6)$; left is west to $(0, 6)$; right from west is north to $(0, 9)$.

52.10 South. Evening shadows fall east. East on the left means facing south.

52.11 South. $45° + 135° = 180°$.

52.12 $2\sqrt{13} \approx 7.21$ **km.** Final position $(6, 4)$: $\sqrt{36 + 16}$.

Chapter 53: Cubes, Dice and Figures

53.1 8; 36; 54; 27.

53.2 96; 512. $12 \cdot 8$; 8^3.

53.3 27; 36. $3 \cdot 3 \cdot 3$; $3 \cdot 3 \cdot 4$.

53.4 8; 12; 4; 0. $(a - 2, b - 2, c - 2) = (2, 1, 0)$. Two faces: $4 \cdot 3$. One face: $2(2 + 0 + 0)$. Total 24. ✓

53.5 61. $125 - 4^3$.

53.6 4. $(n - 2)^3 = 8$.

53.7 F; D.

53.8 13; 27.

53.9 55; 225. $\frac{5 \cdot 6 \cdot 11}{6}$; 15^2.

53.10 15. All faces total 21, and the visible 3, 2, 1 total 6.

Chapter 54: Formal Logic, Truth-Tellers and Weighing

54.1 It did not rain. The contrapositive.

54.2 No. Something else may have cancelled it.

54.3 Some bankers are not rich.

54.4 A is a knight, B is a knave. If A is a knight, B is a knave, and B's claim of "same type" is false, which is consistent. If A is a knave, B is a knight, whose claim of "same type" would be true although they differ: a contradiction.

54.5 II. If exactly one is true, then exactly two are false, which is what II says.

54.6 3; 4. $3^3 = 27$, and 28 exceeds it.

54.7 6; 4. $1, 2, 4, 8, 16, 32$ (sum 63, and five weights reach only 31); $1, 3, 9, 27$.

54.8 Only I. "Only" makes II far too strong.

54.9 Priya had a ticket. No: a ticket is necessary, not sufficient, so it doesn't follow that a ticket-holder got in.

54.10 The box labelled Mixed. It is not mixed, so the fruit tells you what it really holds. Suppose it is an apple: that box is Apples. The box labelled Oranges cannot be Oranges and cannot be Apples, so it is Mixed. The last is Oranges.

Part IX – Probability and Quant Interviews

Chapter 55: Foundations

55.1 5/12. Sums 2, 3, 5, 7, 11 occur $1+2+4+6+2=15$ times out of 36.

55.2 15/28. $\frac{5\cdot 3}{\binom{8}{2}}$.

55.3 0.72. $\frac{648}{900}$.

55.4 0.3; yes. $1-(0.5+0.4-0.2)$; $0.5\times 0.4=0.2=P(A\cap B)$.

55.5 4/13. $\frac{4+13-1}{52}$.

55.6 89/144 ≈ 0.618. $1-\frac{12\cdot 11\cdot 10\cdot 9\cdot 8}{12^5}$.

Chapter 56: Conditional Probability and Bayes

56.1 1/2; 1/3.

56.2 1/2. $\frac{\frac{1}{3}\cdot 1}{\frac{1}{3}\cdot 1+\frac{2}{3}\cdot\frac{1}{2}}$.

56.3 6/17. Defects per 1000 items: A 5, B 6, C 6. $\frac{6}{17}$.

56.4 3/8 (vs 1/4 for staying). The car is behind one of the other 3 doors with probability $\frac{3}{4}$; after the host removes a goat, you pick the right one of the 2 remaining with probability $\frac{1}{2}$.

56.5 1/2. Of the cases where a random door shows a goat, half have the car behind your door.

56.6 19/117 ≈ 16.2%. Per 1000: 20 sick, 19 positive; 980 healthy, 98 positive.

56.7 4/5. $\frac{3+1}{3+2}$.

Chapter 57: Counting-Based Probability

57.1 2162/54145 ≈ 0.040. $\frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}=\frac{6\cdot 17296}{2598960}$.

57.2 5108/2598960 ≈ 0.197%. $4\binom{13}{5}-40$.

57.3 6 dice. Probabilities ≈ 0.665, 0.619, 0.597. (This was Pepys's question to Newton.)

57.4 23/42. $\frac{\binom{4}{2}\binom{6}{2}+\binom{4}{3}\binom{6}{1}+\binom{4}{4}}{\binom{10}{4}}=\frac{90+24+1}{210}$.

57.5 2/7. A's two neighbours are 2 of the 7 seats B could take.

57.6 2/3. $\frac{40}{60}$ (Exercise 3.1).

Chapter 58: Distributions

58.1 3/8. $\binom{4}{2}/16$.

58.2 109/262144 ≈ 0.0004; 2.5. $\sum_{k=8}^{10}\binom{10}{k}(\frac{1}{4})^k(\frac{3}{4})^{10-k}=\frac{405+30+1}{4^{10}}=\frac{436}{1048576}$; mean $np=2.5$.

58.3 25/216. $(\frac{5}{6})^2\frac{1}{6}$.

58.4 $\approx$ 0.224. $e^{-3}\frac{27}{6} = 4.5e^{-3}$.

58.5 6; 30. $\frac{1}{p}$ and $\frac{1-p}{p^2} = \frac{5/6}{1/36}$.

Chapter 59: Expectation

59.1 91/36 $\approx$ 2.53. min + max = sum of the dice, so $\mathbb{E}[\min] = 7 - \frac{161}{36}$.

59.2 $\approx$ 0.537. Each box is empty with probability $(\frac{4}{5})^{10}$; times 5.

59.3 53/5 = 10.6. Expected non-aces before the first ace = $\frac{48}{5}$, plus the ace.

59.4 5.5. Runs = 1+ (number of changes); each of 9 adjacent pairs differs with probability $\frac{1}{2}$.

59.5 21. 6 rolls expected; the 5 non-six rolls average 3 (uniform on 1–5), giving 15, plus 6. (By Wald's identity.)

59.6 $\approx$ 1.19. $\binom{30}{2} \cdot \frac{1}{365} = \frac{435}{365}$.

Chapter 60: First-Step Analysis

60.1 14. $2 + 4 + 8$.

60.2 42. $6 + 36$.

60.3 10; 8. HTH: $2 + 8$. HHT: only the full overlap, 8.

60.4 1/2; 25 rounds. $\frac{5}{10}$; $i(N - i) = 5 \cdot 5$.

60.5 $\approx$ 0.116. $\frac{1-1.5^5}{1-1.5^{10}} = \frac{-6.594}{-56.665}$.

60.6 2/5. Shift by 2: start at 2, absorbing barriers at 0 and 5: $\frac{2}{5}$.

60.7 THH, winning with probability 3/4. Once a T appears, HHT can only appear after an HH that follows a T – but then THH appears first. HHT wins only if the first two tosses are HH ($\frac{1}{4}$).

Chapter 61: Continuous and Geometric

61.1 5/9. $1 - (\frac{40}{60})^2$.

61.2 3/4. The longer piece is uniform on $[\frac{1}{2}, 1]$.

61.3 3/4. $\frac{n}{n+1}$.

61.4 1/2; $\pi/4$. A triangle of area $\frac{1}{2}$; a quarter disc of area $\frac{\pi}{4}$.

61.5 1/2. $\frac{4}{2^3}$.

61.6 0.36. Points more than 0.1 from every side form an inner square of side 0.8: $1 - 0.64$.

61.7 5 minutes; 10 minutes. With regular buses your wait is uniform on $[0, 10]$. With Poisson buses you are more likely to arrive during a long gap, and memorylessness gives an expected wait equal to the full mean gap.

Chapter 62: Optimal Stopping and Games

62.1 14/3. Keep the first roll if it is 5 or 6 (beats 4.25); keep the second if it is 4, 5 or 6 (beats 3.5); otherwise take the third. $\frac{11}{6} + \frac{4}{6} \cdot \frac{17}{4} = \frac{14}{3}$.

62.2 Rs. 3; stop on 4 or more (stopping or continuing on a 3 are equally good). Let V be the game's value before paying for a roll: $V = -1 + \mathbb{E}[\max(X, V)]$. Trying a threshold of 4: $V = -1 + \frac{4+5+6}{6} + \frac{3}{6}V \Rightarrow V = 3$, which is consistent (you keep a roll exactly when it beats $V = 3$).

62.3 (a) 10%; (b) 10%. (a) $0.55 - 0.45 = 0.10$. (b) $0.4 - \frac{0.6}{2} = 0.10$.

62.4 Reject about 37; success probability ≈ 37%.

62.5 10 losses; 1/1024; zero. $1 + 2 + \cdots + 512 = 1023$ covers 10 losses. Ruin needs 10 losses in a row: $\frac{1}{1024}$. Expected profit per cycle $= \frac{1023}{1024} \cdot 1 - \frac{1}{1024} \cdot 1023 = 0$.

Chapter 63: Variance, Covariance and Correlation

63.1 $\frac{35}{12} \approx 2.92$; $\frac{35}{6} \approx 5.83$. $\frac{91}{6} - \frac{49}{4}$. Variances add.

63.2 30; 21; about 4.58. np; $np(1-p)$.

63.3 13. $4 \cdot 4 + 9 \cdot 1 - 2 \cdot 2 \cdot 3 \cdot 1$.

63.4 $\frac{1}{\sqrt{2}} \approx 0.707$. $\mathrm{Cov}(X, X+Y) = \sigma^2$; divide by $\sigma \cdot \sqrt{2}\sigma$.

63.5 $\frac{6}{7} \approx 85.7\%$. $\frac{0.09-0.03}{0.04+0.09-0.06}$.

63.6 1.5. $\frac{15}{\sqrt{100}}$.

63.7 $\frac{1225}{1296} = \left(\frac{35}{36}\right)^2 \approx 0.945$. $\mathbb{E}[\max \cdot \min] = \mathbb{E}[XY] = \frac{49}{4}$; subtract $\frac{161}{36} \cdot \frac{91}{36}$.

63.8 9. $5 + 2^2$.

63.9 9. $\frac{30}{\sqrt{n}} = 10$.

63.10 -1. $Y = 10 - X$ is an exact decreasing linear function of X.

Chapter 64: The Normal Distribution, LLN and CLT

64.1 About 2.3%; about 68%. $z = 2$; within one SD.

64.2 About 68. One SD above the mean.

64.3 About 71%. SD 10. $z = \frac{10.5}{10} = 1.05$; $1 - 2(0.147)$. (Exact: 70.6%.)

64.4 About 3.7%. Mean 350; SD $\sqrt{100 \cdot \frac{35}{12}} \approx 17.08$. $z = \frac{380.5-350}{17.08} \approx 1.79$.

64.5 4 times. Error scales as $\frac{1}{\sqrt{n}}$.

64.6 $55\% \pm 4.9\%$. $1.96\sqrt{\frac{0.55 \cdot 0.45}{400}}$.

64.7 About 15.9%; about 2.3%. $X + Y \sim N(30, 25)$, so $z = 1$. $X - Y \sim N(-10, 25)$, so $z = 2$.

64.8 About 23.8%. $1.5\% \times \sqrt{252}$.

64.9 Yes, just. $z = \frac{60-50}{5} = 2 > 1.96$.

64.10 About 6.7%; about 7. $z = 1.5$.

Chapter 65: Conditional Expectation and Recursion

65.1 12.25. 3.5×3.5.

65.2 7. 2×3.5.

65.3 $\frac{1}{2}$. $\frac{1/3}{1-\frac{2}{3}\cdot\frac{1}{2}}$.

65.4 $\frac{11}{4}$. $\frac{10+1}{3+1}$.

65.5 $\frac{2}{3}$. Each draw has the same chance as the first, by the urn's symmetry.

65.6 $\frac{1}{4}$. Each break halves the expected length.

65.7 Mean Rs. 2,000; SD about Rs. 652. Var $= 10 \cdot 50^2 + 10 \cdot 200^2 = 425000$.

65.8 $\frac{1}{4}$. $\mathbb{E}[U_2 \mid U_1] = \frac{U_1}{2}$.

65.9 1; 5. Before the first 6, each of the faces 1–5 is as likely to come up as the 6 itself, so it occurs on average once.

65.10 $\frac{4}{3}$. Per toss, $P(\text{heads}) = \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 = \frac{2}{3}$. By linearity, $2 \cdot \frac{2}{3} = \frac{4}{3}$.

Chapter 66: Random Walks and Markov Chains

66.1 $\pi = \left(\frac{5}{6}, \frac{1}{6}\right)$. $0.1\pi_A = 0.5\pi_B$.

66.2 $\frac{5}{16}$. $\frac{\binom{6}{3}}{64}$.

66.3 6. Mirror the start to -2. Then go from -2 to $+2$ in 6 steps: 5 ups and 1 down. (Total 15 end at $+2$.)

66.4 0.6; 24. $\frac{6}{10}$; 4×6.

66.5 $\frac{2}{3}$; $\frac{8}{27}$. $\frac{q}{p} = \frac{2}{3}$, and then cube it.

66.6 $\frac{1}{3}$. A fair game: $\frac{10}{30}$.

66.7 8. A regular graph, so π is uniform on the 8 vertices.

66.8 16. $4(8-4)$.

66.9 $\frac{1}{4}$. $\frac{5-3}{8}$.

66.10 25. 5×5.

Chapter 67: Trading, Options and Pricing

67.1 $\frac{35}{36} \approx 0.97$. $S - 7 = 1, \ldots, 5$ with counts $5, 4, 3, 2, 1$: $\frac{5+8+9+8+5}{36}$.

67.2 5; 5. $q = \frac{1}{2}$. Payoffs 10 or 0. By parity, equal at $K = S_0$ with zero rates.

67.3 About 5.88. $q = \frac{1.02-0.9}{0.2} = 0.6$. $\frac{0.6 \cdot 10}{1.02}$.

67.4 9.75. $P = C - S_0 + Kd = 10 - 100 + 99.75$.

67.5 Yes; about 1.95%. $\frac{1}{1.9} + \frac{1}{2.2} \approx 0.981 < 1$. Return $\frac{1}{0.981} - 1$.

67.6 0.5. $\frac{15-5}{20}$.

67.7 0.5; 7%. $4 + 0.5 \cdot 6$.

67.8 20%; about 2.0% (log) per bet. $0.6 - 0.4$. $0.6 \ln 1.2 + 0.4 \ln 0.8 \approx 0.0201$.

67.9 Long call plus short put; $|S_T - K|$; 90 and 110.

67.10 Rs. 4; keep 3 or more (a 3 exactly ties), and reroll 1 or 2. Let V be the value. You keep x whenever $x \geq V - 1$. Trying a threshold of 3: $V = \frac{3+4+5+6}{6} + \frac{2}{6}(V-1)$, so $V = 4$, and $V - 1 = 3$ is consistent.

67.11 Rs. 2; no finite fair price. $\mathbb{E}[n] = 2$. The second is the St Petersburg game, with infinite expectation.

67.12 Adverse selection. Their purchase hints that the first 10 had more heads than the expected 5. Your fair value should move up, and so should your quote.

Chapter 68: Maths Refresher

68.1 $x^x(\ln x + 1)$**;** $2xe^{x^2}$**.**

68.2 2. 2!.

68.3 10.05. $10 + \frac{1}{20}$. (True: 10.0499.)

68.4 1.2; 1.2214. True: 1.2190.

68.5 3 and 1. Trace 4, determinant 3.

68.6 $\frac{1}{e}$**, at** $x = 1$**.** $f' = (1 - x)e^{-x}$.

68.7 e^2**.**

68.8 $\frac{1}{3}$**;** $\frac{1}{3}$**;** $\frac{1}{12}$**.** $\frac{1}{3} - \frac{1}{4}$.

68.9 About 2.5%. $\frac{1}{\sqrt{\pi \cdot 500}} \approx 0.0252$.

68.10 $\frac{1+\sqrt{5}}{2} \approx 1.618$**.** The ratio of successive Fibonacci numbers tends to it.

68.11 About 6.9. 10×0.693.

68.12 About 520. $100H_{100} \approx 100(\ln 100 + 0.577) \approx 518$.

Chapter 69: Brainteasers and Fermi Estimates

69.1 17 minutes. 1 and 2 cross (2); 1 returns (1); 5 and 10 cross (10); 2 returns (2); 1 and 2 cross (2). The idea is to get the two slowest over together.

69.2 Light rope A at both ends and rope B at one end. A finishes in 30 minutes. At that moment, light B's other end: its remaining 30 minutes takes 15 more.

69.3 7. 5 heats, then a race of the 5 winners (6 races). For the seventh, race whoever could still be 2nd or 3rd: the 2nd and 3rd of the winners' race, the 2nd and 3rd in the champion's heat, and the 2nd in the heat of the runner-up.

69.4 98, 0, 1, 0, 1. Backwards. With 2 pirates, the senior takes all (needs just their own vote). With 3: $(99, 0, 1)$. With 4: $(99, 0, 1, 0)$. With 5: $(98, 0, 1, 0, 1)$. Each pirate is bought for one more than they get next round.

69.5 The perfect squares: 10 lockers. A locker is toggled once per divisor, and only squares have an odd number of divisors.

69.6 99. The last in line calls the parity of the black hats they see (a 50% chance for them). Each person ahead can then deduce their own hat.

69.7 14. Drop at floors $14, 27, 39, \ldots$, with gaps that fall by 1. You need the smallest k with $\frac{k(k+1)}{2} \geq 100$.

69.8 10. Number the bottles in binary. Prisoner j drinks from each bottle with bit j set. The deaths spell out the bottle's number, and $2^{10} = 1024 \geq 1000$.

69.9 Fill 5, pour into 3 (leaves 2 in the 5). Empty the 3. Pour the 2 into the 3. Fill 5, and top up the 3 with 1 more. That leaves 4 in the 5.

69.10 Turn switch 1 on for ten minutes, then off. Turn switch 2 on, and go in. The lit bulb is 2; the warm dark bulb is 1; the cold dark bulb is 3.

69.11 2 cuts. Pieces of 1, 2 and 4 sevenths, with change given back. It is binary again.

69.12 Equal. Each glass ends with its original volume, so whatever wine left the wine glass was replaced by an equal amount of water.

69.13 Subtract the sum of the given numbers from 5050. Alternatively, XOR all of them against $1, \ldots, 100$, avoiding overflow.

69.14 Each handshake adds 2 to the total of everybody's counts, so that total is even. Even-count people contribute an even total, so the odd-count people must also total an even number, and only an even number of odd values can do that.

69.15 Tens to a few hundred. For example: 1 crore people is about 25 lakh homes. If 1 in 200 has a piano, that is 12,500 pianos, and each may be tuned once a year. A tuner might do 2 a day for 250 days, or 500 a year. That gives about **25** tuners. A much lower ownership rate in India would make this lower still. Any answer with clear assumptions is fine.

Chapter 70: Famous Puzzles

70.1 1/2. The process ends when someone takes either seat 1 or seat 100, and at every step these two are equally likely to be chosen. So the last passenger finds seat 100 free exactly when seat 1 was taken first: probability $\frac{1}{2}$. (The answer is the same for any number of passengers ≥ 2.)

70.2 1/4; 1/8. No collision only if all ants go the same way: $\frac{2}{2^3}$ and $\frac{2}{2^4}$.

70.3 1/5. $\frac{6-4}{10}$.

70.4 4 socks (3 red, 1 blue); 21 socks (15 red, 6 blue). Need $\frac{r(r-1)}{(r+b)(r+b-1)} = \frac{1}{2}$; check $\frac{3}{4} \cdot \frac{2}{3} = \frac{1}{2}$ and $\frac{15}{21} \cdot \frac{14}{20} = \frac{1}{2}$.

70.5 $\approx$ 31%. All succeed iff the random permutation has no cycle longer than 50: $1 - \sum_{k=51}^{100} \frac{1}{k} \approx 1 - \ln 2 \approx 0.31$. (Opening boxes at random would give 2^{-100}.)

70.6 Toss twice: HT $\to$ "heads", TH $\to$ "tails", HH or TT $\to$ repeat. HT and TH both have probability $p(1-p)$. (Von Neumann's trick.)

70.7 $6/\pi^2 \approx 0.61$**.** A prime p divides both with probability $\frac{1}{p^2}$, so $P = \prod_p (1 - \frac{1}{p^2}) = \frac{1}{\zeta(2)} = \frac{6}{\pi^2}$.

70.8 Families with the older child a Tuesday-boy: 14 (younger child any type). With the younger child a Tuesday-boy: 14. Both: 1. Total 27. Both boys: $7 + 7 - 1 = 13$.

70.9 4 tosses by the simple method. Toss 3 times to get a number from 0 to 7. If it is 6 or 7, repeat. Each round works with probability $\frac{3}{4}$, so the expected number of rounds is $\frac{4}{3}$, and thus 4 tosses. (A cleverer scheme reuses the leftover randomness and takes $\frac{11}{3}$.)

70.10 No. The fraction of reds left is a martingale, so every stopping rule wins exactly $\frac{1}{2}$. Equivalently, saying "stop" and winning on the next card has the same chance as winning on the *last* card, which is $\frac{1}{2}$.

70.11 Deliberately miss. If A hits C, then B shoots A dead. If A hits B, then A faces C with C shooting first, and wins only $\frac{1}{4}$ of the time; overall, aiming at B gives $\frac{11}{36}$. If A misses, B (certain) shoots C, the stronger threat, and A gets the first shot at B: a duel in which A wins $\frac{1}{3}$ at once and otherwise dies. Missing gives A a win chance of $\frac{1}{3}$, the best available.

70.12 About 9 or 10. $\log_2(1000) \approx 10$. There are about 500 runs of heads, and each is longer than k with probability 2^{-k}.

70.13 253. $1 - \left(\frac{364}{365}\right)^n > \frac{1}{2}$ needs $n \geq 253$. Interestingly, this is $\binom{23}{2}$, the number of pairs among 23 people.

70.14 $\frac{1}{4}$. Only rolls of 1, 3, 5 or 6 matter, and 6 is one of those 4.

Part X – Mock Tests

Chapter 71: Mock Test 1

71.1 (b) 102. $84 + 18$.

71.2 (a) 20%. $\frac{25}{125}$.

71.3 (c) 35%. SP $= 540$.

71.4 **(d) Rs. 1,050.** $5000(1.21-1)$.

71.5 **(b)** $8:15$**.** $A:B:C=8:12:15$.

71.6 **(c) 32.** $120-88$.

71.7 **(d) 6.** $\frac{10\cdot 15}{25}$.

71.8 **(c) 25.** $\frac{500}{20}$.

71.9 **(a) 3.** $\frac{30}{10}$.

71.10 **(c) 252.** HCF 42; $\frac{84\cdot 126}{42}$.

71.11 **(b) 3.** $35\equiv 3 \pmod 4$.

71.12 **(c) 3.** $31\equiv 3 \pmod 4$; $2^3=8$.

71.13 **(d) 25.** $49-24$.

71.14 **(c) 670.** $10(5+62)$.

71.15 **(d) 40.** Width 8.

71.16 **(c) 1540.**

71.17 **(a)** $10°$**.** $|120-110|$.

71.18 **(b) Monday.** An ordinary year is 52 weeks and 1 day, so the last day matches the first.

71.19 **(a) 360.** $\frac{6!}{2!}$ (E twice).

71.20 **(b) 60.** $\binom{5}{3}\binom{4}{2}=10\cdot 6$.

71.21 **(b)** $\frac{1}{9}$**.** $(3,6),(4,5),(5,4),(6,3)$.

71.22 **(c) 65.** $2x-1$.

71.23 **(a) 10.**

71.24 **(a) 500.** 200 is 40%.

71.25 **(d) 8.** Milk 24, water 16.

Chapter 72: Mock Test 2

72.1 **48.** $2^3\cdot 3^2\cdot 5\cdot 7$: $4\cdot 3\cdot 2\cdot 2$.

72.2 **31.** $25+5+1$.

72.3 **77.** $24k+5$ divisible by 7: $k=3$.

72.4 **4.** $3^6\equiv 1$; $100\equiv 4 \pmod 6$; $81\equiv 4$.

72.5 **Rs. 11,236.** 10000×1.06^2.

72.6 **Rs. 7,200.** $x\cdot\frac{1.2+1}{1.44}=11000$.

72.7 **10.5%.** $1.3\times 0.85=1.105$.

72.8 **15 days.** 60 units at rates 3, 2, 1. Each 3 days: $3+3+6=12$ units.

72.9 **30 km/h.** $\frac{3}{\frac{1}{20}+\frac{1}{30}+\frac{1}{60}}$.

72.10 **10 s.** 72 km/h = 20 m/s; $\frac{200}{20}$.

72.11 **190.** $0.9^2=0.81$.

72.12 $\frac{5}{3}$**.** $\frac{s}{p}=\frac{5/2}{3/2}$.

72.13 **2.** $2 < x < 5$: 3, 4.

72.14 **3.** The chain gives $\log_2 8$.

72.15 $\frac{1}{3}$**.** $\frac{8}{1-r} = 12$.

72.16 **4.** A right triangle: $\frac{9+40-41}{2}$.

72.17 60π**.** $l = 10$.

72.18 **18.** A diamond with diagonals of 6.

72.19 **1260.** $\frac{7!}{2!2!}$.

72.20 **36.** $\binom{9}{2}$.

72.21 $\frac{4}{7}$**.** $\frac{0.8}{0.8+0.6}$.

72.22 **4.**

72.23 $\sqrt{8} \approx 2.83$**.**

72.24 **About 16%.** $z = 1$.

72.25 $+12.5\%$**.** Profit 80 becomes 90.